

How State-of-the-Art Embeddings Are Generated

A Wide Literature Review on Maximizing the Downstream Benefit of Learned
Representations,
for the Ride-Hailing & Food-Delivery Market-State Embedding Project

212 annotated sources · 197 unique · 11 research areas · live web-verified

Compiled 1 August 2026 · Deep-research compilation (8 parallel area surveys + adversarial completeness audit + 3
gap-fill surveys)

Companion documents: 00_problem_and_head_selection_framework.md · 01_levers_endogeneity_and_control.md

Contents

- 1 Executive Summary
- 2 Scope & Method
- 3 Coverage Audit
- 4 Self-Supervised Representation Learning Foundations (18 sources)
- 5 Time-Series Representation Learning & Foundation Models (19 sources)
- 6 Tabular & Heterogeneous Business Data Embeddings (19 sources)
- 7 Auxiliary Tasks & Multi-Task Learning as Latent-Space Organizers (23 sources)
- 8 Representations for Control & Causal / Counterfactual Inference (19 sources)
- 9 Industrial Embedding Systems at Marketplaces & Two-Sided Platforms (18 sources)
- 10 Spatiotemporal, Urban & Graph Embeddings (22 sources)
- 11 Evaluating Embedding Quality & Operating Embeddings Over Time (19 sources)
- 12 Causal Inference & Experiment Design Under Marketplace Interference gap-fill (13 sources)
- 13 Temporal Distribution Shift: Generalization & Continual Adaptation of the Encoder gap-fill (15 sources)
- 14 Temporal Distribution Shift: Generalization & Continual Adaptation of the Encoder gap-fill (15 sources)
- 15 Master Source Index (197 unique sources)

1 Executive Summary

This report surveys how state-of-the-art embeddings are generated and — just as importantly — how researchers and industry ensure the maximum benefit is actually harvested from them downstream. It was compiled for the market-state embedding programme (replacing hand-engineered features for ride-hailing and food delivery with a learned per-zone, per-time representation consumed by pricing, dispatch, ETA, repositioning, incentive and forecasting systems). Eleven research areas were surveyed with live web verification; every cited source carries the URL at which it was verified. Ten findings cut across all areas:

1. **The field has converged on “predict in latent space + explicit anti-collapse regularization” as the way to generate embeddings.** Reconstruction has been deprecated as the primary objective in every surveyed community — vision SSL (MAE’s weak linear probes vs. the JEPA line), time series (forecasting-as-pretraining), RL (self-predictive ZP losses replacing pixel decoders), and spatiotemporal modeling (masked latent modeling). The programme’s founding skepticism about plain encoder–decoders is independently confirmed by every literature at once. The 2025 frontier (LeJEPA) even derives the isotropic-Gaussian embedding distribution as provably optimal for the linear-probe consumption model this project has committed to.
2. **Auxiliary heads are validated as the central latent-space organizer — and head selection is now a measured problem, not a whiteboard debate.** Theory (Maurer 2016; Tripuraneni 2020; JEPA-auxiliary theory 2025) says benefits require the head set to *span* the function space of downstream tasks (“task diversity”), while near-duplicate heads buy provably little. Practice offers cheap instruments: gradient task-affinity (TAG), linearized subset scoring over hundreds of candidate heads at ~3% of training cost (Grad-TAG), and dense banks of auto-generated heads (Proto-Value Networks). The programme’s “representative heads” question has an operational answer: enumerate candidates, measure affinity in one

instrumented run, select the smallest subset covering a basket of downstream probes under a diversity constraint.

3. **Linear probes are simultaneously the field's constraint, contract, and quality metric.** Frozen-encoder linear probing is the standard evaluation everywhere from tabular foundation models to urban region embeddings; industry goes further and treats “a consumer needs a deep model to extract signal” as a defect of the embedding. For versioned contracts, MDL/description-length probing is more stable than raw probe accuracy, and data-efficiency curves (probe skill at 1%/10% of labels) predict how the embedding serves new consumers.
4. **Action-conditioning is non-negotiable — and it is this project's genuine novelty.** No public time-series, tabular, or urban foundation model conditions on control actions; the machinery must be imported from RL and causal inference: action-conditioned latent transition losses (SPR / TD-MPC2 / BYOL-AC), propensity heads that preserve rather than delete confounding information (DragonNet — *do not* put CFR-style balancing penalties on a shared embedding), and varying-coefficient dose-response heads for the continuous surge lever (VCNet). This corroborates the companion levers/endogeneity analysis (doc 01) from an independent direction.
5. **Every invariance is deleted information.** Augmentation and normalization choices are the true prior: instance normalization — universal in time-series foundation models — strips the absolute demand level that pricing needs, so removed statistics must be re-injected; amplitude augmentations would erase weather extremes. Each candidate invariance must be validated against the downstream probe suite before adoption, because the deletion is permanent.
6. **One multi-task embedding serving many consumers is the shipped industrial standard.** Pinterest (ItemSage, OmniSearchSage), Netflix's personalization foundation model, and Airbnb all converged on a single multi-task embedding over per-team embeddings — better metrics, lower maintenance — distributed as a feature-store artifact with declared refresh tiers, backward-compatible version governance (BCT / stationary target geometries), Matryoshka-style nested dimensions, and per-consumer negative-flip release gates.
7. **Spatial machinery must earn its complexity.** Learned adaptive adjacency beats hand-specified geographic graphs (Graph WaveNet lesson), travel-time and flow-based edges beat geodesic hex adjacency — but STID showed that per-hex and per-time-slot identity embeddings attached to an MLP capture most of the value. The one-week STID ablation is the cheapest de-risking experiment available before committing to a GNN backbone.
8. **Evaluate against the near future, and prefer boring retraining.** Wild-Time shows a ~20% performance cliff between random and temporally-forward splits — forward splits must be the release contract. Across large-scale studies, no temporal-domain-generalization method consistently beats frequent warm-started retraining (with shrink-and-perturb and replay of tail regimes); the “future-decay curve” of frozen probes sets each city's retraining cadence.
9. **The experimentation substrate is part of the representation system.** Marketplace interference (surge in one hex moves drivers and riders in neighbors) breaks unit-level A/B testing and every SUTVA-based causal tool. Clustered switchback designs, always-on lever dithering with logged propensities, and interference-aware estimators (Differences-in-Q's, Horvitz–Thompson, VCDP) are the upstream preconditions for any causal head — and for honestly measuring the embedding's own business lift. The causal content of the embedding is bounded by the randomization present in the logs.
10. **The time bucket itself is a hand-engineered feature.** The raw data is an asynchronous event stream; neural temporal point process intensity heads with censored likelihoods recover latent demand suppressed under surge — the principled fix for the “realized demand understates true demand” problem

— and JODIE-style projection makes embeddings queryable at any timestamp. Verdict from the literature: build event-level machinery as *training signal and input pathway*, not as the serving backbone.

An honest caveat the literature insists on: deep embeddings do not automatically beat strong feature engineering + gradient-boosted trees on tabular data, and complex spatiotemporal models often lose to embarrassingly simple baselines. Every area recommends benchmarking against the incumbent (hand-engineered features, GBDT, STID, persistence forecasts, latest warm-started checkpoint) before believing any lift — and reporting the concatenation condition (features + embedding) to measure what the embedding actually adds.

2 Scope & Method

Eight parallel research agents each surveyed one area of the literature with live web search (August 2026), under instructions to verify every citation against an arXiv/publisher page or an authoritative index before including it, and to cover seminal foundations, the 2024–2026 state of the art, and industrially deployed methods. An adversarial completeness critic then audited the combined coverage against the project’s requirements and identified three genuine gaps — causal inference under marketplace interference, temporal domain generalization / continual encoder adaptation, and continuous-time event-stream encoding — each of which was closed by a dedicated gap-fill survey. The result is **212 annotated sources** across eleven chapters (197 unique after cross-area de-duplication). Each chapter contains: a narrative state-of-the-art summary, the cross-cutting principles of that area, concrete recommendations mapped to the market-state embedding project, and an annotated bibliography. A de-duplicated master index closes the report.

Verification caveat: URLs were checked live at compilation time by the research agents; before quoting any specific claim externally, spot-check the source. Industrial engineering-blog entries document deployed practice but are not peer-reviewed.

3 Coverage Audit (Adversarial Completeness Critique)

The eight areas give strong, largely non-overlapping coverage of encoder objectives (SSL, masked/latent prediction, contrastive), heterogeneous input encoding (tabular, time-series, spatiotemporal-graph), latent-space organization via auxiliary heads, individual-level action-conditioned/causal representation, industrial serving patterns, and evaluation/versioning. Three genuine seams remain, all sitting exactly where the project’s hardest requirements live. (1) Most consequential: every causal tool surveyed (TARNet/DragonNet/VcNet, OPE, switch-level self-prediction) assumes SUTVA — no survey covers causal inference and experiment design under cross-unit interference, even though surge in one hex moves drivers and riders in neighboring hexes and the whole marketplace-experimentation literature (switchbacks, two-sided-platform bias, mean-field equilibrium designs, spatial designs — several built on ride-hailing simulators, ICML 2025) exists precisely for this; without it, both the causal auxiliary heads and every embedding-version A/B will be biased. (2) The evaluation-versioning area covers drift DETECTION and version COMPATIBILITY but no survey covers drift ADAPTATION — temporal domain generalization and continual encoder updating (Wild-Time, DRAIN and successors), which governs retraining cadence and forward-generalization of a continuously retrained market-state encoder. (3) All eight surveys silently assume the input is already a per-zone-per-time-bucket tensor; the bucketing itself is a hand-engineered feature the project claims to eliminate. The continuous-time event-stream literature (JODIE/TGN dynamic graph embeddings, neural temporal point processes) encodes raw trip/request events directly, yields embeddings queryable at arbitrary timestamps, and handles demand censoring — an entire missed subfield.

Items checked and judged NOT top-3 gaps: discrete/VQ latent tokenization, hierarchical forecast reconciliation, embedding privacy/inversion, LLM event-signal encoding — real but second-order relative to the above.

Gap 1: Causal inference and experiment design under marketplace interference (spatial spillovers, equilibrium effects, switchbacks). Surge price/radius in one zone shifts driver supply and rider demand in neighboring zones and over time, violating SUTVA. The control-causal survey's machinery (TARNet, DragonNet, VCNet, standard IPS/DR OPE) and the evaluation survey's A/B-gated promotion process are all unit-level i.i.d. tools; applied naively they give biased estimates of both the surge dose-response the causal heads are supposed to encode and the business lift of any new embedding version. This literature also prescribes WHAT randomization/logging (switchback schedules, cluster/spatial designs, lever dithering) the platform must run so that action-conditioned representations are identifiable at all — an upstream data-collection requirement no survey states. *(closed by its own chapter below)*

Gap 2: Temporal distribution shift: temporal domain generalization and continual adaptation of the encoder (not just drift detection). Market data is relentlessly non-stationary — seasonality, city growth, competitor moves, and the platform's own policy changes — and the encoder will be retrained on a cadence forever. The evaluation-versioning survey covers detecting drift and keeping versions compatible, and the control-causal survey covers policy-induced (performative) shift, but no survey covers the methods for making the encoder itself robust to, or adaptive under, gradual temporal drift: how to train so the embedding generalizes to the NEAR FUTURE (the only distribution that matters at serving time), when to warm-start vs fully retrain, and how to evaluate with temporally forward splits. Wild-Time shows existing methods lose ~20% from in-distribution to future-time data — a directly project-relevant failure mode nobody surveyed. *(closed by its own chapter below)*

Gap 3: Encoding raw event streams: continuous-time dynamic graph embeddings and neural temporal point processes. All eight surveys implicitly accept the per-zone-per-time-bucket aggregation as given — but the bucket width and zone partition are themselves hand-engineered features, the very thing the project set out to eliminate. The underlying data is an asynchronous marked event stream (ride requests, driver session starts/stops, cancellations, deliveries) on a spatial graph. The continuous-time literature — JODIE/TGN-style dynamic graph embeddings and neural temporal point processes — encodes such streams directly, yields embeddings queryable at ANY timestamp (JODIE's projection operator solves the staleness-between-refresh-ticks serving problem), and TPP intensity modeling is the principled way to represent demand under censoring (requests suppressed or abandoned when surge is high — a bias every bucketed count inherits and no survey mentions). *(closed by its own chapter below)*

4 Self-Supervised Representation Learning Foundations

State of the art

Self-supervised representation learning (SSL) evolved through five paradigm families, each solving the same two-part problem — make embeddings invariant to nuisance variation while preventing collapse to trivial solutions — with different machinery. Contrastive methods (CPC 2018 introduced InfoNCE; SimCLR and MoCo scaled it in 2020) pull augmented views together and push negatives apart. Non-contrastive self-distillation (BYOL, SimSiam, DINO, 2020-21) showed negatives are unnecessary: an asymmetric predictor plus stop-gradient/momentum-teacher suffices. Redundancy-reduction methods (Barlow Twins, VICReg, 2021-22) replaced those architectural tricks with explicit statistical regularization of the embedding covariance — decorrelate dimensions, keep per-dimension variance up. Masked modeling, descended from BERT (Devlin et al. 2018, arXiv:1810.04805), reached vision with MAE (2022) and became modality-general with data2vec, which crucially predicts latent teacher representations rather than raw inputs. That idea — predict in representation

space, not input space, so capacity is not wasted modeling unpredictable low-level detail — is the core of LeCun's 2022 position paper "A Path Towards Autonomous Machine Intelligence" (OpenReview, verified via <https://www.semanticscholar.org/paper/775f42ed458b8c5bof2094ea4ff5b64c557b1a34>) and the JEPA line: I-JEPA (2023), V-JEPA 2 (June 2025, which adds an action-conditioned predictor stage for planning), and LeJEPA (Nov 2025), which derives the isotropic Gaussian as the downstream-risk-optimal embedding distribution and replaces all heuristics with a single distribution-matching regularizer. Current vision SOTA is DINOv3 (Aug 2025, 7B params, Gram anchoring), the first SSL model to beat weakly-supervised counterparts broadly — but its recipe is an engineering-heavy fusion of DINO self-distillation + iBOT masked-latent prediction + Koleo/Gram regularization, reinforcing that the families have converged toward "latent prediction + explicit spectral/distributional regularization." Transfer theory explains why these embeddings work downstream: alignment/uniformity (Wang & Isola 2020) and the dimensional-collapse literature (Jing et al. 2022; 2024-25 cures include orthogonality regularization, LDReg, WERank) show that linear-probe utility tracks how uniformly information is spread across embedding dimensions — effective rank is the key health metric. For heterogeneous non-vision data there is no single winning paradigm (survey: Wang et al. 2024, arXiv:2402.01204): masked/predictive objectives tend to win for forecasting- and imputation-shaped downstream tasks (TimeMAE, MOMENT), contrastive wins for instance discrimination, and tabular practice is converging on hybrid masked+contrastive objectives with domain-agnostic corruptions (SCARF); the JEPA latent-prediction principle is the most natural fit for noisy spatiotemporal state data because it never asks the encoder to reconstruct raw stochastic demand. Open problems: augmentation design outside vision, training stability without heuristics (LeJEPA's target), and evaluation beyond linear probes.

Key principles

- Every successful SSL method = an invariance/prediction term + an explicit anti-collapse mechanism (negatives, stop-gradient + predictor, teacher centering, variance/covariance regularization, or distribution matching). Choose the mechanism you can measure and monitor in production; the trend from 2020 to 2026 (SimCLR -> VICReg -> LeJEPA) is from implicit architectural tricks toward explicit, provable regularizers.
- Predict in latent space, not input space (LeCun's core argument, validated by data2vec, I-JEPA, V-JEPA 2): reconstruction objectives spend capacity modeling irreducible input noise and organize the latent space around appearance, not semantics — MAE's weak linear-probe results versus its fine-tuning results are the canonical evidence. For noisy market telemetry, latent prediction keeps only the forecastable structure.
- Views/augmentations are the prior: whatever you make the encoder invariant to is information permanently deleted from the embedding. In vision this is a settled recipe; for heterogeneous tabular/spatiotemporal data it is the main design decision (SCARF marginal-resampling corruption, temporal shifts, spatial-neighbor masking) and each candidate invariance must be validated against downstream probes before adoption.
- Downstream (linear) usefulness is governed by how information is spread across dimensions: alignment + uniformity (Wang & Isola) and high effective rank (Jing et al.) predict transfer; dimensional collapse silently destroys tasks that need the discarded dimensions. Embedding covariance eigenspectrum, effective rank, alignment, and uniformity should be first-class release metrics for every embedding version.
- The projection/predictor head is a buffer that absorbs objective-specific invariance — always publish the pre-projector representation and discard heads (SimCLR finding, reconfirmed across BYOL/VICReg/DINO).
- Temporal predictive objectives (CPC, JEPA over time) produce 'state-like' embeddings: forcing the future to be predictable from the embedding is the self-supervised analog of decision relevance for forecasting, pricing, and dispatch consumers.
- Frozen-encoder evaluation (linear probe + k-NN probe) is the field's standard contract and matches the multi-consumer deployment model: one frozen backbone, many cheap heads (DINOv2/v3 practice). Fine-tuning per consumer breaks embedding compatibility and forfeits the shared-representation benefit.

- Scale-then-distill is the industrial deployment pattern (DINOv2, DINOv3): train the largest affordable teacher once, distill a family of student encoders for different latency budgets; distillation against a fixed teacher also gives a mechanism for cross-version embedding compatibility.
- Pretraining data curation and balancing matter as much as the objective (DINOv2 LVD-142M, DINOv3): dominant clusters (high-volume zones/hours) must be rebalanced or the embedding under-represents the tail states where decisions matter most.
- No paradigm universally wins off vision (tabular SSL survey, arXiv:2402.01204; time-series evidence): masked/latent-predictive objectives transfer best to forecasting/imputation-shaped tasks, contrastive to instance-level discrimination, and the tabular consensus is hybrid masked+contrastive with domain-agnostic corruptions — so a market-state encoder should combine a JEPA-style predictive loss with a distributional regularizer rather than bet on one pure family.
- Heuristic-free stability is a production feature, not an aesthetic one: methods with EMA schedules, warmups, and fragile hyperparameters (classic DINO/BYOL) are risky to retrain continuously; LeJEPA-style single-hyperparameter objectives whose training loss correlates with downstream quality make automated retraining and regression detection tractable.

Recommendations for the market-state embedding project

- Adopt a JEPA-style backbone objective for the market encoder: from a context of observed (zone, time-bucket) cells, predict the latent representations of masked neighboring zones and future time-buckets, conditioned on space-time position (I-JEPA/CPC pattern). Do NOT use raw-value reconstruction as the primary loss — MAE-style objectives demonstrably underperform on the linear-probe interface the project has committed to; keep reconstruction at most as a low-weight auxiliary.
- Use explicit distributional regularization for anti-collapse instead of negative pairs: VICReg's variance+covariance terms or LeJEPA's SIGReg. Negatives are hazardous here because adjacent zones/time-buckets are false negatives, and the explicit regularizers double as monitorable metrics. LeJEPA is the recommended default: single hyperparameter, no EMA/schedule heuristics, and its isotropic-Gaussian target is provably optimal for exactly the linear-probe downstream contract planned.
- Design views with SCARF-style random feature corruption (resample a random feature subset from empirical marginals) plus temporal jitter and spatial-neighbor masking; run an ablation matrix confirming each corruption does not degrade any planned downstream probe before locking it in — each invariance deletes information permanently.
- Handle policy endogeneity with V-JEPA 2's two-stage recipe: (1) pretrain the state encoder action-free (mask/predict objective over logged market data), (2) freeze it and train a separate action-conditioned latent-dynamics predictor that takes surge price/radius as explicit inputs. This keeps control-lever effects out of the state embedding itself while yielding a latent counterfactual simulator ('what does the market state become under surge x') usable by pricing and repositioning.
- Publish the pre-projector embedding; discard projection/predictor heads at export. Version the encoder, and for $v(n+1)$ use distillation against $v(n)$ outputs on an anchor dataset (DINOv2/v3 pattern) to give consumers approximate backward compatibility; distill one large teacher into small students for latency-constrained consumers (dispatch) vs. larger ones for offline forecasting.
- Make four embedding-health metrics release gates for every version: covariance eigenspectrum / effective rank (dimensional collapse check, Jing et al.), alignment on known-equivalent market states, uniformity/Gaussianity of the embedding distribution (SIGReg statistic), and a fixed battery of frozen linear + k-NN probes for each downstream task family (surge accuracy, ETA error, demand forecast, imputation of masked zones).

- Rebalance pretraining data across zones, hours, weather regimes, and event states (DINOv2-style curation): logged data over-represents high-volume downtown weekday states, but decision value concentrates in tail states (storms, events, supply shocks); cluster-and-rebalance before pretraining.
- Keep the planned auxiliary linear prediction heads, but treat them as probes-with-gradient: small linear heads on the frozen-ish trunk predicting decision-relevant targets (completed trips, wait times, conversion) are consistent with SSL theory (they raise linear accessibility) — while ensuring the SSL predictive loss, not the heads, carries most of the gradient, so the embedding does not collapse to only currently-tracked KPIs.
- Benchmark honestly against the incumbent: run every downstream probe with (a) hand-engineered features, (b) SSL embedding, (c) concatenation. Tabular SSL literature (survey arXiv:2402.01204) shows deep SSL does not automatically beat strong feature engineering + GBDTs; the concatenation condition reveals what the embedding adds, and hybrid predictive+contrastive objectives are the current tabular consensus if the pure JEPA objective underperforms.
- Expect the temporal-prediction horizon to be a key hyperparameter (CPC finding): predicting multiple future buckets (e.g., 15/30/60 min) forces the embedding to encode slow market dynamics (supply drift, demand waves) rather than transient noise — align the horizon set with the decision cadences of pricing (minutes) and repositioning (tens of minutes).

Annotated bibliography (18 sources)

2018 van den Oord, A., Li, Y., Vinyals, O. (2018). Representation Learning with Contrastive Predictive Coding. arXiv:1807.03748.

arxiv.org/abs/1807.03748

Introduces InfoNCE: an autoregressive model summarizes past observations into a context vector and is trained to distinguish the true future latent from negatives, maximizing a lower bound on mutual information between context and future. The encoder is generic (conv/RNN), applied to audio, images, text, and RL. Prediction happens in latent space over multiple future steps, so the representation captures slowly-varying 'state' rather than local noise.

Relevance to this project: The closest published template for a market-state encoder: embed a zone's recent history such that the future is linearly predictable from it. Predicting multiple time-buckets ahead in latent space directly optimizes the forecasting-relevance the project's downstream heads need.

2020 Chen, T., Kornblith, S., Norouzi, M., Hinton, G. (2020). A Simple Framework for Contrastive Learning of Visual Representations (SimCLR). ICML 2020 / arXiv:2002.05709.

arxiv.org/abs/2002.05709

Two augmented views of each sample are encoded, passed through a small MLP projection head, and trained with NT-Xent (normalized InfoNCE) using other in-batch samples as negatives. Showed that augmentation composition matters more than architecture, and that the projection head absorbs augmentation-invariance so the pre-projector representation transfers better. Requires large batches for enough negatives.

Relevance to this project: Establishes two rules the project should adopt regardless of paradigm: view/augmentation design is the real prior you impose, and the published embedding should be the pre-projector representation, not the head output.

2020 He, K., Fan, H., Wu, Y., Xie, S., Girshick, R. (2020). Momentum Contrast for Unsupervised Visual Representation Learning (MoCo). CVPR 2020 / arXiv:1911.05722.

arxiv.org/abs/1911.05722

Frames contrastive learning as dictionary lookup: a queue of past embeddings serves as negatives, produced by a momentum-averaged (EMA) copy of the encoder so keys stay consistent as the encoder evolves.

Decouples negative count from batch size, making contrastive training feasible at modest compute.

Relevance to this project: The EMA-teacher trick recurs in BYOL/DINO/data2vec and is the standard way to get stable targets during training — useful for a market encoder retrained continuously on streaming data. The queue idea also suggests how to hold negatives across time-buckets without giant batches.

2020 Wang, T., Isola, P. (2020). Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere. ICML 2020 / arXiv:2005.10242.

arxiv.org/abs/2005.10242

Proves that the InfoNCE loss asymptotically decomposes into alignment (positive pairs map close) and uniformity (features spread uniformly on the hypersphere), and that directly optimizing these two measurable quantities matches contrastive performance. Both metrics are cheap to compute on any embedding batch.

Relevance to this project: Gives the project two concrete, model-agnostic health metrics for every embedding version: alignment against known-equivalent market states and uniformity of the embedding distribution. Uniformity is essentially why linear probes work — information is spread instead of clumped.

2020 Grill, J.-B., Strub, F., Althé, F., et al. (2020). Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning (BYOL). NeurIPS 2020 / arXiv:2006.07733.

arxiv.org/abs/2006.07733

An online network predicts the EMA target network's embedding of a different augmented view of the same input; no negatives at all. Collapse is avoided by the asymmetry: a predictor MLP on the online branch plus the slow-moving target. Matched contrastive SOTA and proved negatives are not fundamental.

Relevance to this project: Removes the biggest contrastive headache for market data: defining negatives when nearby zones/adjacent time-buckets are genuinely similar (false negatives). A BYOL-style setup sidesteps that entirely.

2021 Chen, X., He, K. (2021). Exploring Simple Siamese Representation Learning (SimSiam). CVPR 2021 / arXiv:2011.10566.

arxiv.org/abs/2011.10566

Strips BYOL to its minimum: a shared Siamese encoder, a predictor on one branch, and a stop-gradient on the other — no momentum encoder, no negatives, no large batches. Shows stop-gradient is the essential ingredient preventing collapse, with the predictor implicitly implementing an EM-like alternating optimization.

Relevance to this project: The minimal, cheapest-to-train non-contrastive baseline — a good first ablation for the market encoder before adding heavier machinery. Its lesson: the anti-collapse mechanism, not scale, is what you must get right.

2021 Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., Joulin, A. (2021). Emerging Properties in Self-Supervised Vision Transformers (DINO). ICCV 2021 / arXiv:2104.14294.

arxiv.org/abs/2104.14294

Self-distillation with no labels: a student ViT matches the EMA teacher's softmax output distribution over multi-crop views, with centering and sharpening of teacher outputs preventing collapse. Discovered that SSL ViT features contain emergent semantic structure (segmentation-quality attention maps) and excel at k-NN retrieval without any fine-tuning.

Relevance to this project: Demonstrates that a well-trained SSL embedding supports nonparametric (k-NN) consumption — relevant for 'find historical market states similar to now' use-cases in dispatch/incentive tooling. Multi-crop (global vs. local views) maps naturally to city-wide vs. zone-local market context.

2021 Zbontar, J., Jing, L., Misra, I., LeCun, Y., Deny, S. (2021). Barlow Twins: Self-Supervised Learning via Redundancy Reduction. ICML 2021 / arXiv:2103.03230.

arxiv.org/abs/2103.03230

Computes the cross-correlation matrix between embeddings of two distorted views across the batch and drives it toward identity: diagonal terms enforce invariance, off-diagonal terms decorrelate dimensions so no information is redundantly encoded. Needs no negatives, stop-gradients, momentum encoders, or large batches.

Relevance to this project: Decorrelated dimensions are exactly what makes linear probes and downstream linear heads efficient — each embedding coordinate carries distinct information. A natural objective when the project wants a compact, non-redundant market-state vector consumed by many teams.

2022 Bardes, A., Ponce, J., LeCun, Y. (2022). VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. ICLR 2022 / arXiv:2105.04906.

arxiv.org/abs/2105.04906

Three explicit loss terms: invariance (MSE between views), variance (hinge keeping each embedding dimension's std above a threshold — directly preventing collapse), and covariance (pushing off-diagonal covariance to zero — preventing dimensional collapse). Because the branches need not share weights or architecture, it extends to multimodal/heterogeneous inputs.

Relevance to this project: The most transferable anti-collapse recipe for non-vision data: no augmentation-dependent negatives, and the two branches can be different encoders (e.g., demand-side vs. supply-side towers of the market encoder). Its variance/covariance terms double as monitorable production metrics.

2022 Jing, L., Vincent, P., LeCun, Y., Tian, Y. (2022). Understanding Dimensional Collapse in Contrastive Self-supervised Learning. ICLR 2022 / arXiv:2110.09348.

arxiv.org/abs/2110.09348

Shows that even contrastive methods with negatives suffer dimensional collapse — the embedding covariance spectrum decays so representations occupy a low-rank subspace — driven by strong augmentation and implicit weight-matrix rank regularization. Proposes DirectCLR, which optimizes only a sub-vector of the representation directly, outperforming SimCLR with a trainable projector.

Relevance to this project: Defines the failure mode most dangerous to this project: an embedding that looks fine on one downstream task but has silently discarded the dimensions other tasks need. Mandates tracking the embedding eigenspectrum/effective rank as a release gate for every embedding version.

2022 He, K., Chen, X., Xie, S., Li, Y., Dollár, P., Girshick, R. (2022). Masked Autoencoders Are Scalable Vision Learners (MAE). CVPR 2022 / arXiv:2111.06377.

arxiv.org/abs/2111.06377

Masks 75% of image patches; an encoder sees only visible patches and a lightweight decoder reconstructs the masked pixels, making pretraining 3x+ faster. The asymmetric design keeps the encoder free of mask tokens. Fine-tuning results are excellent, but linear-probe results lag joint-embedding methods — reconstruction targets retain low-level information not organized for linear separability.

Relevance to this project: A cautionary result for the project's plan: pure reconstruction (autoencoding) objectives underperform on linear probes, precisely the consumption interface the project has committed to. Masking is still valuable — as the corruption strategy, not as a pixel/raw-value reconstruction target.

2022 Baevski, A., Hsu, W.-N., Xu, Q., Babu, A., Gu, J., Auli, M. (2022). data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language. ICML 2022 / arXiv:2202.03555.

arxiv.org/abs/2202.03555

One recipe across three modalities: a student sees a masked input and regresses the EMA teacher's contextualized latent representations (averaged over top-K layers) of the full input. Because targets are latent rather than raw (no pixels, no discrete tokens), the same objective works for any input type. Matched or beat modality-specific SSL in speech, vision, and NLP.

Relevance to this project: The strongest evidence that masked-latent-prediction is the modality-agnostic paradigm — directly applicable to a heterogeneous demand/supply/weather feature stack where 'reconstruct the raw input' is ill-posed across mixed continuous/categorical/count data. Mask zones or feature groups, predict their latent state.

2022 Bahri, D., Jiang, H., Tay, Y., Metzler, D. (2022). SCARF: Self-Supervised Contrastive Learning using Random Feature Corruption. ICLR 2022 / arXiv:2106.15147.

arxiv.org/abs/2106.15147

Tabular-native contrastive learning: views are made by corrupting a random subset of features, replacing each with a draw from that feature's empirical marginal distribution, then training with InfoNCE. No domain augmentations needed; evaluated on 69 OpenML-CC18 tabular datasets, improving classification and robustness to label noise and boosting semi-supervised pipelines.

Relevance to this project: Solves the project's hardest SSL question — what is an 'augmentation' for zone-level market features — with a principled, distribution-preserving corruption. Marginal-resampling of feature subsets (e.g., corrupt weather but keep demand) is a drop-in view generator for the market encoder.

2023 Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., Ballas, N. (2023). Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture (I-JEPA). CVPR 2023 / arXiv:2301.08243.

arxiv.org/abs/2301.08243

From one context block of an image, a narrow ViT predictor predicts the EMA target-encoder representations of several masked target blocks, conditioned on their positions — prediction entirely in representation space, with no hand-crafted augmentations and no pixel reconstruction. Learns semantic features faster than MAE and without the invariance priors of view-based methods.

Relevance to this project: The architectural blueprint for a market-state encoder: from observed zones/time-buckets, predict the latent state of unobserved ones conditioned on location/time. Abstracts away unpredictable micro-noise (individual ride requests) while retaining forecastable structure — exactly what pricing/dispatch state should contain.

2023 Oquab, M., Darcet, T., Moutakanni, T., et al. (2023). DINOv2: Learning Robust Visual Features without Supervision. TMLR 2024 / arXiv:2304.07193.

arxiv.org/abs/2304.07193

Combines DINO self-distillation with iBOT masked-latent prediction and KoLeo regularization, trained on LVD-142M, an automatically curated dataset built by retrieval-based deduplication and rebalancing. A 1B-parameter ViT teacher is trained once, then distilled into a family of smaller ViTs. Frozen features beat OpenCLIP across classification, depth, and segmentation without fine-tuning.

Relevance to this project: The industrial template for 'one frozen embedding, many consumers': invest in one large teacher, distill compatible smaller encoders for different latency budgets, and evaluate everything with frozen-backbone probes. Also shows curation/balancing of pretraining data matters as much as the objective — relevant when some city-zones dominate logged data.

2025 Assran, M., et al. (2025). V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. arXiv:2506.09985.

arxiv.org/abs/2506.09985

Stage 1: action-free JEPA pretraining (mask-and-predict in latent space) on 1M+ hours of internet video yields a general visual world model with SOTA motion understanding and action anticipation. Stage 2 (V-JEPA 2-AC): the encoder is frozen and a new action-conditioned predictor is trained on only ~62 hours of unlabeled robot trajectories, then used zero-shot for model-predictive control (pick-and-place on real Franka arms) by planning in embedding space.

Relevance to this project: Directly addresses the project's policy-endogeneity tension: learn the state encoder action-free, then bolt on an action-conditioned latent-dynamics predictor that takes surge price/radius as inputs. That predictor becomes a counterfactual simulator — 'what does the market latent become if we set surge to x' — for pricing and repositioning.

2025 Siméoni, O., Vo, H.V., Seitzer, M., et al. (2025). DINOv3. arXiv:2508.10104.

arxiv.org/abs/2508.10104.

Scales SSL to a 7B-parameter ViT on 1.7B curated images with a constant (schedule-free) training recipe, introducing Gram anchoring: a regularizer that keeps the student's patch-level Gram matrix (pairwise feature similarity structure) close to an earlier checkpoint's, preventing the degradation of dense/local features during long training. Distilled into ViT-S/B/L/H+ and ConvNeXt variants; first SSL model to beat weakly-supervised (CLIP-style) counterparts across a broad suite including dense prediction.

Relevance to this project: Proves SSL embeddings can outperform label/text-supervised ones at scale, and that preserving local (per-zone analog: per-patch) feature quality requires explicit regularization — the market analog is keeping zone-level detail intact while the global city-state objective dominates. Its schedule-free recipe also eases continual retraining.

2025 Balestrieri, R., LeCun, Y. (2025). LeJEPa: Provable and Scalable Self-Supervised Learning Without the Heuristics. arXiv:2511.08544.

arxiv.org/abs/2511.08544.

Proves the isotropic Gaussian is the embedding distribution minimizing worst-case downstream prediction risk, then enforces it with SIGReg (Sketched Isotropic Gaussian Regularization): random 1-D projections of the embedding batch are tested against Gaussian via an Epps-Pulley-style statistic, in linear time and memory. Combined with the JEPa prediction loss it needs one trade-off hyperparameter, no stop-gradient, no teacher-student, no schedulers, ~50 lines of code, and is stable across ResNets/ViTs/ConvNets and domains; training loss correlates strongly with downstream performance.

Relevance to this project: The current best-practice starting point for the project: its optimality target is defined literally by downstream (linear) prediction risk — the project's stated interface — and heuristic-free stability plus loss-tracks-downstream-quality are exactly what a production retraining pipeline needs. SIGReg is also a cheap continuous monitor of embedding-distribution health across versions.

5 Time-Series Representation Learning & Foundation Models

State of the art

Time-series representation learning has moved through three distinct waves. (1) 2021-2023: contrastive/self-supervised methods (TNC, TS-TCC, TS2Vec, CoST) learned embeddings from unlabeled series by defining what should be invariant (augmented views, temporal neighborhoods) and what should be discriminated, evaluated almost universally by LINEAR PROBES on frozen embeddings — TS2Vec's hierarchical timestamp-to-series representations and CoST's causally-motivated seasonal-trend disentanglement are the high-water marks; SimMTM then showed masked reconstruction beats contrast when the reconstruction task is softened (neighbor-aggregated recovery). (2) 2023: PatchTST established patching (sub-series tokens) + instance normalization as the canonical tokenization for transformers, with channel-independence as a robustness trade-off. (3) 2023-2026: the foundation-model wave — Chronos (values-as-language tokens), TimesFM (decoder-only next-patch prediction), Moirai (any-variate, any-frequency masked encoder trained on the 27B-observation LOTSA corpus), MOMENT (embedding-oriented, multi-task), Lag-Llama (lags+calendar covariates), then scaling/specialization via sparse MoE (Time-MoE at 2.4B params on 300B points; Moirai-MoE with token-level expert specialization), and generative/continuous heads (Sundial's flow-matching TimeFlow loss on a 1T-point corpus, ICML 2025 Oral, arXiv:2502.00816; Moirai 2.0's decoder-only multi-quantile simplification, arXiv:2511.11698). The 2025-2026 SOTA frontier on the standard zero-shot benchmarks GIFT-Eval (arXiv:2410.10393) and fev-bench is held by covariate-capable, in-context-learning models: Chronos-2 (group attention over targets+covariates), TabPFN-

TS (tabular prior-fitted network with calendar features), TiRex (35M-param xLSTM, NeurIPS 2025, arXiv:2505.23719) and TiRex-2 (multivariate + streaming with constant per-patch cost, July 2026); industrial deployments include Datadog's Toto trained on its own observability telemetry with the BOOM benchmark (arXiv:2505.14766) and TimesFM 2.5 in production at Google (16k context, continuous quantile head). Open problems directly relevant here: exogenous/known-future covariate handling only became first-class in 2025 (Chronos-2, TabPFN-TS, TiRex-2 — before that, weather-like inputs were ignored by most TSFMs); forecasting skill does not equal embedding quality (MOMENT is the main embedding-first TSFM; 2026 probing work shows intermediate layers beat final layers and that non-stationarity is only partially linearly accessible in TSFM embeddings, with model-specific failure modes — Choi et al. 2026); multivariate dependency modeling vs channel independence remains contested; and none of the public models condition on control actions, so policy endogeneity must be handled by the practitioner via the covariate channel.

Key principles

- Linear probing on frozen embeddings is THE standard protocol of this literature (TS2Vec, TNC, TS-TCC, MOMENT, Mantis all evaluate this way) — the project's plan to force decision-relevant information to be linearly accessible via linear-probe auxiliary heads is exactly aligned with how the field defines representation quality.
- Multi-resolution embeddings maximize downstream reuse: TS2Vec's hierarchical contrast yields representations at timestamp, sub-window, and series level from one encoder; different downstream consumers (dispatch needs per-minute state, forecasting needs per-hour aggregates) can pool the same latent sequence at different granularities.
- The choice of augmentation/invariance IS the choice of what information the embedding discards: scaling/jitter augmentations (TS-TCC) impose amplitude invariance, and instance normalization (PatchTST, all TSFMs) strips level and scale. For market state, absolute demand level and weather extremes are decision-relevant — invariances must be audited, and normalization statistics must be re-injected as features.
- Forecasting-as-pretraining (next-patch prediction in TimesFM, Time-MoE, Moirai 2.0) organizes the latent space around predictive sufficiency rather than reconstruction fidelity — closer to decision value than autoencoding. Mixed objectives (masked reconstruction + forecasting + contrastive, as in MOMENT/SimMTM ablations) consistently beat any single objective.
- Patching is the canonical tokenization: sub-series patches retain local semantics, cut attention cost quadratically, extend usable context, and give per-patch token embeddings that downstream models can attend over rather than a single pooled vector (PatchTST, TimesFM, Moirai).
- Channel independence is robust but lossy; the 2024-2026 SOTA reintroduces cross-variate information through dedicated mechanisms — variate-level cross-attention (TimeXer), any-variate attention with variate encodings (Moirai), group attention across targets+covariates (Chronos-2), asymmetric variate mixers (TiRex-2). Cross-signal structure (demand-supply interaction, weather-to-demand) must flow through explicit attention paths, not be averaged away.
- Covariates need a taxonomy: static (zone geometry), past-observed (realized demand, weather), and known-future (weather FORECASTS, holidays, planned events) — introduced by TFT and only adopted by foundation models in 2025 (Chronos-2, TabPFN-TS, TiRex-2). Known-future inputs are the single largest accuracy lever on covariate-rich benchmarks (Chronos-2's biggest wins are on covariate-informed tasks).
- Multi-horizon probabilistic output heads (TFT's quantile losses, Moirai's mixture distributions, Sundial's flow matching, TimesFM 2.5's continuous quantile head) force the latent to encode uncertainty and horizon-dependent dynamics — downstream pricing/dispatch consume distributions, not point estimates, and quantile heads are cheap auxiliary objectives.

- Disentanglement along generative factors improves shift robustness: CoST's causal argument is that separating seasonal from trend components yields representations stable under intervention/distribution shift — directly relevant when surge policy (an intervention) changes the data-generating process.
- Heterogeneity is handled by conditional capacity, not one dense model: frequency-specific projections proved too coarse, and token-level sparse MoE (Moirai-MoE, Time-MoE) lets different dynamics regimes (rush hour vs overnight, rain vs clear) route to different experts automatically.
- TSFM embeddings are not automatically fit for purpose: 2026 probing shows intermediate-layer representations are more discriminative than final-layer ones, and non-stationarity detectability degrades smoothly with model-specific failure modes (Choi et al. 2026) — embedding quality must be measured per-task, per-layer, not assumed from forecasting leaderboards.
- Evaluation infrastructure is a first-class artifact: the field converged on standardized, leakage-audited, rolling-origin benchmarks spanning domains/frequencies/horizons (GIFT-Eval, fev-bench, BOOM). A versioned company-wide embedding needs the internal equivalent: a fixed probe-task suite that every embedding version must pass before release.

Recommendations for the market-state embedding project

- Architecture: patch-tokenize each zone's multivariate history (PatchTST) but separate endogenous targets (demand, supply, imbalance) from exogenous context (weather, events, holidays) TimeXer-style — patch-level self-attention for targets, variate-level tokens + cross-attention for covariates — and adopt TFT's covariate taxonomy: static zone features, past-observed signals, and KNOWN-FUTURE inputs (weather forecasts, scheduled events, planned incentive campaigns). Use Moirai-style any-variate attention with variate-ID encodings so new signals can be added across embedding versions without retokenizing the world.
- Treat logged control actions (surge multiplier, surge radius, dispatch parameters) as first-class input covariates: past actions as past-observed covariates, planned/hypothetical actions as known-future covariates. No public TSFM conditions on actions, so this is where the bespoke encoder earns its keep — an action-conditioned encoder is the minimal structure needed so the latent represents state rather than a state-policy entanglement, and it enables counterfactual queries ('embedding under multiplier 1.0 vs 1.8') that the causal/RL literature area will build on.
- Objective: use a weighted mixture — (a) multi-horizon multi-quantile forecasting heads for key targets at several horizons (TFT/Moirai-2.0-style; this is simultaneously the auxiliary 'decision-value' objective from the project brief and a probabilistic output downstream pricing needs), (b) SimMTM-style neighbor-aggregated masked-patch reconstruction for completeness of state capture, and (c) TS2Vec-style hierarchical contrast so per-time-bucket latents pool coherently into per-hour/per-day embeddings. Constrain all auxiliary heads to linear/shallow probes on the frozen trunk, as planned — this matches the field's standard representation-quality protocol.
- Normalization: apply reversible instance normalization per zone-window for training stability (universal in TSFMs), but re-inject the removed statistics (level, scale, recent trend of raw demand/supply) as explicit embedding inputs or side-channel features — absolute demand level is decision-critical for pricing and would otherwise be destroyed. Audit every augmentation for invariances that erase decision signal: no amplitude scaling or jitter on demand magnitudes or weather extremes; prefer TS2Vec-style cropping/masking positives and TNC-style temporal-neighborhood positives (and adjacent-zone positives) instead.
- Before building anything bespoke, benchmark three zero-shot baselines on the exact downstream probe tasks: (1) frozen MOMENT embeddings + linear probes, (2) Chronos-2 with zone groups + weather covariates for the forecasting tasks, (3) TabPFN-TS over featurized zone-time rows. These set the bar the custom encoder must clear, and Chronos-2/TiRex-2 fine-tuned on company telemetry may be the fastest path to a strong v1 trunk (TimesFM 2.5 demonstrates LoRA-style cheap adaptation for versioned updates).

- Serving/streaming: for a per-zone per-5-minute embedding computed fleet-wide, prefer either a small recurrent state-passing design (TiRex-2's xLSTM: constant cost per new patch, unbounded context, and the hidden state IS the market state) or sparse MoE capacity (Time-MoE/Moirai-MoE) if transformer-based — token-level expert routing also handles regime heterogeneity (commute vs nightlife vs storm) and yields an interpretable regime label for free.
- Cross-zone structure: implement spatial context via in-context groups (Chronos-2 group attention over a zone and its neighbors) or any-variate attention over stacked neighbor series, rather than a separate graph model in v1 — the 2025-2026 evidence is that attention over related series in-context captures cross-series dependencies zero-shot, and city-level demand shocks (weather fronts, event endings) propagate across zones.
- Evaluation suite as a versioned artifact (GIFT-Eval/fev-bench pattern): a fixed internal benchmark of frozen-encoder linear-probe tasks — multi-horizon demand/supply forecasting, ETA regression, anomaly/incident detection, surge-regime classification — with rolling-origin temporal splits, strict leakage audits, and per-LAYER probing (2026 evidence: intermediate layers are often more discriminative than final layers, so consider exposing a designated 'embedding layer' rather than the last one). Add Choi-et-al.-style controlled non-stationarity injections (synthetic mean/variance/trend shifts in demand) to verify regime shifts remain linearly detectable in every release — surge decisions are precisely non-stationarity detection.
- Cover rare regimes with synthetic data: every SOTA TSFM leans on synthetic pretraining (Chronos KernelSynth, Chronos-2's synthetic multivariate structures, TiRex-2's on-the-fly covariate coupling, Sundial's TimeBench). Generate synthetic extreme-weather, concert-surge, and supply-outage scenarios — including counterfactual action traces — to de-bias the policy-endogenous logged corpus and cover the tail regimes where pricing decisions matter most.
- Version compatibility: freeze and archive each encoder version; gate releases on the probe benchmark (no probe task may regress beyond tolerance); and when the latent space changes, ship a learned linear adapter mapping $v(n)$ embeddings into $v(n+1)$ space so downstream teams migrate asynchronously — the TSFM ecosystem's practice of maintaining stable model families with quantile-head add-ons and LoRA deltas (TimesFM 1.0→2.5, Chronos→Chronos-2, Moirai→2.0) rather than breaking rewrites is the operational pattern to copy.
- Output distributions, not points: whatever trunk is chosen, expose multi-horizon quantile (or sampled-trajectory, Sundial-style flow-matching) forecast heads alongside the embedding — surge pricing and repositioning consume uncertainty (P90 imbalance, probability of undersupply), and the field's uniform move to probabilistic heads (TFT quantiles, Moirai mixtures, TimesFM 2.5 continuous quantile head, Chronos-2 quantiles) reflects that point forecasts are insufficient for control decisions.

Annotated bibliography (19 sources)

2021 Tonekaboni, S., Eytan, D., Goldenberg, A. (2021). **Unsupervised Representation Learning for Time Series with Temporal Neighborhood Coding**. ICLR 2021. arXiv:2106.00750.

arxiv.org/abs/2106.00750

TNC learns embeddings for non-stationary series by exploiting local smoothness of the generative process: windows from the same temporal neighborhood (statistically stationary region, found via an ADF test) are pulled together in latent space, while distant windows are pushed apart using a debiased contrastive objective that treats non-neighbors as unlabeled rather than definitely negative. Encoder is architecture-agnostic (RNN/CNN); evaluated by linear probes for downstream classification and by clusterability of latent states.

Relevance to this project: TNC's premise — that a latent should track the evolving underlying STATE of a non-stationary system — is precisely the market-state concept; its neighborhood construction is a template for defining positives across adjacent time buckets and adjacent geo-zones without labels.

2021 Eldele, E., Ragab, M., Chen, Z., Wu, M., Kwoh, C.K., Li, X., Guan, C. (2021). **Time-Series Representation Learning via Temporal and Contextual Contrasting (TS-TCC)**. IJCAI 2021. arXiv:2106.14112.

arxiv.org/abs/2106.14112

TS-TCC creates two views of each series via weak (jitter-scale) and strong (permutation-jitter) augmentations, then trains a temporal contrasting module in which the context vector from one view must predict the future latents of the other view (a hard cross-view prediction task), plus a contextual contrasting (NT-Xent) module over the resulting contexts. A linear classifier on frozen features approaches fully supervised performance; strong in few-label and transfer settings.

Relevance to this project: Demonstrates that a cross-view FUTURE-PREDICTION task inside the contrastive objective yields more decision-relevant latents than pure instance discrimination — supporting the project's plan to use predictive auxiliary heads rather than reconstruction alone. Its augmentation study is also a cautionary tale about imposing amplitude invariances on demand-level data.

2021 Lim, B., Arik, S.O., Loeff, N., Pfister, T. (2021). **Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting**. International Journal of Forecasting 37(4), 1748-1764. arXiv:1912.09363.

arxiv.org/abs/1912.09363

TFT is the canonical multi-horizon architecture for covariate-rich forecasting: it explicitly partitions inputs into static covariates, past-observed inputs, and KNOWN-FUTURE inputs, processes them with per-type variable-selection networks, gated residual layers, an LSTM for local dynamics, and interpretable multi-head attention for long-range dependencies, and outputs simultaneous quantile forecasts (P10/P50/P90) for all horizons. Widely deployed industrially for retail/energy demand forecasting.

Relevance to this project: The static / past-observed / known-future covariate taxonomy is exactly what the market-state encoder needs (zone features / realized demand-supply-weather / weather forecasts, holidays, and PLANNED CONTROL ACTIONS like scheduled incentives). Its multi-quantile multi-horizon loss is a ready-made auxiliary objective that ties the latent to pricing-relevant uncertainty.

2022 Yue, Z., Wang, Y., Duan, J., Yang, T., Huang, C., Tong, Y., Xu, B. (2022). TS2Vec: Towards Universal Representation of Time Series. AAAI 2022. arXiv:2106.10466.

arxiv.org/abs/2106.10466

TS2Vec performs HIERARCHICAL contrastive learning over two augmented context views (random cropping + timestamp masking of a dilated-CNN encoder's outputs): temporal contrast aligns the same timestamp across views and instance contrast separates different series, applied at every level of a max-pooling hierarchy. This yields a representation per timestamp; any sub-series embedding is obtained by pooling the constituent timestamp representations. A ridge/linear model on frozen features beat then-SOTA on 125 UCR + 29 UEA classification datasets, forecasting, and anomaly detection.

Relevance to this project: The strongest template for a 'universal' market-state embedding: one encoder emitting per-time-bucket latents that different consumers pool at their own granularity (per-15-min for dispatch, per-day for planning). Its contextual-consistency positives (same timestamp under different croppings) avoid the amplitude-distorting augmentations that would corrupt demand-level information.

2022 Woo, G., Liu, C., Sahoo, D., Kumar, A., Hoi, S. (2022). CoST: Contrastive Learning of Disentangled Seasonal-Trend Representations for Time Series Forecasting. ICLR 2022. arXiv:2202.01575.

arxiv.org/abs/2202.01575

CoST argues from a causal graph that forecasting representations should disentangle invariant generative factors: it learns a trend component via time-domain contrastive loss over a mixture of causal-convolution extractors, and a seasonal component via a frequency-domain contrastive loss on DFT coefficients. A simple ridge regressor on the disentangled frozen features improved multivariate forecasting MSE by 21.3% over prior SSL methods, robust across backbone choices.

Relevance to this project: Directly supports the project's causal concern: CoST's argument is that disentangled, causally-stable factors survive distribution shift induced by interventions — surge-policy changes are exactly such interventions. A market-state latent with explicit trend / seasonal (commute cycles) / residual (weather shocks, events) blocks would be more robust and more interpretable to downstream teams.

2023 Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J. (2023). A Time Series is Worth 64 Words: Long-term Forecasting with Transformers (PatchTST). ICLR 2023. arXiv:2211.14730.

arxiv.org/abs/2211.14730

PatchTST tokenizes each univariate channel into sub-series patches fed to a vanilla transformer with channel independence (all channels share weights but are processed separately) and reversible instance normalization. Patching preserves local semantics in each token, reduces attention cost quadratically, and extends usable look-back. It supports both supervised forecasting and self-supervised masked-patch pretraining, where fine-tuning beats supervised training on large datasets.

Relevance to this project: Established the patch-token + instance-norm recipe every subsequent foundation model uses — the market-state encoder should tokenize per-zone series into patches, giving downstream models a sequence of patch embeddings to attend over rather than one pooled vector. Its channel-independence result is also a warning: the project's demand-supply interaction requires deliberately adding cross-variate paths back (see TimeXer/Moirai).

2023 Dong, J., Wu, H., Zhang, H., Zhang, L., Wang, J., Long, M. (2023). SimMTM: A Simple Pre-Training Framework for Masked Time-Series Modeling. NeurIPS 2023 (Spotlight). arXiv:2302.00861.

proceedings.neurips.cc/paper_files/paper/2023/hash/5f9bdfde3685e4ccdb...

SimMTM observes that randomly masking time points destroys the temporal variations that make reconstruction a useful pretext task; it instead reconstructs each series by similarity-weighted aggregation of the representations of MULTIPLE differently-masked neighbor views (manifold-learning view of masked modeling), combining point-wise reconstruction with a series-wise similarity constraint. Achieves SOTA fine-tuning on forecasting and classification, in- and cross-domain.

Relevance to this project: If the project keeps a reconstruction term alongside its linear-probe prediction heads, SimMTM's neighbor-aggregated reconstruction is the modern way to do it — it avoids the failure mode identified in the project brief where raw autoencoding organizes the latent around pixel-level fidelity rather than structure, because reconstruction is forced through cross-view similarity in representation space.

2023 Rasul, K., Ashok, A., Williams, A.R., et al. (2023). Lag-Llama: Towards Foundation Models for Probabilistic Time Series Forecasting. arXiv:2310.08278.

arxiv.org/abs/2310.08278

Lag-Llama is a decoder-only LLaMA-style transformer for univariate probabilistic forecasting whose input tokens are engineered to contain lagged values at calendar-meaningful lags (day, week, year) plus date-time covariates; it outputs Student-t distribution parameters per step and is pretrained autoregressively on a diverse multi-domain corpus. First open general-purpose TSFM; strong zero-shot and SOTA after small-fraction fine-tuning.

Relevance to this project: Historically the first open foundation model, and its lag/calendar tokenization is a reminder that hand-chosen temporal structure (weekly commute cycles, seasonal patterns in ride demand) can be injected into token design cheaply. Its univariate, covariate-poor design also illustrates what the market-state project must avoid.

2024 Das, A., Kong, W., Sen, R., Zhou, Y. (2024). A Decoder-Only Foundation Model for Time-Series Forecasting (TimesFM). ICML 2024. arXiv:2310.10688.

arxiv.org/abs/2310.10688

TimesFM is a 200M-parameter decoder-only transformer pretrained on ~100B real-world time points (Google Trends, Wiki pageviews, synthetic data) with input patching and a longer output patch, trained by next-patch prediction with causal attention. Zero-shot accuracy approaches supervised per-dataset models across domains and granularities; iterated to TimesFM 2.5 in production (16k context, continuous quantile head, LoRA fine-tuning).

Relevance to this project: The industrial proof that next-patch prediction on massive operational data yields transferable temporal representations — the company's own trip/telemetry archive is a comparable pretraining corpus. TimesFM's evolution (quantile heads, long context, cheap LoRA adaptation) is the practical playbook for maintaining a versioned production model.

2024 Ansari, A.F., Stella, L., Turkmen, C., et al. (2024). Chronos: Learning the Language of Time Series. TMLR. arXiv:2403.07815.

arxiv.org/abs/2403.07815

Chronos scales and quantizes real values into a fixed vocabulary of discrete tokens and trains off-the-shelf T5 encoder-decoder language models (20M-710M params) on them with plain cross-entropy, pretraining on public datasets plus Gaussian-process synthetic series (KernelSynth). Sampling tokens autoregressively yields full predictive distributions; strong zero-shot performance on unseen datasets without any time-series-specific architecture.

Relevance to this project: Shows a categorical cross-entropy over quantized values learns flexible multimodal predictive distributions — useful for market states where demand distributions are multimodal (event nights). Its synthetic-data augmentation strategy is directly reusable to cover rare regimes (extreme weather, outages) under-represented in logs.

2024 Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S., Sahoo, D. (2024). Unified Training of Universal Time Series Forecasting Transformers (Moirai). ICML 2024 (Oral). arXiv:2402.02592.

arxiv.org/abs/2402.02592

Moirai is a masked-encoder transformer built for universality: multi-size patch projections handle any frequency, 'any-variate attention' flattens an arbitrary number of variates into one sequence with rotary time encodings plus binary variate encodings (making it permutation-equivariant over channels and able to ingest arbitrary covariates), and a mixture-distribution output head handles heterogeneous scales. Trained on the open LOTSA corpus (27B+ observations, 9 domains). Successors: Moirai-MoE and decoder-only multi-quantile Moirai 2.0 (arXiv:2511.11698).

Relevance to this project: Any-variate attention is the cleanest published mechanism for an encoder that must ingest a VARIABLE set of channels per zone (demand, supply, weather, events, active controls) — new signals can be added without architecture change, which matters for a company-wide embedding that will accrete inputs over versions.

2024 Goswami, M., Szafer, K., Choudhry, A., Cai, Y., Li, S., Dubrawski, A. (2024). MOMENT: A Family of Open Time-series Foundation Models. ICML 2024. PMLR v235.

proceedings.mlr.press/v235/goswami24a.html

MOMENT pretrains T5-style encoders with masked patch reconstruction (patches embedded, ~30% masked, reconstructed via a light head) on the multi-domain 'Time series Pile', deliberately targeting GENERAL-PURPOSE representations rather than only forecasting: the same frozen encoder is evaluated with linear probes / shallow heads on forecasting, classification, anomaly detection, and imputation under limited supervision. Ships an explicit limited-resource evaluation benchmark.

Relevance to this project: The closest existing artifact to what the project wants to build: one frozen encoder whose embeddings serve many heterogeneous downstream tasks, evaluated by linear probing. MOMENT (or its embeddings) should be the baseline the bespoke market-state encoder must beat, and its multi-task probe benchmark is a design template for the internal evaluation suite.

2024 Wang, Y., Wu, H., Dong, J., Qin, G., Zhang, H., Liu, Y., Qiu, Y., Wang, J., Long, M. (2024). TimeXer: Empowering Transformers for Time Series Forecasting with Exogenous Variables. NeurIPS 2024.

proceedings.neurips.cc/paper_files/paper/2024/hash/0113ef4642264adc2e6...

TimeXer gives endogenous and exogenous series different granularities of representation: endogenous targets get fine patch-level tokens with self-attention, exogenous covariates get coarse variate-level tokens, and a learned global endogenous token bridges them — variate-wise cross-attention injects exogenous information into the endogenous patches. SOTA on twelve covariate-rich benchmarks (including electricity price with load covariates).

Relevance to this project: The reference design for the weather problem: weather/events do not need per-timestamp modeling fidelity, they need to MODULATE the demand-supply representation — asymmetric patch-vs-variate tokenization plus cross-attention is the right inductive bias and keeps the covariate set extensible.

2025 Liu, X., Liu, J., Woo, G., Aksu, T., Liang, Y., Zimmermann, R., Liu, C., et al. (2025). Moirai-MoE: Empowering Time Series Foundation Models with Sparse Mixture of Experts. ICML 2025. arXiv:2410.10469.

arxiv.org/abs/2410.10469

Moirai-MoE replaces Moirai's human-designed frequency-specific input/output projections with a single projection plus sparse mixture-of-experts transformer layers, so specialization to different time-series 'dialects' happens at TOKEN level via learned routing (router guided by a clustering of pretrained representations). Outperforms dense Moirai and other TSFMs across 39 datasets with far fewer activated parameters, showing frequency is an unreliable proxy for dynamics regime.

Relevance to this project: Market data mixes regimes within one zone-series (weekday commute, weekend nightlife, storm shocks); token-level expert routing handles such regime heterogeneity automatically, and the learned routing assignments themselves are an interpretable regime signal that could be exposed alongside the embedding.

2025 Shi, X., Wang, S., Nie, Y., Li, D., Ye, Z., Wen, Q., Jin, M. (2025). Time-MoE: Billion-Scale Time Series Foundation Models with Mixture of Experts. ICLR 2025 (Spotlight). arXiv:2409.16040.

arxiv.org/abs/2409.16040

Time-MoE scales decoder-only, point-wise-tokenized autoregressive TSFMs to 2.4B parameters (first billion-scale TSFM) with sparse expert FFNs activating only a subset per token, pretrained on Time-300B (300B+ time points, 9+ domains) with multi-resolution forecast heads supporting arbitrary horizons and 4096 context. Beats dense models of equal activated compute by large margins, establishing scaling laws for sparse TSFMs.

Relevance to this project: Evidence that time-series representations follow scaling laws: with the company's massive telemetry, a modestly-sized sparse encoder could exceed a dense one at equal serving cost — serving cost matters since the embedding will be computed per-zone per-time-bucket company-wide. Its multi-resolution heads align with multi-horizon downstream needs.

2025 Hoo, S.B., Müller, S., Salinas, D., Hutter, F. (2025). From Tables to Time: Extending TabPFN-v2 to Time Series Forecasting (TabPFN-TS). arXiv:2501.02945.

arxiv.org/abs/2501.02945

TabPFN-TS casts forecasting as tabular regression: each timestamp becomes a row of lightweight engineered features (calendar features, running index) plus arbitrary covariate columns, fed to the frozen 11M-parameter TabPFN-v2 prior-fitted network (pretrained on synthetic tabular data), which predicts the full horizon in one non-autoregressive in-context pass with quantiles. No time-series pretraining at all, yet SOTA on covariate-informed forecasting and competitive univariate results on GIFT-Eval and fev-bench.

Relevance to this project: A humbling, ultra-cheap baseline the project should run first: if in-context tabular regression over hand-featurized zone-time rows with weather columns matches a learned encoder on the forecasting probes, the encoder's value must come from elsewhere (cross-zone structure, multi-task reuse, action conditioning). Also the simplest existing route to covariate-aware zero-shot forecasts.

2025 Ansari, A.F., Shchur, O., Küken, J., Auer, A., Han, B., et al. (2025). Chronos-2: From Univariate to Universal Forecasting. arXiv:2510.15821.

arxiv.org/abs/2510.15821

Chronos-2 is a 120M encoder-only model that unifies univariate, multivariate, and covariate-informed zero-shot forecasting: a GROUP ATTENTION mechanism lets information flow in-context across arbitrary sets of related series — variates of a multivariate system, related targets, or targets plus past-only and known-future covariates — with multi-step quantile output. Trained largely on synthetic data imposing diverse multivariate dependency structures; SOTA on fev-bench, GIFT-Eval, and Chronos Benchmark II, with largest margins on covariate-informed tasks, and deployed via AutoGluon/Amazon.

Relevance to this project: The strongest current off-the-shelf option for the project's data shape (multivariate zone series + weather covariates + known-future signals), and its group-attention-as-in-context-learning is the mechanism to copy for cross-zone pooling: a 'group' could be a zone and its neighbors, letting the market-state embedding absorb spatial context without a bespoke graph model.

2026 Podest, P., Pichler, M., Bürger, E., Zólyomi, L., et al., Klambauer, G., Hochreiter, S. (2026). TiRex-2: Generalizing TiRex to Multivariate Data and Streaming. arXiv:2607.01204.

arxiv.org/abs/2607.01204

TiRex-2 extends TiRex (35M xLSTM recurrent TSFM, NeurIPS 2025, arXiv:2505.23719, previous GIFT-Eval leader) with a bidirectional time mixer plus an asymmetric grouped-attention variate mixer that incorporates known-future covariates while keeping targets causal, and a synthetic coupling pipeline that composes multivariate training samples on the fly from univariate corpora. Its recurrent memory gives CONSTANT per-patch compute over unbounded streaming contexts; SOTA zero-shot on GIFT-Eval and fev-bench (July 2026).

Relevance to this project: The 2026 SOTA proves small recurrent state-passing models beat much larger transformers — and a persistent, constant-cost recurrent state updated as each 5-minute bucket arrives is operationally the ideal form for a live market-state embedding (the xLSTM hidden state literally IS a streaming market-state summary). Its synthetic covariate-coupling trick also solves the scarcity of aligned multivariate training data.

2026 Choi, J., Shook, B., Dubrawski, A. (2026). Non-Stationarity in the Embedding Space of Time Series Foundation Models. arXiv:2604.16428.

arxiv.org/abs/2604.16428

Systematically probes the embedding spaces of multiple TSFMs with LINEAR probes under controlled injections of distributional non-stationarity (mean shifts, variance changes, trends) and temporal non-stationarity (persistence/long memory), varying shift strength. Finds these properties are substantially linearly accessible in TSFM embeddings, but detectability degrades smoothly as shifts weaken and each model exhibits distinct, model-specific failure modes.

Relevance to this project: Direct 2026 evidence for the project's central bet — that decision-relevant structure can be made linearly accessible in an embedding — while warning that which structure is accessible is model-specific and must be measured, not assumed. Its controlled-injection linear-probe methodology is a ready-made recipe for auditing whether surge-relevant regime shifts are readable from the market-state embedding.

6 Tabular & Heterogeneous Business Data Embeddings

State of the art

Tabular representation learning evolved in four waves. (1) Entity embeddings of categoricals (Guo & Berkhahn 2016) replaced one-hot encoding with learned Euclidean vectors, becoming standard practice in industrial ML. (2) Transformer-era architectures (TabTransformer 2020, FT-Transformer 2021, SAINT 2021) established the "feature-as-token" paradigm: every column — categorical or numerical — is embedded into a vector and contextualized by attention; Gorishniy et al. (2022) then showed that HOW numerics are embedded (piecewise-linear or periodic encodings rather than raw scalars) is a decisive, architecture-independent performance lever. (3) Tabular self-supervision (VIME 2020, SCARF 2022, SAINT's CutMix/MixUp contrastive pretraining, and later augmentation-free latent-prediction methods like T-JEPA 2024) found that corruption-and-recovery and contrastive objectives beat plain autoencoder reconstruction, though gains on tabular data are more modest and inconsistent than in vision/NLP — which is why auxiliary supervised signal remains important. (4) Tabular foundation models are the current SOTA frontier: cross-table pretraining with shared backbones (XTab 2023), schema-free graph-attention over string-embedded cells (CARTE 2024), LLM-based row serialization (TabuLa-8B 2024), and above all in-context-learning prior-fitted networks — TabPFN v2 (Nature 2025), TabICL (ICML 2025), TabDPT (NeurIPS 2025), TabPFN-2.5 (Nov 2025), and 2026 successors (TabICLv2, LimiX-2M, generalist embedding models like TabEmbed, benchmarked on TabArena). Notably, Ye et al. (2025) showed TabPFN v2 works as a frozen feature extractor whose embeddings are nearly linearly separable — directly validating linear-probe-based embedding design. In parallel, multimodal fusion of tabular + time-series + spatial + weather streams matured from the Temporal Fusion Transformer's typed-covariate gating (2021) to deployed industrial systems: Uber's DeepETA (multi-resolution hashed geospatial embeddings + linear transformer, 2022) and DiDi's MVGR-Net (WWW 2026), which contrastively aligns POI-semantic and mobility views of geo-zones before demand forecasting. Open problems: parity with tuned GBDTs on medium-sized data is still contested (well-tuned MLP ensembles like TabM remain highly competitive), tabular SSL objectives lack a clear winner, context-length/latency limits of ICL models constrain serving, and none of this literature addresses policy endogeneity of logged actions — that must be imported from the causal/RL representation literature.

Key principles

- Feature-as-token is the unifying interface: embed every input — categorical, numerical, spatial, temporal — into a d-dimensional token and let attention contextualize them (TabTransformer, FT-Transformer, SAINT).

This gives heterogeneous business data one uniform representation and a single pooled [CLS]-style vector to publish.

- Numeric features must be explicitly embedded, not fed as raw scalars: piecewise-linear (quantile or target-aware bins) and periodic embeddings deliver large, architecture-independent gains (Gorishniy et al. 2022) and were independently rediscovered in production at Uber (bucketize-and-embed beat raw continuous inputs).
- High-cardinality spatial IDs need multi-resolution + hashing: quantize locations at several grid resolutions and map cells to embeddings via multiple feature hashing — better accuracy, less memory, and graceful handling of unseen cells (Uber DeepETA); pure entity embeddings (Guo & Berkhahn) remain right for moderate-cardinality categoricals.
- Reconstruction alone is the weakest pretraining signal: mask-estimation (VIME), contrastive learning over randomly corrupted feature subsets (SCARF, SAINT), masked-column prediction on real tables (TabDPT), and latent-space prediction (T-JEPA) all organize the latent space better than autoencoding — but tabular SSL gains are smaller and less reliable than in vision/NLP, so auxiliary supervised heads carry real weight.
- Linear separability is both the mechanism and the metric of downstream usefulness: the best tabular encoders produce nearly linearly separable embedding spaces (Ye et al. on TabPFN v2), and linear probing is the standard way to verify that decision-relevant information is cheaply accessible — validating the project's linear-probe auxiliary-head design.
- Transfer across heterogeneous tables requires decoupling schema from backbone: either per-table featurizers around a shared trunk (XTab), schema-free string/graph encodings (CARTE), or distribution-aware column embeddings (TabICL). This is what makes one encoder serve many cities, business lines, and evolving schemas.
- Context conditioning beats static encoding: ICL foundation models make a row's representation depend on a reference set (TabPFN, TabICL, TabDPT), and intersample attention (SAINT) lets rows borrow from similar rows — for market state this means embeddings can be conditioned on the current regime rather than a fixed global mapping.
- Typed multimodal fusion beats naive concatenation: separate static / observed-past / known-future input pathways with variable-selection gating (TFT), and contrastive alignment of complementary views of the same spatial unit (DiDi MVGR-Net POI + mobility), consistently outperform flat feature stacking.
- Pretraining data mix matters and scaling laws hold: synthetic causally-structured priors give broad robustness (TabPFN), but real in-domain tables add signal synthetics miss and improve downstream generalization (TabDPT); performance scales as power laws in model and data size.
- Serve via distillation and frozen encoders: foundation-scale teachers can be distilled into compact MLP/tree students with orders-of-magnitude lower latency (TabPFN-2.5), and frozen encoders with lightweight task heads preserve generalization while keeping the embedding contract stable for many consumers.
- One embedding space can serve many consumption modes — prediction, retrieval, clustering, similarity search — if trained with multi-task contrastive objectives and hard negatives (TabEmbed 2026), but it must be evaluated on a benchmark covering all intended uses, not just one task's accuracy.

Recommendations for the market-state embedding project

- Front-end: adopt an FT-Transformer-style feature tokenizer as the encoder's input layer — periodic or piecewise-linear embeddings (quantile bins; consider target-aware bins keyed to a core task like conversion) for all numeric streams (demand counts, supply, ETAs, precipitation, temperature); entity embeddings for moderate-cardinality categoricals (time bucket, day type, event category); and multi-resolution geohash grids with multiple feature hashing for zone identity, following Uber DeepETA.

- Pretraining objective: use a composite of masked-feature modeling with mask estimation (VIME) and SCARF-style contrastive corruption over unlabeled zone-hour rows, NOT a plain autoencoder; weight these below the planned auxiliary supervised heads, since the literature shows tabular SSL alone underdelivers — the linear-probe prediction heads are what will organize the space by decision value.
- Structure the encoder's temporal fusion like TFT: three typed pathways — static zone attributes, observed-past series (demand, supply, completed trips), and known-future inputs (weather forecasts, scheduled events, holidays) — with variable-selection gating; this also yields per-stream attributions that build trust with downstream teams. Treat the surge price and surge radius levers as explicit action tokens in the observed pathway so the representation is action-conditioned rather than silently policy-confounded (full causal correction must come from the causal-representation workstream).
- Make linear probing the official evaluation harness: for every downstream task (pricing, dispatch, ETA, repositioning, incentives, forecasting) maintain a frozen-encoder linear probe suite, and gate encoder releases on probe metrics — this mirrors both the project's own auxiliary-head design and the field's standard (Ye et al. 2025 shows the best encoders make tasks linearly solvable).
- Baseline before building: extract embeddings from TabPFN v2 / TabPFN-2.5 (the API exposes them) and from a tuned GBDT on hand-engineered features, and require the custom encoder to beat both on the linear-probe suite; use TabArena-style rigorous protocols to avoid the field's chronic weak-baseline problem.
- Pretrain on your own logged tables, not just public/synthetic data: TabDPT shows real in-domain rows materially improve transfer; the company's historical zone-hour logs across all cities are exactly the large real corpus the scaling laws reward.
- For multi-city and cold-start: share the transformer backbone across cities/business lines with lightweight per-city featurizers (XTab pattern), and consider CARTE-style string-grounded column encoding so new signal feeds and new-market schemas onboard without schema matching or encoder retraining.
- Add a multi-view zone-pretraining stage as DiDi does in production: contrastively align each zone's static-semantic view (POI composition, road network) with its dynamic-mobility view (trip flows), so the market-state embedding inherits both; inject discrete external events as first-class inputs rather than hoping the encoder infers them.
- Serving and versioning: keep the published encoder frozen between versions with lightweight adapters/heads per consumer; for request-time latency, distill the full encoder into a compact MLP student (TabPFN-2.5's distillation-engine pattern); on each new version, train with an alignment/distillation loss to the previous embedding geometry and re-run the frozen linear-probe suite of every consumer before rollout to guarantee backward compatibility.
- Consider context-conditioned inference for regime awareness: an ICL-style pathway (TabICL's row-embedding-then-context architecture) that conditions today's zone-hour embedding on a reference window of recent city-wide data would let the same encoder adapt to demand shocks and seasonal regime shifts without retraining — evaluate it against the static encoder on shock periods (storms, concerts, holidays).

Annotated bibliography (19 sources)

2016 Guo, C., Berkahn, F. (2016). Entity Embeddings of Categorical Variables.

arXiv:1604.06737.

arxiv.org/abs/1604.06737

Maps each categorical variable into a learned low-dimensional Euclidean space via an embedding layer trained end-to-end inside a supervised neural network (originally for the Rossmann store-sales Kaggle task, placing 3rd with simple features). Similar category values land close together in embedding space, revealing intrinsic structure (e.g., German states arranged roughly by geography), reducing memory versus one-hot, and improving generalization on sparse, high-cardinality features. The learned embeddings can be exported and fed to other models (including GBDTs).

Relevance to this project: The foundational technique for encoding geo-zone IDs, time buckets, day-of-week, and event types in the market-state encoder; establishes that embeddings learned under a supervised objective can be reused as features by other downstream models.

2020 Huang, X., Khetan, A., Cvitkovic, M., Karnin, Z. (2020). TabTransformer: Tabular Data Modeling Using Contextual Embeddings. arXiv:2012.06678.

arxiv.org/abs/2012.06678

Embeds each categorical feature as a token and passes tokens through Transformer self-attention layers to produce contextual embeddings (continuous features bypass the transformer and are concatenated at the MLP head). Beat prior tabular DL by $\geq 1.0\%$ mean AUC on 15 datasets and added an unsupervised masked-pretraining + fine-tuning recipe giving $+2.1\%$ AUC in semi-supervised settings. Contextual embeddings proved markedly robust to missing and noisy features.

Relevance to this project: Demonstrates that attention-contextualized embeddings of business categoricals are robust to the missing/noisy telemetry common in marketplace data; its pretrain-then-finetune recipe is a template for using abundant unlabeled zone-hour records.

2021 Gorishniy, Y., Rubachev, I., Khrulkov, V., Babenko, A. (2021). Revisiting Deep Learning Models for Tabular Data. NeurIPS 2021.

proceedings.neurips.cc/paper/2021/hash/9d86d83f925f2149e9edboac3b49229...

Introduces FT-Transformer: a Feature Tokenizer converts EVERY feature — numerical and categorical — into a d -dimensional token (numericals via learned per-feature affine maps), prepends a [CLS] token, and runs standard Transformer blocks; the final [CLS] representation is the row embedding used for prediction. In a rigorous unified benchmark it was the most universal architecture, outperforming ResNet-style and bespoke tabular models across the widest range of tasks.

Relevance to this project: FT-Transformer's tokenizer + [CLS] design is the canonical backbone for a market-state encoder: one uniform token interface for heterogeneous demand/supply/weather features and a single pooled embedding vector to publish company-wide.

2022 Gorishniy, Y., Rubachev, I., Babenko, A. (2022). On Embeddings for Numerical Features in Tabular Deep Learning. NeurIPS 2022.

proceedings.neurips.cc/paper_files/paper/2022/hash/9e9foffc3d836836ca9...

Shows that the scalar-to-vector embedding module for numerical features is a critical, underexplored degree of freedom. Proposes two schemes: piecewise-linear encoding (bin the value using quantile- or target-aware tree-based bins, encode fractional bin fill) and periodic activation embeddings (learned-frequency sine/cosine features followed by a linear layer). Both are universally beneficial across architectures — with proper numeric embeddings even plain MLPs match attention-based SOTA and close the gap to GBDTs; adopted as a core component in TabR, TabM, RealMLP, and ModernNCA.

Relevance to this project: Directly applicable to the encoder's raw inputs (demand counts, supply levels, temperature, rain intensity, price multipliers): embedding these with PLE/periodic schemes rather than feeding raw scalars is one of the cheapest, best-validated ways to increase how much decision-relevant signal the latent space captures.

2021 Somepalli, G., Goldblum, M., Schwarzschild, A., Bruss, C.B., Goldstein, T. (2021). SAINT: Improved Neural Networks for Tabular Data via Row Attention and Contrastive Pre-Training. arXiv:2106.01342.

arxiv.org/abs/2106.01342

Embeds all features into tokens and alternates self-attention over columns with intersample (row) attention over a batch, letting each row's embedding borrow information from similar rows — improving robustness to missing/noisy entries. Adds a contrastive self-supervised pretraining stage where views are built by CutMix in input space and MixUp in embedding space (features masked with prob 0.3), for label-scarce regimes. Outperformed XGBoost/CatBoost/LightGBM on average across benchmarks.

Relevance to this project: Intersample attention is a precursor to context-conditioned market embeddings — a zone-hour's representation can attend to comparable zone-hours (same weather regime, similar events), and its mixed-space contrastive recipe is a proven tabular augmentation strategy.

2020 Yoon, J., Zhang, Y., Jordon, J., van der Schaar, M. (2020). VIME: Extending the Success of Self- and Semi-supervised Learning to Tabular Domain. NeurIPS 2020.

proceedings.neurips.cc/paper/2020/hash/7d97667a3e056acab9aaf653807b4a0...

Self-supervised stage: corrupt a row by masking random features and imputing them from the empirical marginals, then train an encoder with two pretext heads — reconstruct the original feature vector AND predict which features were masked (mask estimation). Semi-supervised stage: use the frozen encoder to generate masked/imputed augmentations and enforce prediction consistency across them alongside supervised loss. Beat SOTA baselines on genomics and clinical tabular datasets.

Relevance to this project: The mask-estimation pretext is exactly the kind of objective that forces an encoder to internalize inter-feature dependency structure (e.g., how supply co-moves with weather and demand) rather than merely compressing — a direct upgrade over the pure-autoencoder baseline the project already distrusts.

2022 Bahri, D., Jiang, H., Tay, Y., Metzler, D. (2022). SCARF: Self-Supervised Contrastive Learning using Random Feature Corruption. ICLR 2022 (Spotlight).

arxiv.org/abs/2106.15147

InfoNCE contrastive pretraining for tabular data where the positive view of a row is created by corrupting a random subset (~60%) of its features with draws from each feature's empirical marginal distribution; encoder + projection head are pretrained, then the encoder is fine-tuned. On 69 OpenML-CC18 datasets it improved fully-supervised accuracy, and was notably robust under label noise and in semi-supervised regimes; explicitly outperformed autoencoder pretraining and composes with other objectives.

Relevance to this project: The strongest simple recipe for pretraining the market-state encoder on millions of unlabeled zone-hour rows; its finding that contrastive corruption beats reconstruction directly supports the project's move away from pure autoencoding.

2023 Zhu, B., Shi, X., Erickson, N., Li, M., Karypis, G., Shoaran, M. (2023). XTab: Cross-table Pretraining for Tabular Transformers. ICML 2023.

arxiv.org/abs/2305.06090

Pretrains a shared Transformer backbone across many heterogeneous tables by giving each table its own lightweight featurizer and projection head while the backbone is jointly trained (via federated-style updates) with reconstruction, contrastive, or supervised objectives. Handles inconsistent column types/counts across tables; on 84 AMLB tasks it consistently boosted generalizability, convergence speed, and final performance of multiple tabular transformer architectures.

Relevance to this project: The shared-backbone/per-table-featurizer split is the right pattern for one market-state encoder serving many cities and business lines (rides vs. food delivery) whose schemas and feature sets differ, and for onboarding new markets without training from scratch.

2024 Kim, M.J., Grinsztajn, L., Varoquaux, G. (2024). CARTE: Pretraining and Transfer for Tabular Learning. ICML 2024.

arxiv.org/abs/2402.16785

Represents each table row as a star graph whose nodes carry string embeddings of column names and cell values (an open vocabulary), then runs a graph-attentional network that contextualizes each entry with its column name and neighboring entries; pretrained on a large knowledge base (YAGO) with no schema matching required. Outperformed tree-based baselines on an extensive benchmark and enables joint learning across tables with unmatched columns.

Relevance to this project: Schema-free, string-grounded encoding means new signal sources (a new weather vendor, new event feeds, a new city's zone taxonomy) can be ingested without re-aligning columns — valuable for a long-lived, company-wide embedding service whose upstream schemas will churn.

2024 Gardner, J., Perdomo, J.C., Schmidt, L. (2024). Large Scale Transfer Learning for Tabular Data via Language Modeling (TabuLa-8B). NeurIPS 2024.

arxiv.org/abs/2406.12031

Fine-tunes Llama-3 8B for tabular prediction by serializing rows as text with a novel row-packing and attention-masking scheme, trained on 2.1B rows from 4.2M tables (filtered TabLib corpus) with standard language-modeling loss over target tokens. Achieves zero-shot accuracy on unseen tables >15pp above random and beats XGBoost/TabPFN by 5-15pp in 1-32-shot settings even when those are trained on up to 16x more rows.

Relevance to this project: Represents the LLM-serialization branch of tabular foundation models: strong few-shot transfer and natural handling of free-text features (event descriptions), but token-costly for the dense numeric streams and latency budgets of a real-time market-state service — a useful contrast informing the project's architecture choice toward native tabular encoders.

2025 Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S.B., Schirrmeister, R.T., Hutter, F. (2025). Accurate predictions on small data with a tabular foundation model (TabPFN v2). Nature 637, 319-326.

nature.com/articles/s41586-024-08328-6

A transformer pretrained once on millions of synthetic datasets sampled from a structural-causal-model prior; at inference the entire labeled training set plus query rows are fed as context and predictions come out in a single forward pass (in-context learning, no gradient training per task). Uses two-way attention across rows and features to handle arbitrary table shapes; on datasets up to ~10K rows it outperformed baselines tuned for 4 hours in 2.8 seconds, and it also supports embedding extraction, density estimation, and synthetic data generation.

Relevance to this project: The watershed tabular foundation model (Nature 2025, massive adoption): pretraining on causally-structured synthetic priors yields representations that transfer to arbitrary business tables — an immediate strong baseline/feature-extractor to benchmark the custom market-state encoder against before committing to bespoke pretraining.

2025 Ye, H.-J., Liu, S.-Y., Chao, W.-L. (2025). A Closer Look at TabPFN v2: Understanding Its Strengths and Extending Its Capabilities. arXiv:2502.17361.

arxiv.org/abs/2502.17361

Analysis paper showing TabPFN v2 can be repurposed as a frozen feature extractor that maps tabular instances into a nearly linearly separable embedding space — a linear model on the extracted features matches the full in-context learner's accuracy. Also shows the model infers attribute relationships even from randomized attribute tokens, and introduces a test-time divide-and-conquer strategy extending TabPFN v2 to high-dimensional, many-class, and large-scale data without retraining.

Relevance to this project: The clearest published validation of the project's core design bet: a good tabular encoder should make decision-relevant structure LINEARLY accessible, and linear-probe accuracy is the right acceptance test for embedding quality.

2025 **Qu, J., Holzmüller, D., Varoquaux, G., Le Morvan, M. (2025). TabICL: A Tabular Foundation Model for In-Context Learning on Large Data. ICML 2025, PMLR 267:50817-50847.**
arxiv.org/abs/2502.05564

Two-stage architecture for scalable tabular ICL: distribution-aware column-wise embeddings, then a context-aware row-interaction module compresses each row into a fixed-dimensional representation, and a final transformer performs in-context learning over these row vectors — decoupling cost from feature count and scaling to 500K samples on affordable hardware. Pretrained on synthetic datasets; surpasses TabPFNv2 and CatBoost on 53 datasets with >10K samples; TabICLv2 (2026, arXiv:2602.11139) adds a more diverse synthetic engine and regression support.

Relevance to this project: Its explicit 'row-embedding-first' factorization — build a fixed-size embedding per row, then let a downstream model consume it — is architecturally the same contract as the market-state embedding service, and shows distribution-aware column embeddings improve the row representation.

2025 **Grinsztajn, L., Flöge, K., Key, O., et al., Hollmann, N., Hutter, F. (Prior Labs) (2025). TabPFN-2.5: Advancing the State of the Art in Tabular Foundation Models. arXiv:2511.08667.**
arxiv.org/abs/2511.08667

Scales the prior-fitted ICL approach to 50K rows x 2K features (20x more cells than TabPFNv2) with improved synthetic priors and architecture; leads the industry-standard TabArena benchmark, beating tuned tree ensembles and matching a 4-hour AutoGluon ensemble. Crucially, ships a distillation engine that compresses the foundation model into a compact MLP or tree ensemble, preserving most accuracy at orders-of-magnitude lower latency for plug-and-play production deployment.

Relevance to this project: The distillation-for-deployment pattern answers the project's serving constraint: pretrain/teach with a heavy encoder, then distill into a low-latency student that pricing and dispatch can call at request time — and TabArena is the evaluation harness to copy.

2025 **Ma, J., Thomas, V., Hosseinzadeh, R., Labach, A., Kamkari, H., Cresswell, J.C., Golestan, K., Yu, G., Caterini, A.L., Volkovs, M. (2025). TabDPT: Scaling Tabular Foundation Models on Real Data. NeurIPS 2025.**
arxiv.org/abs/2410.18164

Trains a row-based ICL transformer on REAL tables (not synthetic priors) using self-supervised masked-column prediction: random columns are treated as targets, and retrieval assembles similar rows into training contexts. Shows real data carries signal synthetic priors miss, that mixing real data speeds training and improves generalization, and that performance follows power-law scaling in model and data size; SOTA on CC18/CTR23 with no per-task tuning.

Relevance to this project: Direct evidence that pretraining on the company's own massive logged marketplace tables (rather than only synthetic or public data) should yield better embeddings, and that masked-column prediction over real rows is an effective large-scale objective for the market-state encoder.

2021 Lim, B., Arık, S.Ö., Loeff, N., Pfister, T. (2021). Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting* 37(4), 1748-1764.

arxiv.org/abs/1912.09363

Attention-based forecaster with a typed-input contract: static covariates get a dedicated encoder that conditions the whole network, known-future inputs (e.g., weather forecasts, planned events) and historically-observed series are processed by variable-selection networks and gated residual layers that learn to suppress irrelevant features, with LSTM local processing plus interpretable multi-head attention for long-range dependencies. Widely adopted in industry for demand/retail/energy forecasting.

Relevance to this project: The reference design for fusing exactly the project's input mix — static zone attributes, observed demand/supply history, and known-future weather/event signals — inside one encoder, with variable-selection gates providing built-in attribution for which streams drive the market state.

2026 Hao, X., Li, G., Wu, D., Guo, X., Zhu, Y., Zou, Z., Zhen, P., Yao, Y., Liang, Y. (2026). Enhancing Ride-Hailing Forecasting at DiDi with Multi-View Geospatial Representation Learning from the Web (MVGR-Net). *The Web Conference (WWW) 2026*.

arxiv.org/abs/2602.10502

Two-stage industrial system on DiDi production data: a pretraining stage builds geo-zone representations by fusing web-sourced Points-of-Interest semantics with temporal mobility patterns (multi-view alignment capturing both what a region IS and how it MOVES), then a forecasting stage consumes these representations via prompt-based fine-tuning of an LLM that also injects external event information for demand forecasting.

Relevance to this project: The closest published analogue to the project — a ride-hailing company learning reusable geo-zone embeddings from heterogeneous views and consuming them downstream — validating multi-view pretraining of zone representations and explicit event-stream injection at a direct competitor's scale.

2022 Uber Engineering (2022). DeepETA: How Uber Predicts Arrival Times Using Deep Learning. *Uber Engineering Blog*.

uber.com/us/en/blog/deepeta-how-uber-predicts-arrival-times/

Production ETA system embedding essentially all features: locations are quantized into multi-resolution lat/long grids and mapped to embeddings via multiple feature hashing (several independent hash functions into compact bins — more accurate and smaller than exact indexing), discretized continuous features get embedded buckets, and a linear (kernel-trick) transformer learns feature interactions at millisecond latency, followed by a calibrated fully-connected decoder serving global traffic.

Relevance to this project: Battle-tested answers for the encoder's spatial front end (multi-resolution geohash + hashed embeddings for hundreds of thousands of zones) and for the latency discipline a company-wide, request-time embedding service must meet; notably Uber found bucketized-and-embedded continuous features beat raw scalars — converging with Gorishniy's academic result.

2026 Qiang, M., et al. (2026). TabEmbed: Benchmarking and Learning Generalist Embeddings for Tabular Understanding. arXiv:2605.04962.

arxiv.org/abs/2605.04962

First generalist tabular embedding model unifying classification and retrieval in one shared embedding space: diverse tabular tasks are reformulated as semantic matching and the encoder is trained with large-scale contrastive learning using positive-aware hard-negative mining to capture fine-grained structural and numerical nuances. Introduces TabBench (311 classification datasets + 30K retrieval queries) and significantly outperforms SOTA text-embedding models, which fail on numerical semantics.

Relevance to this project: State-of-the-art 2026 evidence that a single embedding space can serve heterogeneous downstream uses (prediction, similarity search, retrieval) — the same multi-consumer contract the market-state embedding must satisfy — and supplies a benchmark methodology for evaluating general-purpose tabular embeddings.

7 Auxiliary Tasks & Multi-Task Learning as Latent-Space Organizers

State of the art

Auxiliary-task and multi-task learning as a way to ORGANIZE a latent space began with UNREAL (Jaderberg et al., 2016), which showed that a few cheap prediction heads sharing an encoder dramatically improve representations when the primary signal is weak — exactly the project's bet. The field then split into four strands that have now converged. (1) THEORY: Maurer et al. (2016) and Tripuraneni et al. (2020) proved that a shared representation trained on T diverse tasks cuts per-task sample complexity, with "task diversity" — the heads must span the function space downstream tasks live in — as the load-bearing condition; Lyle et al. (2021) characterized WHICH auxiliary tasks induce which latent geometry (value-like heads pull representations toward the transition operator's top eigenspace), and 2025 JEPA theory (Yu et al.) proved auxiliary regression heads provably prevent representation collapse in predictive encoders. (2) WHICH TASKS TRAIN TOGETHER: Standley et al. (2020) showed semantic relatedness does not predict training compatibility; TAG (Fifty et al., 2021) reduced grouping to measurable inter-task gradient affinity in one run; Grad-TAG (KDD 2024) scores all task subsets via linearization at ~3% of full-training cost up to 500 tasks; and ICLR 2025 work (Jeong & Yoon) exploits proximal affinity online during training. (3) GRADIENT CONFLICT: uncertainty weighting (Kendall et al., 2018), GradNorm (2018), PCGrad (2020) and CAGrad (2021) remedy scale imbalance and destructive interference, but Kurin et al. (2022) showed well-regularized plain loss summation often matches them — measurement before machinery. (4) WHAT A REPRESENTATION SHOULD RETAIN AND HOW TO CHECK: the information-bottleneck view (Aleml et al., 2017) frames the embedding as a minimal sufficient statistic for the chosen prediction targets; Rakelly et al. (2021) proved popular self-supervised objectives can be provably INSUFFICIENT for control, and Khetarpal et al. (2024/25) showed action-conditional self-prediction (BYOL-AC) recovers per-action distinctions that policy-averaged objectives lose — directly relevant to policy-endogenous surge levers; probing (Alain & Bengio 2016; MDL probing, Voita & Titov 2020) supplies the evaluation contract. The project's key open question — choosing a SMALL set of heads to shape a general-purpose embedding — remains only partially solved: Proto-Value Networks (Farebrother et al., 2023) showed hundreds of auto-generated successor-measure heads yield frozen embeddings that linear heads can decode to near-DQN performance, while automated selection is attacked by meta-gradient discovery of GVF questions (Veeriah et al., 2019, arXiv:1909.04607), bilevel weighting (AuxiLearn, 2021; Auto-Lambda, TMLR 2022, arXiv:2202.03091), evolutionary loss search (A2LS, NeurIPS 2022, arXiv:2210.06041), generate-and-test discovery (Rafiee et al., arXiv:2210.14361), and RL-based task generation (RL-AUX, 2025, arXiv:2510.22940). No a-priori theory yet picks the minimal head set for a given downstream portfolio; the practical SOTA is cheap measured affinity plus diversity-constrained subset selection, validated by probe scorecards.

Key principles

- Auxiliary heads shape representations through their GRADIENTS, not their outputs: the value of a head is which distinctions it forces the shared encoder to preserve. Recent JEPA theory (Yu et al. 2025) makes this precise — auxiliary heads act as anchors that provably prevent collapse of exactly the distinctions they predict, so heads should be chosen per downstream decision family.
- Task diversity is the load-bearing theoretical condition (Tripuraneni et al. 2020; Maurer et al. 2016): benefits require the head set to SPAN the function space downstream consumers live in. A small set of near-duplicate targets (e.g., several demand proxies) buys provably little; coverage of distinct decision-relevant function directions matters more than head count.
- Sufficiency must be defined relative to downstream decisions, and observational prediction is not enough under policy endogeneity: plausible self-supervised/MI objectives can be provably insufficient for control (Rakelly et al. 2021), and policy-averaged forecasting collapses per-action distinctions that action-conditional prediction preserves (Khetarpal et al. 2024/25). When the logged data is generated by control levers, at least some heads must be action-conditioned.
- Semantic relatedness does not predict training compatibility (Standley et al. 2020); affinity is an empirical, regime-dependent quantity. It is now cheap to measure: inter-task gradient affinity in one instrumented run (TAG 2021), linearized subset scoring at ~3% of compute for hundreds of tasks (Grad-TAG, KDD 2024), and online proximal affinity during training (ICLR 2025). Head selection should be an experiment, not a whiteboard argument.
- Loss balancing first, gradient surgery second: uncertainty weighting (Kendall 2018) and GradNorm (2018) fix scale imbalance cheaply; PCGrad/CAGrad fix directional conflict; but a well-regularized plain sum often matches all of them (Kurin et al. 2022). Measure gradient conflict at the shared trunk before adopting machinery, and prefer the simplest optimizer that passes evaluation — it also aids reproducibility of versioned embeddings.
- Many cheap, automatically generated heads scale surprisingly well: banks of random GVF/successor-measure prediction tasks yield frozen embeddings decodable by linear heads at near end-to-end performance (Proto-Value Networks 2023), and theory links such heads to the transition operator's dominant eigenspace (Lyle et al. 2021). A dense bank of cheap structural heads complements, rather than competes with, a few hand-picked decision heads.
- Probing is both the training constraint and the evaluation contract: linear probes (Alain & Bengio 2016) operationalize 'linearly accessible', but accuracy-only probing is unstable — MDL/codelength probing (Voita & Titov 2020) measures how EASILY information is extractable and gives stable version-over-version comparisons, which is what a company-wide embedding SLA needs.
- The information-bottleneck view supplies the compression side: the embedding should approach a minimal sufficient statistic for the UNION of its heads' targets (Alemi et al. 2017), with the bottleneck strength as an explicit dial between compactness/robustness and downstream coverage.
- Head selection itself is learnable when hand-design stalls: bilevel/implicit-differentiation weighting of heads against a validation objective (AuxiLearn 2021, Auto-Lambda 2022), meta-gradient discovery of prediction questions (Veeriah et al. 2019), and evolutionary auxiliary-loss search with cross-environment transfer (A2LS 2022) form a maturity ladder from cheap (learned weights over fixed heads) to expensive (searching loss space).

Recommendations for the market-state embedding project

- Treat auxiliary-head selection as a measured task-affinity problem, not a design debate: enumerate ~20-30 candidate heads, run ONE instrumented all-heads training with TAG-style inter-task affinity logging, then

use Grad-TAG-style linearized scoring to rank head SUBSETS against a held-out basket of downstream probe tasks (pricing, dispatch, ETA, repositioning, forecasting). Select the smallest subset that covers the basket, with a Tripuraneni-style diversity constraint rejecting near-duplicate heads.

- Structure the head set in four families: (a) exogenous forecasting heads (next-bucket demand, supply arrivals, weather-conditioned volumes); (b) ACTION-CONDITIONED transition heads predicting next market state given the applied surge price/radius and incentive levels (BYOL-AC-style) — this is the non-negotiable answer to policy endogeneity, since policy-averaged forecasting provably collapses per-action structure; (c) decision-value heads (completion rate, pickup ETA, driver idle minutes, conversion at offered price) constrained to linear probes; (d) a cheap dense bank of random GVF/occupancy heads over the zone-time process (Proto-Value-Network-style) to densify latent geometry beyond what few hand-picked heads anchor.
- Weighting and optimization: start with unitary scalarization plus Kendall-style uncertainty weighting to handle scale mismatch across heads. Log pairwise gradient cosines at the shared trunk throughout training; adopt PCGrad or CAGrad only if persistent conflict coincides with probe-scorecard degradation that a tuned equal-weight baseline cannot fix (per Kurin et al.'s counterpoint). If conflict is transient/regime-dependent, consider ICLR-2025-style selective task-group updates instead of permanent gradient surgery.
- Make the evaluation contract MDL-based, not accuracy-based: version every embedding with a scorecard of online (prequential) codelengths for each downstream label family, computed with frozen encoder and standardized probe budgets. Gate releases on 'no codelength regression on any consumer's task family' — this is a stable, effort-aware backward-compatibility test that plain probe accuracy is not.
- Add explicit sufficiency audits from the Rakelly framework: verify that decision-critical latent variables (price-elasticity proxies, supply-repositioning potential, cancellation risk drivers) are recoverable from the embedding, and never accept reconstruction loss as evidence of representation quality — reconstruction can be perfect while decision-relevant structure is linearly inaccessible.
- Use a VIB-style stochastic bottleneck on the embedding with a modest β , swept against the MDL scorecard, to control dimensionality and noise robustness; this also gives a principled knob when downstream teams request smaller embeddings for latency-critical services.
- Plan a maturity ladder for head selection automation: phase 1, fixed hand-designed heads with measured affinity-based pruning; phase 2, AuxiLearn/Auto-Lambda bilevel meta-learning of head weights against a multi-downstream validation objective (adapting their single-main-task formulation to a weighted basket); phase 3 (only with large spare compute), A2LS-style evolutionary search over the auxiliary-loss space for the marketplace domain.
- Re-measure affinities on regime changes: Standley et al. showed task groupings shift with data volume and network capacity, so re-run the affinity analysis when onboarding new cities, changing zone granularity, or scaling the encoder — do not assume the head set chosen at launch remains optimal, and fold this re-measurement into the embedding versioning cadence.

Annotated bibliography (23 sources)

2016 Jaderberg, M., Mnih, V., Czarnecki, W.M., Schaul, T., Leibo, J.Z., Silver, D., Kavukcuoglu, K. (2016). Reinforcement Learning with Unsupervised Auxiliary Tasks. ICLR 2017 / arXiv:1611.05397.

arxiv.org/abs/1611.05397

Augments an A3C agent with auxiliary pseudo-reward heads — pixel control, reward prediction, and value replay — all sharing one CNN-LSTM encoder, with auxiliary heads trained off-policy from a replay buffer. The auxiliary gradients continue shaping the shared representation even when extrinsic reward is absent. Delivered ~10x learning speedup on Labyrinth and 880% mean human-normalized score on Atari.

Relevance to this project: The canonical demonstration that a handful of cheap auxiliary prediction heads on a shared encoder radically improve representation quality when the primary signal is sparse — the direct ancestor of the project's auxiliary-head design.

2016 Maurer, A., Pontil, M., Romera-Paredes, B. (2016). The Benefit of Multitask Representation Learning. JMLR 17(81):1-32.

jmlr.org/papers/v17/15-242.html

Theory paper giving excess-risk bounds for learning a representation shared across T tasks, in both multitask and learning-to-learn settings. Shows the per-task sample-complexity burden of learning the representation is amortized across tasks (benefit terms shrinking with T), instantiated for linear feature learning and half-space learning, with explicit regimes (sample size, task count, intrinsic dimension) where sharing provably helps.

Relevance to this project: The formal justification for the project's premise: one encoder trained against many marketplace tasks needs less data per task and transfers to new downstream consumers — but only with enough tasks and genuine shared structure.

2016 Alain, G., Bengio, Y. (2016). Understanding Intermediate Layers Using Linear Classifier Probes. arXiv:1610.01644.

arxiv.org/abs/1610.01644

Introduces linear probes: freeze the network, train linear classifiers on each layer's features entirely independently of the model, and measure how linearly decodable a target is at each depth. Applied to Inception v3 and ResNet-50, showing linear separability of class information increases monotonically with depth.

Relevance to this project: The origin of the project's central instrument — constraining auxiliary heads to linear probes operationalizes 'decision-relevant information must be linearly accessible', and frozen-encoder probes double as a regression-test suite for versioned embeddings.

2017 Alemi, A.A., Fischer, I., Dillon, J.V., Murphy, K. (2017). Deep Variational Information Bottleneck. ICLR 2017 / arXiv:1612.00410.

arxiv.org/abs/1612.00410

Derives a variational bound on the information-bottleneck objective $\max I(Z;Y) - \beta I(Z;X)$, parameterized by a stochastic neural encoder trained with the reparameterization trick. The resulting Deep VIB models generalize better and are markedly more robust to adversarial perturbation than deterministically regularized nets.

Relevance to this project: Provides the compression dial for the market-state embedding: with the auxiliary heads defining the Y's, a VIB objective retains decision-relevant structure while discarding zone-level noise, and β trades embedding compactness against downstream coverage.

2018 Kendall, A., Gal, Y., Cipolla, R. (2018). Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics. CVPR 2018 / arXiv:1705.07115.

arxiv.org/abs/1705.07115

Learns per-task homoscedastic uncertainty parameters that set each task's loss weight as $1/(2\sigma^2)$ with a log σ penalty, derived from Gaussian and softmax likelihoods. A shared encoder with per-task decoders simultaneously learns depth regression, semantic segmentation, and instance segmentation despite losses of different units and scales.

Relevance to this project: The simplest principled way to weight heads whose losses live on incommensurate scales (ETA regression vs surge classification vs completion-rate) — the right default weighting scheme before reaching for gradient surgery.

2018 Chen, Z., Badrinarayanan, V., Lee, C.-Y., Rabinovich, A. (2018). GradNorm: Gradient Normalization for Adaptive Loss Balancing in Deep Multitask Networks. ICML 2018, PMLR 80.

proceedings.mlr.press/v80/chen18a.html

Dynamically tunes per-task loss weights so that each task's gradient norm at the shared trunk tracks a common average modulated by its relative training rate (current-to-initial loss ratio), controlled by a single asymmetry hyperparameter α . Weight updates come from an auxiliary L1 loss on gradient norms, replacing exhaustive grid search over task weights.

Relevance to this project: Prevents any single high-variance auxiliary head (e.g., a demand-forecast head during weather shocks) from dominating the shared market encoder; an industry-standard balancing baseline for the project's multi-head training.

2019 Veeriah, V., Hessel, M., Xu, Z., Rajendran, J., Lewis, R.L., Oh, J., van Hasselt, H., Silver, D., Singh, S. (2019). Discovery of Useful Questions as Auxiliary Tasks. NeurIPS 2019 / arXiv:1909.04607.

arxiv.org/abs/1909.04607

Uses non-myopic meta-gradients to learn general value function (GVF) 'questions' — cumulant and discount functions — such that learning their answers as auxiliary tasks induces representations that improve the main RL task. The auxiliary tasks themselves are parameterized and discovered, removing the need to hand-design environment-matched auxiliary objectives.

Relevance to this project: The seminal answer to the project's key open question from the discovery side: instead of hand-picking heads, meta-learn which prediction questions most improve a validation objective — directly adaptable to a downstream-task validation basket.

2020 Standley, T., Zamir, A.R., Chen, D., Guibas, L., Malik, J., Savarese, S. (2020). Which Tasks Should Be Learned Together in Multi-task Learning? ICML 2020 / arXiv:1905.07553.

arxiv.org/abs/1905.07553

Systematically trains many pairwise and groupwise task combinations on Taskonomy, measures inter-task transfer, and solves an assignment of tasks to a small set of networks under an inference-time budget. Key empirical finding: which tasks cooperate in joint training is NOT predicted by semantic relatedness or transfer-learning affinity, and shifts with data regime and network capacity.

Relevance to this project: A direct warning for head selection: intuitively related marketplace signals (e.g., demand and surge) may conflict in training while unrelated ones cooperate — compatibility must be measured, and re-measured when data scale or encoder capacity changes.

2020 Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., Finn, C. (2020). Gradient Surgery for Multi-Task Learning. NeurIPS 2020 / arXiv:2001.06782.

arxiv.org/abs/2001.06782

Identifies the tragic triad of conflicting task gradients (negative cosine similarity), high curvature, and large magnitude differences, and proposes PCGrad: project each task's gradient onto the normal plane of any gradient it conflicts with before applying the update. Model-agnostic, improving data efficiency and final performance across multi-task supervised, RL, and goal-conditioned RL settings.

Relevance to this project: The standard remedy if pricing-aligned and ETA-aligned heads send conflicting gradients into the shared encoder; its conflict measurements (pairwise gradient cosines) also double as a cheap head-affinity diagnostic for the project.

2020 Tripuraneni, N., Jordan, M.I., Jin, C. (2020). On the Theory of Transfer Learning: The Importance of Task Diversity. NeurIPS 2020 / arXiv:2006.11650.

arxiv.org/abs/2006.11650

General framework where tasks share a representation h composed with task-specific functions f_t ; introduces a task-diversity condition and a chain rule for Gaussian complexities yielding excess-risk bounds of order $C(H)/(nT) + C(F)/m$ for a new task. The training tasks must be diverse enough to 'span' the space in which downstream task-specific functions live for transfer to provably work.

Relevance to this project: The single most important theoretical criterion for CHOOSING the project's auxiliary heads: heads must be functionally diverse enough to span what pricing, dispatch, ETA, and repositioning models need — several correlated demand proxies buy provably little.

2020 Voita, E., Titov, I. (2020). Information-Theoretic Probing with Minimum Description Length. EMNLP 2020.

aclanthology.org/2020.emnlp-main.14/

Recasts probing as transmitting labels given representations and measures the minimum description length via variational coding or online/prequential coding. MDL jointly captures final probe quality and the 'effort' (data/model complexity) needed to reach it, and is empirically more stable across probe hyperparameters than accuracy-only probing.

Relevance to this project: The right metric for the project's embedding scorecard: online codelength measures how EASILY each downstream signal is extractable from a candidate embedding, giving stable, comparable numbers for version-over-version regression gates.

2021 Fifty, C., Amid, E., Zhao, Z., Yu, T., Anil, R., Finn, C. (2021). Efficiently Identifying Task Groupings for Multi-Task Learning. NeurIPS 2021 (spotlight) / arXiv:2109.04617.

arxiv.org/abs/2109.04617

Task Affinity Grouping (TAG): train all tasks together once, and during training measure inter-task affinity as the degree to which one task's gradient update on shared parameters lowers another task's loss; then select groupings maximizing total affinity. Cuts Taskonomy test loss 10% versus training everything together, ~11.6x faster than prior grouping search.

Relevance to this project: The cheapest practical instrument for the project's head-selection question — one instrumented training run over ~20-30 candidate heads yields the full affinity matrix from which a small cooperating subset can be chosen.

2021 Lyle, C., Rowland, M., Ostrovski, G., Dabney, W. (2021). On the Effect of Auxiliary Tasks on Representation Dynamics. AISTATS 2021, PMLR 130 / arXiv:2102.13089.

arxiv.org/abs/2102.13089

Analyzes temporal-difference learning as dynamics in representation space and connects the representations induced by auxiliary tasks (value ensembles, random cumulants, additional policies) to the spectral decomposition of the environment's transition operator. Uses the theory to predict and select which auxiliary tasks help in sparse-reward deep RL.

Relevance to this project: The closest thing to a theory of WHICH auxiliary heads induce WHICH latent geometry: predicting many cumulants of the market process pushes the embedding toward the transition operator's dominant eigenspace — precisely the structure forecasting and repositioning consumers need.

2021 Rakelly, K., Gupta, A., Florensa, C., Levine, S. (2021). Which Mutual-Information Representation Learning Objectives are Sufficient for Control? NeurIPS 2021 / arXiv:2106.07278.

arxiv.org/abs/2106.07278

Formalizes sufficiency of a state representation for learning and representing the optimal policy, then audits popular mutual-information objectives (forward info, state-only transition info, inverse dynamics) through that lens. Proves that two widely used objectives yield provably insufficient representations under mild, common MDP assumptions, with empirical confirmation in simulation.

Relevance to this project: A precise warning for the project: plausible self-supervised objectives can silently discard decision-critical state — sufficiency must be defined relative to the downstream decision family, which is exactly what decision-aligned auxiliary heads enforce.

2021 Navon, A., Achituve, I., Maron, H., Chechik, G., Fetaya, E. (2021). Auxiliary Learning by Implicit Differentiation (AuxiLearn). ICLR 2021 / arXiv:2007.02693.

arxiv.org/abs/2007.02693

Formulates auxiliary-task usage as bilevel optimization: the inner loop trains on the main loss plus weighted auxiliary losses, while the outer loop optimizes a combination network (allowing nonlinear interactions among losses) against main-task validation loss via the implicit function theorem. When no good auxiliary task is known, the same machinery learns to GENERATE a useful auxiliary labeling.

Relevance to this project: Turns 'how much should each head shape the encoder' into a differentiable problem: with a validation basket of downstream marketplace tasks as the outer objective, head weights can be meta-learned rather than hand-tuned. Successor Auto-Lambda (arXiv:2202.03091) extends this to dynamic task relationships.

2021 Liu, B., Liu, X., Jin, X., Stone, P., Liu, Q. (2021). **Conflict-Averse Gradient Descent for Multi-task Learning**. **NeurIPS 2021 / arXiv:2110.14048**.

arxiv.org/abs/2110.14048

CAGrad seeks the update direction that maximizes the worst-case local improvement over all tasks within a ball around the average gradient, generalizing gradient descent and MGDA. Unlike heuristic gradient surgery it provably converges to a stationary point of the AVERAGE loss while automatically balancing per-task progress; improves over prior gradient-manipulation methods in multi-task supervised and RL benchmarks.

Relevance to this project: The conflict remedy with a convergence guarantee on the mean objective — appropriate for the project's setting where no single downstream vertical's head should be quietly sacrificed to help the others.

2022 Kurin, V., De Palma, A., Kostrikov, I., Whiteson, S., Mudigonda, P.K. (2022). **In Defense of the Unitary Scalarization for Deep Multi-Task Learning**. **NeurIPS 2022 / arXiv:2201.04122**.

arxiv.org/abs/2201.04122

Shows across supervised MTL and multi-task RL benchmarks that plain unitary scalarization (summing task losses) combined with standard single-task regularization and stabilization matches or beats specialized multi-task optimizers (PCGrad, CAGrad, MGDA-style methods). Offers analysis suggesting the specialized optimizers act largely as implicit regularizers.

Relevance to this project: The essential counterweight before the project invests in gradient machinery: run a well-tuned equal-weight baseline first, and prefer the simplest optimizer that passes the probe scorecard — simplicity also aids reproducibility of versioned embeddings.

2022 He, T., Zhang, Y., Ren, K., Liu, M., Wang, C., Zhang, W., Yang, Y., Li, D. (2022). **Reinforcement Learning with Automated Auxiliary Loss Search (A2LS)**. **NeurIPS 2022 / arXiv:2210.06041**.

arxiv.org/abs/2210.06041

Defines a general space of $\sim 7.5 \times 10^{20}$ auxiliary loss functions over trajectory elements (observations, actions, rewards across horizons, with different similarity measures) and searches it with an evolutionary outer loop wrapped around inner-loop RL training. The discovered loss 'A2-winner' transfers robustly to unseen image-based and state-based continuous-control environments, beating hand-designed auxiliary losses.

Relevance to this project: Proof-of-concept that auxiliary-loss CHOICE itself is searchable and that searched losses generalize across environments — a compute-heavy but viable fallback if the project's hand-designed head sets plateau.

2023 Farebrother, J., Greaves, J., Agarwal, R., Le Lan, C., Goroshin, R., Castro, P.S., Bellemare, M.G. (2023). Proto-Value Networks: Scaling Representation Learning with Auxiliary Tasks. ICLR 2023 / arXiv:2304.12567.

arxiv.org/abs/2304.12567

Generates large families of auxiliary tasks from the successor measure — predicting random indicator functions of future state occupancy — and scales the number of auxiliary heads jointly with network size. The resulting frozen representations, evaluated by training only linear value heads on top, yield performance comparable to full DQN training on Atari.

Relevance to this project: The strongest evidence for the project's target architecture: many cheap, automatically generated heads produce a general-purpose FROZEN embedding that linear downstream readers can decode — and it argues for a dense bank of random occupancy/GVF heads alongside the hand-picked decision heads.

2024 Li, D., Sharma, A., Zhang, H.R. (2024). Scalable Multitask Learning Using Gradient-based Estimation of Task Affinity (Grad-TAG). KDD 2024 / arXiv:2409.06091.

arxiv.org/abs/2409.06091

Trains a single base model on all tasks, then estimates the fine-tuned loss of ANY task combination via linearization around the base model: logistic regression over low-dimensionally projected gradient features predicts each combination's performance without retraining. Task clusters are then chosen by semidefinite programming on the estimated affinity matrix; achieves within 2.7% of true affinities at ~3% of full-training compute, scaling to 500 tasks.

Relevance to this project: Makes near-exhaustive head-subset evaluation tractable for the project: score all candidate subsets of auxiliary heads for a few percent of the compute of one full run before committing an encoder version — the current practical SOTA for task-affinity prediction.

2024 Khetarpal, K., Guo, Z.D., Avila Pires, B., Tang, Y., Lyle, C., Rowland, M., Heess, N., Borsa, D., Guez, A., Dabney, W. (2025). A Unifying Framework for Action-Conditional Self-Predictive Reinforcement Learning. AISTATS 2025, PMLR 258 / arXiv:2406.02035.

arxiv.org/abs/2406.02035

Analyzes BYOL-style latent self-prediction with continuous-time ODE tools, comparing policy-averaged prediction (BYOL-II) with action-conditional prediction (BYOL-AC). Characterizes their distinct fixed points — action-conditioning preserves per-action distinctions that policy-averaged objectives collapse — and unifies the two through a variance-like third objective, with validation in linear and deep RL settings.

Relevance to this project: Directly addresses the project's policy-endogeneity tension: training the encoder to predict the next market state CONDITIONED on the surge/incentive action learns representations that support counterfactual comparison of control levers, which policy-averaged forecasting provably cannot.

2025 Jeong, W., Yoon, K.-J. (2025). Selective Task Group Updates for Multi-Task Optimization. ICLR 2025 / arXiv:2502.11986.

arxiv.org/abs/2502.11986

Introduces proximal inter-task affinity, tracked online during training, and uses it to adaptively partition tasks into groups that are updated sequentially within each batch rather than jointly. Theoretical analysis shows sequential group updating improves task-specific parameter learning and reduces negative transfer; outperforms prior multi-task optimization approaches across architectures and task counts.

Relevance to this project: 2025 state of the art showing affinity can be exploited DURING training, not just for offline grouping — applicable when the market encoder trains with its full head set and some heads interfere only transiently.

2025 Yu, J., Chen, S., Liu, M., Horiuchi, N., Braverman, V., Xu, Z., Haramati, D., Balestriero, R. (2025). Why and How Auxiliary Tasks Improve JEPA Representations. arXiv:2509.12249.

arxiv.org/abs/2509.12249

Theoretically analyzes a JEPA variant that trains latent transition dynamics jointly with an auxiliary regression head. Proves a 'No Unhealthy Representation Collapse' theorem: in deterministic MDPs, driving both the latent-consistency loss and the auxiliary loss to zero forces all non-equivalent observations to distinct latents — the auxiliary head acts as an anchor defining which distinctions the encoder must preserve. Controlled experiments confirm joint training beats post-hoc separate training.

Relevance to this project: The most recent theory validating the project's exact architecture: auxiliary heads are what prevent a predictive/reconstruction-style market encoder from collapsing decision-relevant distinctions, and the anchoring argument justifies one head per downstream decision family.

8 Representations for Control & Causal / Counterfactual Inference

State of the art

Representation learning for control has evolved through three phases. First came reconstruction-based encoders (autoencoders on pixels), which were displaced around 2019-2021 by "self-predictive" objectives — DeepMDP, DBC-style bisimulation metrics, and SPR — that train an encoder so the latent state, conditioned on the ACTION, predicts the next latent state and the reward, discarding decision-irrelevant detail. In parallel, value-equivalent world models (MuZero, Dreamer, TD-MPC2) showed that an encoder need only preserve what is required to predict reward, value and policy under action sequences, and that planning in such latents is state-of-the-art for control; successor features separately showed how to factor dynamics from reward so one representation serves many reward functions. A consolidation phase (Ni et al. ICLR 2024; large-scale 2025 studies; the 2025 Echchahed & Castro survey, arXiv 2506.17518) established that the action-conditioned latent self-prediction ("ZP") loss plus reward prediction, with stop-gradient/EMA targets, is the common core that actually works, while pure metric losses are brittle. The causal thread runs alongside: because logged actions depend on the state (policy endogeneity), TARNet/CFR introduced balanced representations for counterfactual outcome prediction, DragonNet replaced balance with propensity-sufficiency, VCNet extended this to continuous treatments (directly analogous to a continuous surge multiplier), Delphic offline RL handles non-identifiable hidden confounding, and the OPE literature (Uehara-Shi-Kallus) specifies the logging (propensities, overlap/randomization) that any downstream evaluation needs. The 2025-2026 frontier is exactly this project's tension: learned world models degrade under policy-induced distribution shift (a formalized Lucas critique — performative prediction; DSGE-Gym 2026 shows off-path tail error ~40x on-path for Dreamer/transformer world models, largely repaired by training on structurally simulated counterfactual data). Open problems: objective mismatch between model loss

and decision quality, representation collapse in self-prediction, the balance-vs-information tradeoff in shared embeddings, and OPE for continuous action spaces.

Key principles

- Reconstruction is neither necessary nor sufficient for control: the field converged on action-conditioned latent self-prediction (predict z_{t+1} from z_t AND a_t) plus reward prediction as the core encoder objective (DeepMDP, SPR, Ni et al. 2024). A pure autoencoder organizes the latent by pixel/feature variance, not decision value — exactly the failure this project already anticipates.
- Value equivalence is the strongest form of decision-awareness: MuZero/TD-MPC2 encoders keep ONLY what predicts reward, value and action outcomes, and this outperforms observation-predictive models for control; Dreamer-style observation prediction is a useful regularizer when rewards are sparse or many downstream tasks exist. For a shared company-wide embedding, a middle point (self-prediction + multiple KPI heads) beats either extreme.
- Actions must be inputs to the representation learner, never treated as observed features: every control-grade objective conditions the transition on the action. If surge enters the encoder as just another input feature, the embedding confounds "market was hot" with "we priced high" and cannot answer counterfactuals.
- Decouple dynamics from reward for cross-team transfer: successor features $\psi(s,a)$ with task-specific linear weights w give provable zero-shot transfer across reward functions and yield exactly linear value readouts — the mathematically grounded version of the project's linear-probe constraint.
- How the objective is optimized matters as much as the objective: stop-gradient/EMA target encoders prevent latent collapse (SPR, Ni et al. 2024); multi-step (K-step) latent rollouts beat one-step prediction; bisimulation-metric losses are brittle as the sole training signal and should at most be auxiliary (Luo et al. 2025 large-scale study).
- Under policy-endogenous logging, the representation must support counterfactual queries, and there is a tradeoff: CFR-style balancing reduces policy-selection bias but destroys information; DragonNet's propensity-sufficiency view says instead to preserve exactly the confounding information and adjust with it. For a multi-consumer embedding, keep the shared representation information-rich (including a propensity head) and do balancing inside downstream causal heads.
- Continuous control levers need continuous-treatment machinery: discretizing surge into treatment buckets produces discontinuous dose-response estimates; varying-coefficient heads (VCNet) keep the price-response curve smooth and identifiable.
- Downstream benefit is certified by off-policy evaluation, not by reconstruction or forecasting error: IPS/DR estimators require logged propensities and overlap (some randomization of the levers). An embedding version should ship with an OPE-based scorecard of fixed reference policies plus a linear-probe suite on decision-relevant targets.
- Lucas critique / performativity is a first-class design constraint: models fit on on-policy data are only disciplined where data were seen and collapse under policy regime shift (Perdomo et al. 2020; DSGE-Gym 2026). Mitigations that work: structural/simulator-generated counterfactual training data, explicit uncertainty over world models compatible with the data (delphic uncertainty), bounded exploration in the levers, and monitoring for performative drift after each policy change.
- Pretrain-then-finetune with a frozen encoder is viable and industrially proven (SGI; DiDi's deployed dispatching value functions), but the probe/finetune protocol and the evaluation suite must be versioned together with the encoder weights.

Recommendations for the market-state embedding project

- Train the encoder with a TD-MPC2/SPR-style composite objective, not an autoencoder: (1) K-step action-conditioned latent self-prediction with an EMA target encoder and stop-gradient ($K \approx 4-8$ time buckets), (2) reward/KPI prediction heads (completed trips, GMV, supply hours, cancellations), (3) optional TD value heads. Keep any observation decoder only as an offline diagnostic, never as the organizing loss.
- Treat surge price, surge radius, incentives, and repositioning nudges as ACTIONS entering the latent transition model — $z_{t+1} = f(z_t, a_t)$ — and never as covariates inside the state encoder. This is what makes the embedding answer counterfactuals ('demand at 1.4x surge') instead of memorizing the historical policy's correlations.
- Add a DragonNet-style propensity head to the shared encoder: predict the logged surge/incentive action from z_t . It (a) certifies that confounder information survived compression, (b) supplies propensities for doubly-robust OPE, and (c) is itself a useful linear probe. For the continuous surge multiplier use a VCNet varying-coefficient outcome head to get smooth dose-response curves; do NOT discretize surge into buckets.
- Do not put CFR-style balancing penalties on the shared embedding — balancing provably discards information (Shalit et al.'s bound) that other consumers (ETA, forecasting) need. Keep the shared z information-rich; apply balancing/weighting inside each downstream causal head that estimates effects.
- Add a successor-feature head $\psi(z, a)$ over a fixed basis of market features so any team's value function is a linear readout $\psi \cdot w$. This operationalizes the linear-probe constraint, gives provable transfer when reward priorities change (e.g., growth vs profitability quarters), and lets you compare embedding versions by SF-probe quality.
- Fix the logging contract now: log the propensity/parameters of every surge and incentive decision, and run bounded randomization (small jitter on surge in a rotating subset of zone-hours) to guarantee overlap. Per Uehara-Shi-Kallus, without logged propensities and overlap, no future OPE of embedding versions or policies is statistically valid.
- Evaluate embeddings by decisions, not reconstructions: each versioned embedding ships with (a) a linear-probe suite on decision-relevant targets (short-horizon demand/supply, price elasticity, matching rate), (b) OPE scores (doubly-robust) of a fixed set of reference policies computed through the embedding, and (c) a denoising-factor-style diagnostic (Luo et al. 2025) checking that known-irrelevant telemetry is filtered out.
- Defend against the Lucas critique explicitly (DSGE-Gym 2026 result): maintain a structural marketplace simulator (queueing/matching + simple rider/driver choice models) and mix its counterfactual regimes (extreme surge, supply shocks, new-policy rollouts) into encoder training data; report off-path prediction error separately from on-path error in every embedding release; monitor performative drift after each downstream policy launch and schedule retraining accordingly (Perdomo et al.).
- Quantify confounding risk with delphic-uncertainty-style ensembles: train several world models consistent with the logged data and flag zone-hours where their counterfactual predictions diverge; downstream pricing should fall back to conservative policies there, and those zone-hours are the highest-value targets for randomized exploration.
- Sequence the rollout like SGI/DiDi: pretrain the encoder offline on historical zone-time trajectories, validate with the probe+OPE suite, freeze and version it, let downstream teams consume via linear heads first (cheap, comparable, auditable), and only later allow selective fine-tuning through decision-focused (regret) losses for the highest-value consumers such as surge optimization.

Annotated bibliography (19 sources)

2019 Gelada, C., Kumar, S., Buckman, J., Nachum, O., Bellemare, M. G. (2019). DeepMDP: Learning Continuous Latent Space Models for Representation Learning. ICML 2019.
proceedings.mlr.press/v97/gelada19a.html

Trains an encoder by minimizing two tractable latent losses: predict the immediate reward from (z_t, a_t) and predict the distribution of the next latent state z_{t+1} — i.e., learn a latent MDP rather than reconstruct observations. Proves these losses bound the value difference between the latent MDP and the true MDP (via a connection to bisimulation), so the embedding is guaranteed to preserve decision-relevant structure. Used as an auxiliary task on Atari it substantially improves model-free RL.

Relevance to this project: The theoretical charter for the project's planned auxiliary heads: reward-prediction + action-conditioned next-latent-prediction is provably sufficient to make a market-state latent 'decision-complete' without any reconstruction loss.

2021 Zhang, A., McAllister, R., Calandra, R., Gal, Y., Levine, S. (2021). Learning Invariant Representations for Reinforcement Learning without Reconstruction (DBC). ICLR 2021.
arxiv.org/pdf/2006.10742

Learns an encoder whose latent L1 distances match a bisimulation metric: two states are close iff they yield similar immediate rewards and similar transition distributions under all actions. Trained jointly with a latent dynamics model and reward model, with no decoder, so task-irrelevant distractors (backgrounds, noise) are provably discarded. Demonstrated on control tasks with natural-video distractors.

Relevance to this project: Defines the right notion of 'similar market states' for a company-wide embedding — similarity in reward/transition consequences, not in raw feature space (two zones with identical weather but different elasticity should embed far apart). Caveat: use as an auxiliary shaping loss, not the sole objective (see Luo et al. 2025).

2021 Schwarzer, M., Anand, A., Goel, R., Hjelm, R. D., Courville, A., Bachman, P. (2021). Data-Efficient Reinforcement Learning with Self-Predictive Representations (SPR). ICLR 2021.
arxiv.org/pdf/2007.05929

Adds a self-supervised objective to a Q-learning agent: an online encoder plus action-conditioned transition model must predict, K steps ahead, the representations produced by an EMA target encoder (BYOL-style cosine loss, stop-gradient), with augmentations. No reconstruction and no negative samples; the multi-step action-conditioned consistency loss alone drives the representation. Set the standard on Atari-100k; its successor SGI (Schwarzer et al., NeurIPS 2021, proceedings.neurips.cc/paper/2021/hash/69eba34671b3ef1ef38ee85caae6b2a1) showed the same objectives work for offline PRE-training of an encoder that is then finetuned cheaply on downstream tasks.

Relevance to this project: The concrete recipe for the encoder's training loop: K -step latent rollout consistency conditioned on surge/incentive actions, EMA target network to prevent collapse. SGI's pretrain-then-finetune result is the direct template for shipping one pretrained market encoder to many teams.

2024 Ni, T., Eysenbach, B., Seyedsalehi, E., Ma, M., Gehring, C., Mahajan, A., Bacon, P.-L. (2024). **Bridging State and History Representations: Understanding Self-Predictive RL.** ICLR 2024. arxiv.org/abs/2401.08898

Unifies SPR, DBC, TD-MPC, EfficientZero and POMDP history abstractions under one condition: the latent must predict its own next value given the action (the 'ZP' condition), organizing all methods into Q*-irrelevance, self-predictive, and observation-predictive abstraction classes. Provides theory for why stop-gradient targets avoid collapse and distills a minimalist algorithm; validated on MDPs with distractors and sparse-reward POMDPs.

Relevance to this project: The market state is partially observed (latent driver intent, competitor actions), and this paper is the authoritative guide to learning a self-predictive HISTORY representation — plus the design rules (stop-gradient, which losses to combine) the project should follow rather than rediscover.

2025 Luo, Z., Ni, T., Bacon, P.-L., Precup, D., Si, X. (2025). **Understanding Behavioral Metric Learning: A Large-Scale Study on Distracting Reinforcement Learning Environments.** arXiv:2506.00563. arxiv.org/abs/2506.00563

Benchmarks five behavioral-metric (bisimulation-style) representation methods, unified as isometric embeddings, across 370 task configurations with task-irrelevant distractions, introducing a 'denoising factor' to measure how well an encoder filters distractors and an isolated setting where only the metric loss shapes the encoder. Finds metric losses alone are unreliable; simple self-prediction plus reward signals carry most of the benefit.

Relevance to this project: The most current empirical evidence on which auxiliary losses to actually deploy in the market encoder, and its denoising-factor diagnostic is directly reusable as an embedding-quality metric (does the embedding ignore decision-irrelevant telemetry?).

2020 Schrittwieser, J., Antonoglou, I., Hubert, T., Simonyan, K., Sifre, L., Schmitt, S., Guez, A., Lockhart, E., Hassabis, D., Graepel, T., Lillicrap, T., Silver, D. (2020). **Mastering Atari, Go, Chess and Shogi by Planning with a Learned Model (MuZero).** Nature 588 / arXiv:1911.08265. arxiv.org/abs/1911.08265

Learns an encoder and recurrent latent dynamics trained end-to-end so that unrolled latents predict only reward, value, and policy — never observations ('value equivalence'). MCTS plans directly in this latent space; the representation is shaped purely by what improves planning targets. Achieved SOTA on Atari while matching AlphaZero on board games without knowing the rules.

Relevance to this project: The purest existing answer to 'organize the latent space by decision value': the encoder is trained by decision-relevant heads only. It also warns that a purely value-equivalent latent overfits to the current reward/policy — a shared multi-team embedding should hedge with several KPI heads.

2023 Hafner, D., Pašukonis, J., Ba, J., Lillicrap, T. (2023; Nature 2025). Mastering Diverse Domains through World Models (DreamerV3). arXiv:2301.04104; Nature (2025).

arxiv.org/abs/2301.04104

An RSSM world model encodes inputs into discrete categorical latents and is trained with three losses: observation reconstruction/prediction, reward and continuation prediction, and a KL 'dynamics vs representation' consistency between the action-conditioned prior and the posterior; an actor-critic is then trained entirely inside imagined latent rollouts. Robustness tricks (symlog, two-hot returns, free bits) let one hyperparameter set master 150+ domains; published in Nature in 2025.

Relevance to this project: The strongest template for a market SIMULATOR: train policy candidates (surge, repositioning) in imagination before touching the real marketplace, and its stochastic latent (posterior vs prior) cleanly separates 'what we can infer from data' from 'what our actions cause'.

2024 Hansen, N., Su, H., Wang, X. (2024). TD-MPC2: Scalable, Robust World Models for Continuous Control. ICLR 2024; arXiv:2310.16828.

arxiv.org/abs/2310.16828

A decoder-free implicit world model: the encoder and latent dynamics are trained jointly with multi-step latent self-consistency, reward prediction, and TD value losses (no reconstruction), and control is done by short-horizon trajectory optimization (MPPI) in the latent space. A single hyperparameter set works across 104 continuous-control tasks, and one 317M-parameter model handles 80 tasks across embodiments and action spaces.

Relevance to this project: Current SOTA for continuous action spaces — surge multiplier and radius are continuous levers, and TD-MPC2's recipe (latent consistency + reward + value heads, planning over candidate action sequences) maps directly to evaluating candidate surge trajectories per zone. Its multi-task scaling result supports one encoder serving many cities/tasks.

2017 Barreto, A., Dabney, W., Munos, R., Hunt, J. J., Schaul, T., van Hasselt, H., Silver, D. (2017). Successor Features for Transfer in Reinforcement Learning. NeurIPS 2017; arXiv:1606.05312.

arxiv.org/abs/1606.05312

Decomposes the value function as $Q^\pi(s,a) = \psi^\pi(s,a) \cdot w$, where successor features ψ are the expected discounted sum of state features under the policy (learned by TD) and w encodes the reward function linearly. With generalized policy improvement, policies trained under old reward weights transfer to any new w with performance guarantees, before any new learning.

Relevance to this project: The exact mechanism the project needs for 'one embedding, many downstream objectives': pricing, dispatch, and incentive teams differ mainly in reward weights, and an SF head makes each team's value function a LINEAR readout of the shared representation — the principled version of the linear-probe constraint.

2024 Mandi, J., Kotary, J., Berden, S., Mulamba, M., Bucarey, V., Guns, T., Fioretto, F. (2024). **Decision-Focused Learning: Foundations, State of the Art, Benchmark and Future Opportunities**. *JAIR* 80 (2024); arXiv:2307.13565.

arxiv.org/abs/2307.13565

Survey and benchmark of decision-focused learning: training predictive models end-to-end through a downstream constrained-optimization layer so the loss is regret in DECISION quality, not prediction accuracy; covers smoothed/surrogate gradients of argmax operators, perturbation methods, and learned losses, with standardized benchmarks.

Relevance to this project: Surge setting and dispatch are optimization layers on top of the embedding; DFL supplies the gradients to fine-tune the encoder (or its heads) against realized decision regret rather than forecasting error — the strongest known way to 'harvest maximum downstream benefit'.

2025 Echchahed, A., Castro, P. S. (2025). **A Survey of State Representation Learning for Deep Reinforcement Learning**. arXiv:2506.17518 (TMLR).

arxiv.org/abs/2506.17518

Up-to-date taxonomy of state-representation learning for control, organizing methods into six classes (reconstruction-based, self-predictive/latent-model, metric-based, contrastive, auxiliary-task, and pretrained/foundation encoders), detailing mechanisms, benefits, limitations, and — importantly — techniques for assessing representation quality independently of final returns.

Relevance to this project: The best single onboarding document for the team, and its representation-evaluation section provides ready-made probes and diagnostics for the embedding's versioned eval suite.

2017 Shalit, U., Johansson, F. D., Sontag, D. (2017). **Estimating Individual Treatment Effect: Generalization Bounds and Algorithms (TARNet/CFR)**. *ICML 2017*.

proceedings.mlr.press/v70/shalit17a/shalit17a.pdf

Learns a shared representation $\Phi(x)$ with per-treatment outcome heads (TARNet); CFR adds an integral-probability-metric (Wasserstein/MMD) penalty that makes the distribution of Φ over treated and control units similar. Proves the ITE error is bounded by factual prediction error plus the representation-induced distance between treatment groups — the founding result of balanced representation learning.

Relevance to this project: Directly applicable to estimating 'what would demand/completion be under a different surge level' from logged, policy-endogenous data; the bound also formalizes the danger — balancing trades off predictive information, so it belongs in downstream causal heads, not the shared embedding.

2019 Shi, C., Blei, D. M., Veitch, V. (2019). **Adapting Neural Networks for the Estimation of Treatment Effects (DragonNet)**. *NeurIPS 2019*; arXiv:1906.02120.

arxiv.org/abs/1906.02120

Three-headed architecture: a shared representation feeds two outcome heads and a propensity head predicting the logged treatment, exploiting the sufficiency of the propensity score — the representation is pushed to retain exactly the confounding information needed for adjustment. Targeted regularization adds a TMLE-inspired term giving asymptotically doubly-robust estimates end-to-end.

Relevance to this project: The most directly reusable causal design for this project: add a propensity head that predicts the logged surge action from the market embedding, guaranteeing confounders survive compression and enabling doubly-robust downstream effect estimates and OPE.

2021 Nie, L., Ye, M., Liu, Q., Nicolae, D. (2021). VCNet and Functional Targeted Regularization for Learning Causal Effects of Continuous Treatments. ICLR 2021 (Oral); arXiv:2103.07861.
arxiv.org/abs/2103.07861

For continuous treatments, replaces per-treatment-bucket heads with a varying-coefficient network whose weights are smooth functions of the treatment value, producing a continuous average dose-response curve from a shared representation, plus a functional targeted regularization giving double robustness for the whole curve.

Relevance to this project: Surge price is a continuous treatment: VCNet is the right head architecture for estimating smooth demand/supply response curves as a function of surge level on top of the market embedding, instead of discretizing surge into buckets.

2024 Pace, A., Yèche, H., Schölkopf, B., Rätsch, G., Tennenholtz, G. (2024). Delphic Offline Reinforcement Learning under Nonidentifiable Hidden Confounding. ICLR 2024; arXiv:2306.01157.
arxiv.org/abs/2306.01157

Defines 'delphic uncertainty' — variation across world models that are all compatible with the confounded logged data — as distinct from epistemic and aleatoric uncertainty, without assuming confounders are identifiable. Estimates it with an ensemble of compatible models and penalizes it pessimistically in offline RL, reducing confounding bias on real (medical) logged data.

Relevance to this project: Logged marketplace data is confounded by whatever ops/policy logic set surge historically (including unlogged manual overrides); delphic uncertainty gives a practical way to flag zone-hours where the embedding's counterfactual predictions are not trustworthy and the policy should stay conservative.

2022 Uehara, M., Shi, C., Kallus, N. (2022). A Review of Off-Policy Evaluation in Reinforcement Learning. arXiv:2212.06355.
arxiv.org/abs/2212.06355

Comprehensive review of OPE: importance sampling, direct/model-based methods, doubly-robust and semiparametrically efficient estimators, marginalized IS for long horizons, and OPE under partial observability/confounding, including efficiency bounds and the overlap/propensity-logging assumptions each estimator needs.

Relevance to this project: Defines the evaluation contract for the embedding program: which estimators the company can use to score new embedding versions and candidate policies offline, and exactly what must be logged (action propensities, sufficient overlap) for those scores to mean anything.

2020 Perdomo, J. C., Zrnic, T., Mendler-Dünner, C., Hardt, M. (2020). **Performative Prediction.** ICML 2020; arXiv:2002.06673.

arxiv.org/abs/2002.06673

Formalizes settings where deploying a predictor shifts the distribution it predicts (the ML formalization of the Lucas critique), introduces performative stability and performative optimality, and analyzes when repeated retraining converges to a stable point, with tools from statistics, game theory, and causality.

Relevance to this project: Surge predictions move riders and drivers, so the market-state embedding is performative by construction: retraining on data generated under the embedding's own downstream policies can drift; this framework tells you when naive retraining converges and motivates explicit policy-shift monitoring in the versioning process.

2022 Eshkevari, S. S., Tang, X., Qin, Z., Mei, J., Zhang, C., Meng, Q., Xu, J. (2022). **Reinforcement Learning in the Wild: Scalable RL Dispatching Algorithm Deployed in Ridehailing Marketplace.** KDD 2022; arXiv:2202.05118.

arxiv.org/abs/2202.05118

DiDi's deployed dispatching system: learns spatiotemporal state-value functions online via an uncertainty-adapted temporal-difference update, uses them inside a matching utility for driver-order assignment, and adds bandit-based graph pruning to cut cancellations — running at full production scale with reported gains in A/B tests.

Relevance to this project: Proof-of-adoption that learned value representations over zone-time market state can be the shared currency of a ride-hailing decision system, and a template for the online-TD infrastructure and A/B validation the embedding program will need.

2026 Xu, W. (2026). **DSGE as a Structured World Model: Benchmarking Counterfactual Generalization in Economic Worlds.** arXiv:2607.03144.

arxiv.org/html/2607.03144v1

Introduces DSGE-Gym, eight economic environments with counterfactual (policy-regime-shift) test cases, and benchmarks Dreamer-style and transformer world models against structural DSGE models. Learned world models fit on-policy trajectories well but collapse off-path (tail error ~40x on-path); augmenting training with synthetic data from the structural model — spanning rare and counterfactual regimes — cuts tail error ~50% and improves policy-regime error 10-280x.

Relevance to this project: The clearest 2026 evidence for this project's core risk: a market-state world model trained only on logged surge policies will mis-predict under a new pricing policy, and the demonstrated fix — blend structural marketplace-simulator counterfactual data into encoder training — is directly actionable.

9 Industrial Embedding Systems at Marketplaces & Two-Sided Platforms

State of the art

Industrial embedding systems at marketplaces evolved in three waves. Wave 1 (2016-2019) established that entity embeddings trained on interaction logs, served via ANN retrieval, ship real revenue: Airbnb's listing embeddings (skip-gram over click sessions with booking-aware objectives), YouTube's two-tower retrieval, and Pinterest's PinSage (graph convolutions over billions of nodes) — all consumed as precomputed vectors from a feature store. In parallel, ride-hailing platforms (Uber, DiDi) shipped spatiotemporal representations: bucketize-and-embed geo/time features (DeepETA), coarse-coded hierarchical spatial encodings in dispatch value

networks, and LSTM-autoencoder "city state" features feeding demand forecasters — the direct ancestors of a market-state embedding. Wave 2 (2020-2023) industrialized the operating model: feature stores (Uber Michelangelo Palette, Lyft Feature Service, DoorDash's Redis-based store) became the distribution fabric for embeddings with offline/online parity, defined refresh cadences, and entity-keyed serving; Pinterest's ItemSage proved that one multi-task embedding serving many surfaces beats per-team embeddings on both metrics and maintenance cost; and Stanford/Pinterest's Learning Backward Compatible Embeddings formalized version governance so consumer teams need not retrain when the producer retrains. Wave 3 (2024-2026) is consolidation into foundation-model-style central representations: OmniSearchSage (multi-entity, multi-task, co-trained for compatibility with existing embedding spaces, 300k QPS), DoorDash's DashCLIP (LLM-curated labels replacing policy-endogenous engagement supervision), Netflix's personalization foundation model (one autoregressive model whose embeddings, fine-tunes, and direct predictions are consumed company-wide with explicit stability handling across retrains), and Airbnb's 2026 production embedding-based retrieval handling dynamic two-sided inventory. Open problems that recur across these systems and map directly onto this project: keeping the latent space organized by decision value rather than reconstruction fidelity, de-biasing training signal that was generated under the platform's own control policies, and managing representation churn/version compatibility for dozens of downstream consumers.

Key principles

- Objective beats architecture: every successful industrial embedding modified its training objective to encode the business event that downstream decisions hinge on (Airbnb's booked-listing global context and in-market negatives; ItemSage's purchase heads; Yi et al.'s bias-corrected softmax). Pure reconstruction or generic co-occurrence consistently underperforms decision-aligned objectives downstream.
- Multi-task, multi-surface training is the shipped standard for embeddings with many consumers: a single embedding trained against several downstream objectives simultaneously (ItemSage, OmniSearchSage, Netflix FM) outperforms per-team embeddings, helps sparse-label tasks most, and slashes maintenance cost — validating auxiliary prediction heads as the central design element, not an add-on.
- Linear accessibility is engineered on purpose: two-tower dot-product compatibility, linear probes, and ANN-consumable geometries are chosen so that downstream consumption is cheap and simple; if a consumer needs a deep model to extract signal from the embedding, industry treats that as a defect of the embedding.
- Bias in the training signal must be corrected explicitly or it is inherited by every consumer: sampling-bias correction (YouTube), position/selection bias handling, causal separation of forecasting from lever-effect estimation (DoorDash), and LLM-curated labels replacing engagement logs (DashCLIP) are all responses to the same problem this project faces with surge-endogenous data.
- Spatiotemporal inputs are quantized then embedded: production systems (DeepETA, DiDi value networks) bucketize continuous features and embed the buckets, and encode location at multiple spatial resolutions simultaneously (multi-precision geohash/H3, CMAC coarse coding) — this gives robustness, smoothness, and cross-city transfer that raw continuous features and single-resolution zone IDs do not.
- Embeddings are feature-store artifacts, not model internals: they are keyed by entity, stored with offline/online parity and point-in-time-correct history for training joins, refreshed on a declared cadence (daily batch for slow entities, streaming for fast market state), and accessed config-driven by consumer teams (Palette, Lyft Feature Service, Uber TTE).
- Version governance is a first-class engineering problem: retraining rotates the latent space and silently breaks consumer models, so shipped systems use learned backward-alignment transforms (BC-Aligner), co-training new versions for compatibility with existing spaces (OmniSearchSage), incremental training with warm starts to limit churn (Netflix), and parallel serving of adjacent versions during migration.

- Evaluation is two-layer and consumer-centric: intrinsic metrics (retrieval hit-rate, probe accuracy on downstream labels) gate iteration speed, but promotion requires wins in consumer-team A/B tests; embeddings survive organizationally only when multiple downstream teams can demonstrate lift from them.
- Cold start and regime shift are handled by content/attribute-based encoding rather than pure ID memorization: Airbnb's attribute averaging, PinSage/ItemSage content features, Netflix metadata tokens — for market state this means encoding zones by observable features (demographics, POI, weather, historical dynamics) so new zones, cities, and regimes generalize.
- Serving constraints shape the encoder: millisecond latency budgets drove linear attention (DeepETA), precomputation plus ANN lookup (all retrieval systems), and offline batch inference at billions-of-entities scale (PinSage MapReduce); the embedding's refresh latency must match the fastest consumer's decision cadence.

Recommendations for the market-state embedding project

- Make multi-task heads the primary training signal, not auxiliary: train the zone-time encoder jointly on demand forecast, supply/availability forecast, conversion rate, ETA residual, and incentive/surge response heads (ItemSage/OmniSearchSage pattern), keeping heads linear to enforce linear accessibility. Weight tasks so sparse-but-valuable labels (incentive response) borrow strength from dense ones (demand). Retain a small reconstruction term only as regularization.
- Encode inputs the way deployed ride-hailing models do: bucketize all continuous features and embed the buckets; represent location with multi-resolution hierarchical embeddings (e.g., H3 at 2-3 resolutions, summed or concatenated) rather than a flat zone-ID table; add calendar/event/weather embeddings; add a functional-similarity graph over zones (DMVST-Net semantic view) so behaviorally similar zones share statistics. This buys robustness, smoothness, and cross-city transfer (DiDi's CMAC evidence).
- Treat action-conditioning as a schema requirement from day one: log surge price, surge radius, and incentive levers alongside every state snapshot; train action-conditioned heads (predict outcome given state AND action) rather than marginal heads, so the embedding separates market state from policy artifacts. Follow DoorDash's split: embedding feeds forecasts, causal lever effects are estimated separately (with experiments or de-confounding), and decisions come from an optimization layer consuming both. Consider LLM/synthetic labeling (DashCLIP pattern) for supervision channels not contaminated by the control policy.
- Publish the embedding as a feature-store artifact with declared refresh tiers: entity key = (zone, time-bucket); a streaming pipeline (Flink-style) refreshes fast-moving state at minutes cadence for pricing/dispatch consumers, a batch pipeline snapshots history with point-in-time correctness for consumer training joins; guarantee training-serving consistency by sharing the exact featurization code path (Palette Transformer pattern).
- Institute version governance before the first consumer onboards: semantic-versioned releases; parallel serving of version N and N-1 during migrations; a learned backward-compatibility transform (BC-Aligner, code available at [snap-stanford/bc-emb](#)) so consumers pinned to old versions keep working across producer retrains; incremental training with warm starts and an explicit churn/stability metric between consecutive versions (Netflix practice) so the space drifts slowly by construction.
- Build a standing two-layer evaluation harness: (1) a frozen linear-probe suite — one probe per downstream domain (pricing, dispatch, ETA, repositioning, incentives, forecasting) on held-out labels — run on every candidate embedding version as a promotion gate and as ongoing drift monitoring; (2) shadow-mode and consumer A/B tests with at least two downstream teams before any version becomes default. No version promotes on intrinsic metrics alone.
- Plan Netflix-style tiered integration modes and document them: default mode is embeddings-as-features via the feature store; high-value consumers (pricing/surge) may fine-tune the encoder or get dedicated heads

served from the central model; forbid ad-hoc distillation of the embedding into consumer-side copies, which recreates the version-skew problem.

- Avoid pure zone-ID memorization for cold start and regime shift: ensure the encoder can produce embeddings for a brand-new zone or city from observable features alone (Airbnb cold-start and Netflix metadata patterns), and validate this with leave-one-city-out probes — this is also your best defense against COVID-style regime breaks that invalidated memorized historical patterns (DoorDash's documented retraining lesson).

Annotated bibliography (18 sources)

2018 Grbovic, M., & Cheng, H. (2018). Real-time Personalization using Embeddings for Search Ranking at Airbnb. KDD 2018, pp. 311-320.

dl.acm.org/doi/10.1145/3219819.3219885

Trains listing embeddings with skip-gram (word2vec) over in-session click sequences, modified for the marketplace: the booked listing is inserted as a global context token for every window in a booking session, and negatives are sampled within the same market so the geometry encodes within-market substitutability. Cold-start listings get embeddings by averaging attribute-similar neighbors; longer-horizon user-type and listing-type embeddings are trained in a shared space from booking sequences. Embeddings are recomputed daily offline and consumed in real time as similarity features inside a GBDT search-ranking model and for similar-listing recommendations.

Relevance to this project: The seminal proof that modifying a generic embedding objective with the business event you actually care about (bookings, not clicks) reorganizes the latent space by decision value — the exact motivation for this project's decision-aware auxiliary heads over pure reconstruction.

2019 Yi, X., Yang, J., Hong, L., Cheng, D. Z., Heldt, L., Kumthekar, A., Zhao, Z., Wei, L., & Chi, E. (2019). Sampling-Bias-Corrected Neural Modeling for Large Corpus Item Recommendations. RecSys 2019.

dl.acm.org/doi/10.1145/3298689.3346996

The canonical two-tower recipe: a query tower and an item content-feature tower trained with in-batch sampled softmax, where the key contribution is a streaming item-frequency estimator that corrects the sampling bias of in-batch negatives under skewed item distributions. Deployed as YouTube's neural retrieval system over tens of millions of videos, with item embeddings precomputed and queried via ANN.

Relevance to this project: Defines the industrial standard for producing co-embedded context/item vectors whose dot product is directly consumable downstream, and demonstrates that explicitly correcting bias in the training signal (here sampling bias; for this project, policy endogeneity) is what makes the embedding trustworthy.

2018 Ying, R., He, R., Chen, K., Eksombatchai, P., Hamilton, W. L., & Leskovec, J. (2018). Graph Convolutional Neural Networks for Web-Scale Recommender Systems (PinSage). KDD 2018.

arxiv.org/abs/1806.01973

Generates item embeddings with GraphSAGE-style localized graph convolutions where neighborhoods are defined by random-walk importance sampling instead of full-graph Laplacian operations, trained with a max-margin ranking loss on 7.5B examples and curriculum hard negatives. Batch MapReduce inference produces embeddings for 3 billion nodes, which are then served to multiple recommendation surfaces.

Relevance to this project: The template for offline batch generation of embeddings for billions of entities consumed by many downstream products; its importance-weighted neighborhood aggregation suggests how a zone encoder could pool information from neighboring/functionally-similar geo-zones.

2022 Baltescu, P., Chen, H., Pancha, N., Zhai, A., Leskovec, J., & Rosenberg, C. (2022). ItemSage: Learning Product Embeddings for Shopping Recommendations at Pinterest. KDD 2022.

arxiv.org/abs/2205.11728

A single transformer-based product embedding aggregating text and image features, trained with multi-task learning across multiple engagement types (clicks, saves, purchases) and multiple query surfaces (home feed, search, closeup) simultaneously. The multi-task objective improves sparse-label objectives with no penalty on data-rich ones, and one unified embedding replaces several per-surface embeddings, explicitly reducing infrastructure and maintenance cost.

Relevance to this project: The clearest industrial validation of this project's core bet: one centrally trained embedding with multiple decision-relevant task heads serves many consumers better than per-team representations, and sparse downstream tasks (e.g., incentive response) benefit most from shared training.

2024 Agarwal, P., Sk, M. I., Pancha, N., Hazra, K. S., Xu, J., & Rosenberg, C. (2024). OmniSearchSage: Multi-Task Multi-Entity Embeddings for Pinterest Search. WWW 2024 Companion.

arxiv.org/abs/2404.16260

Jointly learns a unified query embedding with pin and product embeddings in one space via multi-task training, enriching entity text with LLM-generated image captions, historical engagement, and user-curated board signals. Deployed across the whole Pinterest search stack (retrieval through ranking) at 300k requests/second; embeddings are deliberately co-trained to remain compatible with pre-existing embedding spaces so existing consumer systems keep working. Produced >8% relevance and >7% engagement gains.

Relevance to this project: Current SOTA for the 'one embedding system, many entity types, many consumers' pattern, and shows compatibility with existing embedding spaces can be engineered into the training objective — directly relevant to versioned, company-wide market-state embeddings.

2022 Hu, W., Bansal, R., Cao, K., Rao, N., Subbian, K., & Leskovec, J. (2022). Learning Backward Compatible Embeddings. KDD 2022.

arxiv.org/abs/2206.03040

Formalizes the producer/consumer embedding problem: an embedding team retrains for its intended task while consumer teams have models trained on old versions for unintended tasks. Their BC-Aligner learns a light transformation that maps the latest embedding version back into any historical version's space, so the producer can retrain frequently while consumer models continue working without retraining, with near-zero intended-task sacrifice across multiple version updates. Code released (snap-stanford/bc-emb); motivated by Pinterest-style deployments.

Relevance to this project: The single most on-point paper for the project's versioning requirement: it provides the concrete mechanism (learned backward alignment maps plus version-pinned consumption) for shipping embedding $v(N+1)$ without breaking pricing/dispatch/ETA models trained on vN .

2022 Hu, X., Cirit, O., Binaykiya, T., & Hora, R. (2022). DeepETA: An ETA Post-processing System at Scale. arXiv:2206.02127 (companion to Uber's DeepETA engineering blog, 2022).

arxiv.org/abs/2206.02127

Uber's production ETA system: a deep residual network refines routing-engine ETAs using an encoder-decoder with linear self-attention to meet millisecond latency, trained on $\sim 1.4B$ ETA requests. Its representation recipe is notable: every continuous and categorical feature is bucketized then embedded, and origin/destination locations are embedded as multi-resolution geohashes via feature hashing, which outperformed raw continuous features. Serves all 4-wheel ETA requests at Uber globally across rides and delivery.

Relevance to this project: A direct, deployed blueprint for encoding exactly the kind of geo/time/continuous marketplace inputs this project's encoder will ingest (quantize-then-embed, multi-resolution spatial embeddings), under hard serving-latency constraints.

2023 Uber Engineering (2023). Innovative Recommendation Applications Using Two Tower Embeddings at Uber. Uber Engineering Blog.

uber.com/blog/innovative-recommendation-applications-using-two-tower-e...

Uber's first production two-tower embedding system for Uber Eats: an eater tower encoding a time-sorted bag of past store interactions and a store tower, with layer sharing between towers to reduce cardinality problems. Embeddings are generated offline, registered in the Michelangelo/Palette feature store, and served both online and offline through a config-driven integration so multiple recommendation applications consume the same vectors.

Relevance to this project: Shows the exact operational pattern this project will run at a ride-hailing/delivery company: embeddings as first-class feature-store entities with unified online/offline access for multiple consuming applications.

2019 Nene, A. (2019). Michelangelo Palette: A Feature Engineering Platform at Uber. QCon/InfoQ presentation (with Uber's Palette Meta Store blog).

infoq.com/presentations/michelangelo-palette-uber/

Palette is Uber's centralized feature store organized by entity and feature group, with dual storage — Hive offline for training joins, Cassandra online for single-digit-millisecond serving — and automatic synchronization between them. Features are created via batch Hive/Spark, near-real-time Flink streaming, or bring-your-own microservice patterns, and a shared Transformer framework executes identical transformation logic offline and online to guarantee training-serving consistency.

Relevance to this project: The canonical architecture for distributing per-entity (here, per zone-time) features and embeddings to many consumer teams with point-in-time-correct training joins and training/serving consistency — the infrastructure substrate the market-state embedding must be published into.

2017 Laptev, N., Yosinski, J., Li, L. E., & Smyl, S. (2017). Time-series Extreme Event Forecasting with Neural Networks at Uber. ICML 2017 Time Series Workshop.

cs.columbia.edu/~lierranli/publications/TSW2017_paper.pdf

Pretrains an LSTM autoencoder over city-level trip time series to extract an embedding summarizing complex temporal dynamics, then concatenates that embedding with new inputs (including external variables like weather and holidays) into an LSTM forecaster for demand during extreme events. Deployed at Uber for demand forecasting around holidays, weather, and events, with uncertainty estimation.

Relevance to this project: Literally an early 'market-state embedding' at Uber — an unsupervised encoder feeding a demand forecaster — and its reliance on pure reconstruction pretraining is precisely the limitation (latent not organized by decision value) that this project's auxiliary linear-probe heads are designed to fix.

2018 Xu, Z., Li, Z., Guan, Q., Zhang, D., Li, Q., Nan, J., Liu, C., Bian, W., & Ye, J. (2018). Large-Scale Order Dispatch in On-Demand Ride-Hailing Platforms: A Learning and Planning Approach. KDD 2018.

dl.acm.org/doi/10.1145/3219819.3219824

DiDi's deployed dispatch system: offline temporal-difference learning over historical trip data estimates a spatiotemporal value function $V(\text{grid-cell}, \text{time})$ representing expected future driver earnings from each market state; online, a Kuhn-Munkres bipartite matcher uses these values as edge weights to assign orders to drivers at tens of millions of matches per day. A/B tests showed significant gains in driver income and answer rate.

Relevance to this project: Demonstrates the 'market state to long-run value' mapping that dispatch and repositioning consumers will apply to this project's embeddings, and that the consumer is often an optimization layer, not a model — the embedding must expose value-relevant state, not just reconstruct inputs.

2019 Tang, X., Qin, Z., Zhang, F., Wang, Z., Xu, Z., Ma, Y., Zhu, H., & Ye, J. (2019). A Deep Value-network Based Approach for Multi-Driver Order Dispatching. KDD 2019.

dl.acm.org/doi/10.1145/3292500.3330724

Extends DiDi's dispatch with a deep value network over spatiotemporal state, using cerebellar (CMAC-style) hierarchical coarse coding to embed geographic coordinates at multiple overlapping resolutions, plus Lipschitz regularization for robust, smooth value estimates and transfer techniques to reuse representations across cities. Modeled as a semi-MDP; large-scale online A/B tests on DiDi's platform showed significant improvements in total driver income and user experience metrics.

Relevance to this project: Shows that hierarchical multi-resolution spatial encoding and smoothness regularization are what make learned market-state representations robust and transferable across cities — key design choices for a multi-city zone-time encoder.

2018 Yao, H., Wu, F., Ke, J., Tang, X., Jia, Y., Lu, S., Gong, P., Ye, J., & Li, Z. (2018). Deep Multi-View Spatial-Temporal Network for Taxi Demand Prediction. AAAI 2018.

ojs.aaai.org/index.php/AAAI/article/view/11836

DMVST-Net (a DiDi collaboration, evaluated on large-scale DiDi data) predicts per-region taxi demand with three fused views: a local CNN over each region's spatial neighborhood, an LSTM over its temporal history, and a semantic view that embeds regions via a graph embedding of functional similarity in their demand patterns, so distant but behaviorally similar regions share information.

Relevance to this project: Introduces the idea that geo-zones should be embedded by functional similarity of their market dynamics, not just spatial adjacency — a useful auxiliary structure or pretraining signal for the project's zone encoder.

2021 DoorDash Engineering (2021). Managing Supply and Demand Balance Through Machine Learning. DoorDash Engineering Blog.

careersatdoordash.com/blog/managing-supply-and-demand-balance-through-...

DoorDash forecasts marketplace-health metrics per region-time unit, defines undersupply metrics from those forecasts, and feeds them into mobilization levers (Dasher incentives and messaging) chosen by an optimization layer; the broader program (see also their causal-ML and joint-optimization posts) explicitly separates forecasting models from causal estimates of lever effects because logged outcomes are shaped by past interventions.

Relevance to this project: The closest published analog of this project's downstream stack — region-time state representations feeding forecasting plus control levers — and an explicit industrial acknowledgment that policy-endogenous logs require causal treatment, mirroring the surge price/radius endogeneity concern.

2025 Gurjar, O., Liu, K. S., Kolli, P., Kumar, U., & Raturkar, M. (2025). DashCLIP: Leveraging multimodal models for generating semantic embeddings for DoorDash. arXiv:2504.07110.

arxiv.org/abs/2504.07110

A CLIP-style joint training framework aligning uni-modal and multi-modal product encoders via contrastive learning on product image-text pairs, plus a query encoder trained on an LLM-curated relevance dataset — deliberately eliminating dependence on engagement history for supervision. The resulting embeddings are published to multiple downstream teams (product categorization, relevance prediction, personalized ads), with deployed CTR/conversion lifts, outperforming off-the-shelf CLIP/BLIP/FLAVA.

Relevance to this project: A 2025 delivery-marketplace example of a central team producing embeddings for many consumers, and its LLM-curated supervision is a practical tactic to break dependence on policy-endogenous engagement logs — transferable to labeling market regimes.

2021 Kakade, V. (2021). ML Feature Serving Infrastructure at Lyft. Lyft Engineering Blog.

eng.lyft.com/ml-feature-serving-infrastructure-at-lyft-d30bf2d3c32a

Lyft's Feature Service defines features as SQL keyed by entity ID, computed via batch (Spark/Airflow) and streaming (Flink/Kafka) pipelines, and served uniformly for training and online inference from DynamoDB with Redis caching (embeddings later in OpenSearch), at millions of requests per minute with single-digit-millisecond latency and 99.99%+ availability across thousands of features and pricing, matching, and fraud models.

Relevance to this project: Ride-hailing-specific evidence of the serving fabric a market-state embedding must plug into: entity-keyed, batch-plus-streaming refresh, uniform train/serve access — with pricing and dispatch as consuming teams, exactly this project's consumers.

2025 Netflix Technology Blog (2025). Foundation Model for Personalized Recommendation; and Integrating Netflix's Foundation Model into Personalization Applications. Netflix Tech Blog.

netflixtechblog.medium.com/integrating-netflixs-foundation-model-into-...

Netflix replaced many specialized personalization models with one LLM-inspired autoregressive transformer trained on hundreds of billions of tokenized user interactions, using sparse attention, metadata-based cold-start handling, and incremental training with warm starts to control churn between retrains. Downstream integration happens in three explicit modes: embeddings published as features for consumer models, fine-tuning the foundation model for specific applications, and direct use of its predictions.

Relevance to this project: The strongest recent template for a centrally trained representation consumed company-wide: its three integration modes, incremental-training stability practices, and cold-start handling are directly transplantable to a market-state foundation encoder.

2026 Abdool, M., Banerjee, S., Paul, M., Kim, D., Liu, X., Xu, B., Yu, T., Gao, H., Ouyang, K., Gao, H., He, L., Moyerman, S., & Katariya, S. (2026). Applying Embedding-Based Retrieval to Airbnb Search. arXiv:2601.06873.

arxiv.org/abs/2601.06873

Airbnb's production embedding-based retrieval: two-tower embeddings trained on booking outcomes with contrastive learning, engineered around two-sided marketplace constraints — highly dynamic inventory (availability changes constantly), a multi-stage ranking funnel the retrieval layer must stay consistent with, and multiple product surfaces (regular search, flexible dates, promotional campaigns). Launched to production with statistically significant booking-conversion gains in A/B tests.

Relevance to this project: The newest (2026) evidence of how two-sided marketplaces operationalize embeddings when the underlying inventory/state is fast-moving — its handling of dynamic state and multi-stage consumer pipelines parallels the fast-refresh requirements of a market-state embedding.

10 Spatiotemporal, Urban & Graph Embeddings

State of the art

Spatiotemporal and urban/graph embedding research has evolved along three converging tracks that map directly onto a per-hex-per-time market-state object. (1) Spatiotemporal GNN forecasting began with DCRNN, STGCN, and Graph WaveNet (2018-19), which established that a city is a graph and that spatial dependency should be baked in via graph convolution/diffusion over an adjacency structure — with Graph WaveNet's self-adaptive adjacency showing the decisive lesson that the *learned* dependency graph beats the hand-specified one. DiDi's ST-MGCN carried this into deployed ride-hailing demand forecasting with multiple graphs (spatial neighborhood, functional similarity, transport connectivity). A counter-current then emerged: STID (CIKM 2022) and STAEformer (CIKM 2023) showed that most of the value of these architectures is captured by learned per-node and per-time-slot *identity embeddings* attached to simple MLPs/vanilla Transformers, while BigST (VLDB 2024) made STGNNs linear-complexity and scalable to 100k nodes — i.e., citywide hex grids. (2) Urban region embedding grew from random-walk/skip-gram methods on mobility flow graphs (HDGE 2017, ZE-Mob 2018 — the latter using PMI of origin-destination co-occurrence in taxi trips) through multi-graph attention fusion (MGFN 2022) to task-adaptation-aware methods: HREP (AAAI 2023) introduced per-task prompt embeddings on top of a shared heterogeneous-relation embedding, and the 2025-26 generation (GURPP, FlexiReg, UrbanVerse) does graph-based pre-training plus prompting for cross-task and cross-city transfer; hex2vec/srai provides the production-grade H3-hexagon static-feature embedding tooling. (3) Trajectory embeddings (t2vec 2018, TrajCL 2023) contributed the seq2seq-denoising and contrastive-augmentation recipes for encoding movement, and spatiotemporal self-supervision (ST-SSL, AAAI 2023) showed auxiliary contrastive tasks that explicitly model spatial and temporal heterogeneity improve robustness on skewed distributions — exactly the surge-tail regime. Since 2024 the frontier is urban foundation models: UniST (masked ST modeling + memory-based prompts), UrbanGPT (ST encoder + LLM instruction tuning), OpenCity (graph-Transformer pre-trained across cities, zero-shot with scaling laws), and 2026's UrbanFM (billion-scale corpus, minimal-inductive-bias attention, largest ST benchmark to date). Open problems: policy-endogeneity of logged urban data is barely addressed (models treat demand as exogenous, not action-conditioned), evaluation is still per-task rather than probe-based, and dynamic (time-varying) region embeddings remain much less mature than static ones.

Key principles

- Spatial dependency should enter the encoder as an explicit graph operator (diffusion/message passing over the hex adjacency), but the adjacency should be partly LEARNED: Graph WaveNet, STAEformer, and

OpenCity all show self-adaptive/attention-derived dependency matrices beat purely geographic ones because true influence (e.g., surge spillover, substitutable pickup zones) is not purely distance-based. Best practice is hybrid: fixed hex-neighbor graph + learned adaptive graph + optionally functional-similarity and connectivity graphs (ST-MGCN).

- Learned identity embeddings are load-bearing: STID and STAEformer demonstrate that per-node (hex) and per-time-slot (time-of-day, day-of-week) embedding tables, jointly trained, capture most spatial-temporal distinguishability — any market-state encoder should include them, and they double as a cheap versioned 'static hex embedding' product.
- Reconstruction alone under-organizes the latent space; auxiliary self-supervised tasks that model HETEROGENEITY (spatial: different hexes have skewed distributions; temporal: peak vs off-peak regimes) reshape representations toward decision-relevant structure (ST-SSL). Contrastive objectives with domain-appropriate augmentations (TrajCL) and masked spatiotemporal modeling (UniST) are the two dominant recipes.
- Multi-view / multi-graph fusion beats any single data source: region embeddings built from mobility flows + POI/land-use + neighbor structure (HDGE, MGFN, HREP, hex2vec) transfer to more downstream tasks than single-view embeddings; attention-based fusion with per-view embeddings preserved until late fusion is the standard architecture.
- Task adaptation should be cheap and explicit: the field converged on 'frozen shared embedding + small per-task prompt/adaptor' (HREP, GURPP, FlexiReg, UniST) rather than per-task fine-tuning — this is precisely the mechanism that keeps a company-wide embedding backward compatible while letting pricing vs dispatch vs ETA each get a tailored view.
- Downstream evaluation of region embeddings is standardized as LINEAR probes / shallow regressors on held-out urban indicators (crime, check-ins, land use, flow prediction) — validating the project's plan to constrain auxiliary heads to linear probes so decision-relevant information is linearly accessible.
- Long-history context can be precomputed: BigST separates an offline long-sequence temporal encoder (thousands of past steps compressed into low-dim representations) from a cheap online spatial propagation step — a serving pattern that matches an embedding service with strict latency budgets.
- Cross-city / zero-shot generalization comes from scale plus weak inductive bias: OpenCity and UrbanFM show pre-training across many cities with normalized inputs yields transferable spatiotemporal representations with observable scaling laws, while UrbanVerse shows region-conditioned + task-conditioned generation transfers across both cities and tasks.
- Trajectory-level encoders (t2vec, TrajCL) show robustness must be trained in, not hoped for: denoising/augmentation-invariance objectives (downsampling, noise, distortion) produce embeddings stable under data-quality variation — the analogue for market state is augmentation over missing zones, sensor dropouts, and event shocks.

Recommendations for the market-state embedding project

- Architect the encoder as: (a) learned hex-ID + time-of-day + day-of-week embedding tables (STID/STAEformer lesson), (b) a temporal module over each hex's recent history, and (c) a spatial mixing layer over a HYBRID graph — fixed H3 neighbor adjacency plus a Graph WaveNet-style learned adaptive adjacency plus optional functional-similarity and road-connectivity graphs (ST-MGCN). The learned adjacency is where non-geographic surge spillovers (driver reachability, substitutable zones) get captured; publish it as an interpretable artifact for the pricing team.
- Ablate ruthlessly against the STID baseline (hex-ID + time-ID embeddings + MLP): if the GNN does not beat it on downstream probes, the spatial machinery is not earning its complexity. This is a one-week experiment that de-risks the whole architecture choice.

- Adopt ST-SSL-style heterogeneity-aware auxiliary tasks alongside the planned linear-probe prediction heads: a spatial contrastive task (cluster hexes into soft regimes) and a temporal regime-discrimination task (peak/off-peak/surge). These specifically prevent the failure mode you identified — reconstruction averaging away tail regimes that matter most for surge decisions.
- Use masked spatiotemporal modeling (UniST's masking menu: random, block/'tube', temporal, and spatial-region masking) as the core pre-training objective rather than plain autoencoding; it doubles as robustness training for missing zones and telemetry gaps, and pairs naturally with your linear-probe auxiliary heads.
- Handle policy-endogeneity explicitly, because this literature does not: log surge price and surge radius as INPUTS to the encoder (action-conditioned representation) or train with them masked and probed, so the embedding does not silently entangle 'high demand' with 'we surged here'. None of the surveyed ST models condition on control levers — this is your novelty and your biggest risk; borrow action-conditioning from decision-focused RL representation work rather than expecting it from ST forecasting papers.
- Serve with the BigST split: precompute each hex's long-horizon temporal summary offline (daily/hourly batch), run only linear-complexity spatial propagation plus latest-bucket fusion online. This matches an embedding-service latency budget at 10k-100k hex scale.
- Distribute via the freeze-and-prompt pattern (HREP/GURPP/FlexiReg/UniST): version-freeze the backbone embedding, and give each consumer team a small learned prompt/adaptor head trained against the frozen embedding. Backward compatibility then reduces to keeping the frozen backbone stable, and per-task prompts absorb drift — re-train prompts, not the backbone, between minor versions.
- Build the evaluation harness as a battery of frozen-embedding linear probes over ALL downstream tasks (demand/supply forecasting at multiple horizons, ETA residuals, completion rate, incentive response, plus static probes like POI/land-use recovery), in the spirit of the region-embedding literature's standard protocol and UrbanFM's EvalST; gate embedding version releases on probe-regression tests rather than a single reconstruction metric.
- Concatenate a static channel from hex2vec/srai (OSM tags on your exact H3 grid) with the learned dynamic state; it is nearly free, adds urban-function priors the trip data cannot see for new/sparse zones, and srai accelerates prototyping.
- Pre-train across all your cities jointly with per-city input normalization (OpenCity/UrbanFM recipe) and validate zero-shot transfer to a held-out city — this both improves data efficiency in small markets and gives you a launch-day embedding for new cities; consider warm-starting from OpenCity's open weights for the traffic-like channels.
- If driver/rider trajectories are available, add a pooled trajectory-embedding channel per hex-time bucket using the TrajCL contrastive recipe with augmentations chosen to encode invariances downstream models should have (GPS noise, sampling-rate variation) — supply behavior (cruising vs repositioning) is market state that OD counts miss.

Annotated bibliography (22 sources)

2018 Li, Y., Yu, R., Shahabi, C., & Liu, Y. (2018). **Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting. ICLR 2018 (arXiv:1707.01926).**

arxiv.org/abs/1707.01926

Models traffic as a diffusion process on a directed sensor graph: spatial dependency is captured by bidirectional random-walk diffusion convolutions, temporal dependency by GRUs whose matrix multiplications are replaced with diffusion convolutions, trained seq2seq with scheduled sampling. The encoder hidden state is a per-node spatiotemporal representation. Follow-up work (graph-partitioned DCRNN) deployed it to the entire California highway network of 11,160 sensors.

Relevance to this project: The seminal template for the project's encoder: per-zone hidden states that mix information from graph neighbors via diffusion — the canonical way to make surge-radius-style neighbor spillovers first-class citizens of the embedding.

2018 Yu, B., Yin, H., & Zhu, Z. (2018). **Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. IJCAI 2018 (arXiv:1709.04875).**

arxiv.org/abs/1709.04875

Fully convolutional 'sandwich' blocks: Chebyshev/1st-order graph convolution for space between gated temporal convolutions (GLU) for time — no recurrence, so training is fast and parallel. Each block outputs per-node feature maps that serve as spatiotemporal embeddings feeding an output regression layer.

Relevance to this project: Establishes the cheap, parallelizable conv-only alternative for encoding hex-time tensors; its temporal-conv + graph-conv block remains the standard efficient backbone for latency-sensitive production encoders.

2019 Wu, Z., Pan, S., Long, G., Jiang, J., & Zhang, C. (2019). **Graph WaveNet for Deep Spatial-Temporal Graph Modeling. IJCAI 2019 (arXiv:1906.00121).**

arxiv.org/abs/1906.00121

Combines dilated causal temporal convolutions with graph convolutions over a self-adaptive adjacency matrix computed as $\text{softmax}(\text{ReLU}(E_1 \cdot E_2^T))$ from two learned node-embedding tables, trained end-to-end alongside (or instead of) the given graph. The learned node embeddings capture hidden inter-node dependencies missing from the physical graph.

Relevance to this project: Directly addresses the project's spillover problem: the hexes that influence each other under surge are not only geographic neighbors, and Graph WaveNet shows how to learn that hidden dependency structure — its node-embedding tables are themselves reusable static hex embeddings.

2019 Geng, X., Li, Y., Wang, L., Zhang, L., Yang, Q., Ye, J., & Liu, Y. (2019). **Spatiotemporal Multi-Graph Convolution Network for Ride-Hailing Demand Forecasting**. **AAAI 2019**.

ojs.aaai.org/index.php/AAAI/article/view/4247

Encodes region correlations as multiple graphs — spatial neighborhood, functional (POI) similarity, and road connectivity — applies ChebNet convolution on each, and aggregates with a contextual gated RNN that reweights each region's history by global context. Trained on DiDi ride-hailing GPS data from Xi'an and Chengdu, beating baselines by 10%+.

Relevance to this project: The closest deployed-industry ancestor of the project (DiDi, ride-hailing demand on city zones); its multi-graph design is the proven way to encode 'neighbor effects' that are functional and connective, not merely adjacent — important because surge spillovers follow driver-reachability, not straight-line distance.

2022 Shao, Z., Zhang, Z., Wang, F., Wei, W., & Xu, Y. (2022). **Spatial-Temporal Identity: A Simple yet Effective Baseline for Multivariate Time Series Forecasting**. **CIKM 2022 (arXiv:2208.05233)**.

arxiv.org/abs/2208.05233

Diagnoses the core bottleneck of ST forecasting as sample indistinguishability and solves it by attaching three learned embedding tables — spatial identity (per node), time-of-day identity, and day-of-week identity — to a plain MLP, matching or beating STGNNs at a fraction of the cost. No graph operations at all.

Relevance to this project: A crucial ablation lesson for the encoder design: a large share of 'market state' is just (which hex, which time slot) identity, so the project should include learned hex-ID and time-ID embeddings and measure how much the GNN adds beyond them before paying for graph complexity.

2023 Liu, H., Dong, Z., Jiang, R., Deng, J., Deng, J., Chen, Q., & Song, X. (2023). **STAEformer: Spatio-Temporal Adaptive Embedding Makes Vanilla Transformer SOTA for Traffic Forecasting**. **CIKM 2023 (arXiv:2308.10425)**.

arxiv.org/abs/2308.10425

Feeds a vanilla Transformer (attention along time then along space) with a concatenation of feature projection, periodicity embeddings, and a learned spatio-temporal adaptive embedding tensor that is indexed by node and input time step; this adaptive embedding alone lifts a vanilla Transformer to SOTA on METR-LA/PEMS benchmarks.

Relevance to this project: Confirms at SOTA level that adaptive embeddings, not exotic architecture, carry decision-relevant structure — supporting a design where the market-state encoder invests capacity in rich learned hex-time embeddings with spatial attention providing the neighbor mixing.

2024 Han, J., Zhang, W., Liu, H., Tao, T., Tan, N., & Xiong, H. (2024). **BigST: Linear Complexity Spatio-Temporal Graph Neural Network for Traffic Forecasting on Large-Scale Road Networks**. **PVLDB 17(5):1081-1090**.

vldb.org/pvldb/vol17/p1081-han.pdf

Two-part design for scale: an offline scalable long-sequence feature extractor compresses thousands of past time steps per node into low-dimensional representations (precomputed), and an online linearized global spatial convolution (kernelized attention) distills a time-varying graph and propagates messages in $O(N)$. Scales to road networks with ~100k nodes.

Relevance to this project: The serving blueprint for a citywide hex embedding service: precompute long-horizon temporal context per hex offline, run cheap linear-complexity spatial propagation online — and its authors sit in the DiDi ecosystem (co-authors from Didi Chuxing), evidencing industrial fit.

2023 Ji, J., Wang, J., Huang, C., Wu, J., Xu, B., Wu, Z., Zhang, J., & Zheng, Y. (2023). **Spatio-Temporal Self-Supervised Learning for Traffic Flow Prediction**. AAAI 2023 (arXiv:2212.04475). arxiv.org/abs/2212.04475

Adds two auxiliary self-supervised tasks to an ST encoder: adaptive augmentation of the traffic-flow graph at attribute and structure level, then (i) a spatial-heterogeneity-aware contrastive task that clusters regions into soft categories and (ii) a temporal-heterogeneity-aware task distinguishing time regimes, jointly optimized with the forecasting loss so representations reflect skewed spatial and temporal distributions.

Relevance to this project: The direct precedent for the project's auxiliary-head plan: SSL tasks targeting heterogeneity make embeddings robust exactly where a marketplace needs them (peak/surge regimes, atypical zones) instead of letting reconstruction average them away.

2017 Wang, H., & Li, Z. (2017). **Region Representation Learning via Mobility Flow**. CIKM 2017. dl.acm.org/doi/10.1145/3132847.3133006

HDGE builds a flow graph from taxi trip counts between zones (time-sliced) plus a spatial-proximity graph, runs joint random walks across both, and learns zone vectors with word2vec-style skip-gram so that zones with similar mobility relations and locations embed nearby.

Relevance to this project: The founding paper of mobility-based region embedding: proves the company's own trip matrix is the richest signal for a hex embedding, and that spatial-graph regularization should be co-trained rather than post-hoc.

2018 Yao, Z., Fu, Y., Liu, B., Hu, W., & Xiong, H. (2018). **Representing Urban Functions through Zone Embedding with Human Mobility Patterns**. IJCAI 2018. ijcai.org/proceedings/2018/545

ZE-Mob treats origin and destination zones co-occurring in taxi trips as context pairs and learns zone embeddings whose inner products approximate shifted PMI of OD co-occurrence, weighting by mobility direction, departure/arrival time, destination attraction, and travel distance. Evaluated on NYC taxi data for land-use clustering and popularity prediction.

Relevance to this project: Shows a second, complementary factorization of the same trip data (co-occurrence semantics rather than flow-graph walks); useful as the cheapest static-hex-function baseline the project should beat with its learned encoder.

2022 Wu, S., Yan, X., Fan, X., Pan, S., Zhu, S., Zheng, C., Cheng, M., & Wang, C. (2022). **Multi-Graph Fusion Networks for Urban Region Embedding**. IJCAI 2022 (arXiv:2201.09760). ijcai.org/proceedings/2022/321

MGFN builds one mobility graph per hour (24 graphs), clusters them into a few 'mobility patterns' via graph-structure distance, fuses each cluster with a mobility-graph fusion module, and applies multi-level cross-attention (intra- and inter-pattern) to learn region embeddings used for crime, check-in, and land-use prediction.

Relevance to this project: Demonstrates how to compress time-varying mobility into a small set of regime-conditioned region embeddings — a template for the project's time-bucketed market states where weekday-peak, weekend-night, etc. form distinct spatial regimes.

2023 Zhou, S., He, D., Chen, L., Shang, S., & Han, P. (2023). Heterogeneous Region Embedding with Prompt Learning. AAI 2023.

ojs.aaai.org/index.php/AAAI/article/view/25625

HREP builds a heterogeneous region graph over mobility, POI, and geographic-neighbor relations, learns shared region embeddings with relation-aware aggregation, then concatenates small learnable prompt embeddings per downstream task (crime, check-in, land use) — the prompt, not the backbone, adapts the representation to the task.

Relevance to this project: The prompt-per-consumer pattern the project needs for company-wide distribution: pricing, dispatch, and ETA teams each get a tiny learned prompt against a frozen, versioned market-state embedding, preserving backward compatibility.

2021 Woźniak, S., & Szymański, P. (2021). hex2vec: Context-Aware Embedding H3 Hexagons with OpenStreetMap Tags. 4th ACM SIGSPATIAL GeoAI Workshop (GEOAI '21).

dl.acm.org/doi/10.1145/3486635.3491076

Divides cities into Uber H3 hexagons, aggregates curated OpenStreetMap tag counts (land use, buildings, functions) per hexagon, and trains a skip-gram-with-negative-sampling model where a hexagon's spatial context (neighboring hexes) predicts its tag vector, yielding static urban-function embeddings. Maintained in the open-source srai library (kraina-ai/srai) alongside other region embedders.

Relevance to this project: The exact H3-hexagon geometry the marketplace runs on, with production-grade tooling: hex2vec/srai embeddings are a ready-made static-feature channel to concatenate with the learned dynamic market state, and srai is useful scaffolding for experimentation.

2025 Jin, J., Song, Y., Kan, D., Zhu, H., Sun, X., Li, Z., Sun, X., & Zhang, J. (2025). Urban Region Pre-training and Prompting: A Graph-based Approach. KDD 2025 (arXiv:2408.05920).

arxiv.org/abs/2408.05920

GURPP constructs an urban region graph down to fine-grained spatial entities (e.g., store brands), pre-trains a subgraph-centric model to capture heterogeneous, transferable entity-interaction patterns, then serves downstream tasks via graph prompts that dynamically assemble relevant entities/relations per region into task-specific representations.

Relevance to this project: Current SOTA of the pre-train-then-prompt paradigm on region graphs; its subgraph-centric pre-training suggests encoding each hex's market state from its local subgraph (neighbors + connected zones), which is naturally aligned with surge-radius locality.

2026 Sun, F., Tanin, E., Karunasekera, S., Li, Z., Salim, F. D., & Qi, J. (2026). UrbanVerse: Learning Urban Region Representation Across Cities and Tasks. arXiv:2602.15750.

arxiv.org/abs/2602.15750

Represents regions as graph nodes and runs random walks emphasizing local neighborhood structure (rather than whole-city patterns) to generate region sequences; a cross-task module (HCondDiffCT) injects region-conditioned priors and task-conditioned semantics into a diffusion process serving multiple downstream predictors, achieving up to 35.9% gains across six tasks in cross-city settings.

Relevance to this project: The 2026 frontier of region2vec-style embeddings: shows that locality-biased structural encoding plus task-conditioned decoding transfers across cities — relevant for launching the embedding in new markets without per-city retraining.

2018 Li, X., Zhao, K., Cong, G., Jensen, C. S., & Wei, W. (2018). Deep Representation Learning for Trajectory Similarity Computation. ICDE 2018.

dblp.org/rec/conf/icde/LiZCJW18.html

t2vec learns fixed-dimensional trajectory embeddings with a GRU seq2seq model trained to reconstruct a high-quality trajectory from a downsampled/noisy version, using a spatial-proximity-aware loss over discretized cells, making embeddings robust to non-uniform, low sampling rates and noise.

Relevance to this project: The denoising-autoencoder recipe for movement data: driver/rider trajectory embeddings built this way can be pooled per hex-time bucket as an additional supply-behavior channel, robust to GPS quality variation.

2023 Chang, Y., Qi, J., Liang, Y., & Tanin, E. (2023). Contrastive Trajectory Similarity Learning with Dual-Feature Attention. ICDE 2023 (arXiv:2210.05155).

arxiv.org/abs/2210.05155

TrajCL is a self-supervised contrastive framework: four trajectory augmentations (e.g., point shifting, masking, truncation, simplification) generate positive pairs, and a dual-feature self-attention encoder jointly models spatial locations and structural patterns; embeddings can be fine-tuned to approximate classical similarity measures with strong accuracy/speed gains.

Relevance to this project: State-of-the-art contrastive recipe for movement embeddings and a catalogue of domain-appropriate augmentations — the analogue for market state is designing augmentations (zone dropout, temporal jitter, demand-scale perturbation) whose invariances match what downstream decisions should ignore.

2024 Yuan, Y., Ding, J., Feng, J., Jin, D., & Li, Y. (2024). UniST: A Prompt-Empowered Universal Model for Urban Spatio-Temporal Prediction. KDD 2024 (arXiv:2402.11838).

arxiv.org/abs/2402.11838

A Transformer pre-trained on grid-based ST data from many cities/domains with diverse masking strategies (random, tube, block, temporal) for generative context modeling, then equipped with memory-based spatio-temporal knowledge pools from which prompts are retrieved and injected at inference, aligning shared knowledge to each scenario; strong few-shot/zero-shot gains (~10% over best per-dataset baselines).

Relevance to this project: Masked ST pre-training plus retrieval-style prompting is a direct architectural candidate for the market-state encoder, and its elaborate masking menu is a ready-made curriculum for making hex-time embeddings robust to missing data.

2024 Li, Z., Xia, L., Tang, J., Xu, Y., Shi, L., Xia, L., Yin, D., & Huang, C. (2024). UrbanGPT: Spatio-Temporal Large Language Models. KDD 2024 (arXiv:2403.00813).

arxiv.org/abs/2403.00813

Couples a dedicated spatio-temporal dependency encoder (multi-scale temporal convolutions) with an LLM via instruction tuning: ST embeddings are projected into the LLM token space alongside text instructions describing region, time, and POIs; trained on NYC taxi/bike/crime data, it performs zero-shot prediction in unseen regions by exploiting the LLM's semantic reasoning.

Relevance to this project: Shows how a learned market-state embedding can be made LLM-consumable — a forward-looking consumer class (ops copilots, incident triage) for the same versioned embedding, beyond the numeric downstream models.

2024 Li, Z., Xia, L., Shi, L., Xu, Y., Yin, D., & Huang, C. (2024). **OpenCity: Open Spatio-Temporal Foundation Models for Traffic Prediction**. **arXiv:2408.10269 (ACM TIST 2025)**.

arxiv.org/abs/2408.10269

Pre-trains a Transformer integrated with graph neural networks on large heterogeneous traffic corpora from multiple cities, normalizing diverse data characteristics so the model learns transferable spatio-temporal dependency patterns; exhibits strong zero-shot prediction on unseen regions/cities and promising scaling laws. Code and weights are open (HKUDS/OpenCity).

Relevance to this project: Evidence that one pre-trained ST backbone can serve many cities zero-shot — the multi-city pre-training strategy the marketplace should adopt so the market-state encoder generalizes to newly launched cities, plus open weights to bootstrap from.

2026 Chen, W., Wu, Y., Chen, J., Zhou, X., & Liang, Y. (2026). **UrbanFM: Scaling Urban Spatio-Temporal Foundation Models**. **arXiv:2602.20677**.

arxiv.org/abs/2602.20677

Scales urban ST foundation models along three axes: WorldST, a billion-scale standardized corpus from 100+ global cities; MiniST units that discretize continuous ST fields into learnable computational units unifying grid- and sensor-based observations; and a minimalist self-attention architecture with deliberately limited inductive biases, letting scale learn the dependencies. Establishes EvalST, the largest urban ST benchmark, and shows strong zero-shot generalization to unseen cities and tasks.

Relevance to this project: The 2026 state of the art in the foundation-model direction: its unit-discretization idea maps cleanly onto hex-time buckets, and EvalST-style broad benchmarking is a model for the project's embedding-evaluation harness.

2024 Zhang, W., Han, J., Xu, Z., Ni, H., Lyu, T., Liu, H., & Xiong, H. (2024, rev. 2026). **Towards Urban General Intelligence: A Review and Outlook of Urban Foundation Models**. **arXiv:2402.01749**.

arxiv.org/abs/2402.01749

Comprehensive survey (maintained through 2026) that taxonomizes urban foundation models by data modality (language, trajectory/ST series, vision, multimodal) and adaptation mechanism (pre-training objectives, prompting, fine-tuning), and proposes a framework for versatile UFM; companion repository (usail-hkust/Awesome-Urban-Foundation-Models) tracks the field.

Relevance to this project: The map of the whole design space the project is entering; its pre-training-objective and adaptation-technique taxonomy is a checklist for choosing the market-state encoder's SSL objectives and per-team adaptation interfaces.

11 Evaluating Embedding Quality & Operating Embeddings Over Time

State of the art

This area answers two operational questions: (1) how do you know an embedding is good before and after downstream teams consume it, and (2) how do you keep shipping new encoder versions without breaking every consumer. Evaluation evolved from single-task linear probes (the de facto SSL standard) to critiques showing probe accuracy is protocol-sensitive and conflates "information exists" with "information is accessible" — driving information-theoretic reframings (MDL probes, readout model switching) that measure description length and data-efficiency rather than raw accuracy. In parallel, a family of label-free metrics emerged that predict

downstream performance from embedding geometry alone: alignment/uniformity (2020), effective rank / RankMe (2023), eigenspectrum decay alpha-ReQ (2022), LiDAR (2024), and intrinsic dimension (2019); 2025 work (including a Dec-2025 persistent-homology metric) shows each fails in some regimes — especially class imbalance and domain shift — so the current SOTA position is a dashboard of complementary metrics validated against a fixed multi-task benchmark suite, the pattern institutionalized by VTAB and MTEB and scaled by MMTEB (ICLR 2025) with correlation-based task downsampling. The operations track began with Backward-Compatible Training (Shen et al., CVPR 2020), which made backfill-free encoder upgrades possible, and progressed through forward-compatible training with side-information (Apple, CVPR 2022), regression/negative-flip minimization (Amazon, CVPR 2021), theoretically grounded stationary representations via fixed simplex classifiers (CVPR 2024), and 2025 work from Snap/Yale using hyperbolic entailment cones and from the continual-learning community using query-drift compensation to keep old indexes usable without re-embedding. Drift monitoring rests on Failing Loudly (NeurIPS 2019) — two-sample tests on dimensionality-reduced representations plus domain classifiers — now productized in Evidently/Arize/WhyLabs. Matryoshka Representation Learning (NeurIPS 2022), adopted by OpenAI and Google embedding APIs, resolved the dimensionality trade-off by nesting granularities in one vector. Open problems: no label-free metric is universally reliable; the compatibility-plasticity trade-off (alignment to old embeddings caps improvement); and principled evaluation under policy-endogenous distribution shift remains essentially unstudied — directly relevant when surge control both consumes and perturbs the embedding.

Key principles

- Probe accuracy conflates existence with accessibility of information: MDL-based measures (description length, prequential coding, readout model switching) quantify how cheaply information can be extracted — which is what downstream consumers actually pay for — and are far better at separating real signal from probe memorization than raw linear-probe accuracy.
- Label-free geometric/spectral metrics (alignment-uniformity, effective rank/RankMe, eigenspectrum decay alpha, LiDAR, intrinsic dimension, topological persistence) are cheap continuous training-time monitors that catch representation collapse before any downstream model exists — but every one of them fails in some regime (imbalance, domain shift, non-vision data), so they are guardrails and selection heuristics, never training objectives or sole release gates.
- Representation quality is only defined relative to a fixed, versioned, diverse multi-task benchmark (VTAB/MTEB pattern): frozen encoder, constrained adaptation budget, limited-label splits, and correlation-based task downsampling to keep evaluation cheap enough to run on every candidate — the benchmark suite is the embedding's contract with the organization.
- Report data-efficiency curves, not point accuracies: how well a probe does with 1% and 10% of labels (equivalently, area under the online-coding learning curve) is the property that transfers to new downstream use-cases and small-sample geo-zones.
- Compatibility is a trainable property, achieved three ways with different costs: train it in (BCT influence loss, fixed stationary/simplex target geometry), transform it in (old-to-new mappings with side-information, FCT), or compensate at serving time (query-drift projection). The success metric is cross-version performance (old heads on new embeddings) and per-consumer negative-flip rate — not new-model average accuracy.
- Regression is a first-class release metric: a version that improves on average can still silently break individual consumers; negative flips must be measured per downstream task and gated per consumer.
- Capacity should be nested, not fixed: Matryoshka-style training makes one embedding serve consumers with wildly different latency/storage budgets via prefix truncation, and makes the dimension count part of a graceful versioned contract rather than a breaking change.

- Drift monitoring belongs on the embedding itself: two-sample tests on (reduced) embedding dimensions, domain classifiers, and centroid distances detect distribution shift without labels — but statistical drift is not harm; alerts must be paired with downstream proxy metrics, and in systems where actions feed back into data (surge), policy-induced drift must be distinguished from environment-induced drift.
- No single metric is universal (metric correlations reverse across datasets): operate a validated panel — spectral + probe-based + drift — and re-validate the panel against your own downstream tasks whenever data distribution or policy changes.

Recommendations for the market-state embedding project

- Build a frozen 'Market-State Eval Suite' (VTAB/MTEB pattern) before shipping v1: 8-12 fixed downstream tasks with locked splits (surge-outcome regression, demand and supply forecasting at multiple horizons, ETA error, dispatch match quality, incentive uplift, anomaly/event detection), evaluated with frozen-encoder linear probes; score every encoder candidate on the full suite and prune redundant tasks via inter-task correlation (MMTEB's downsampling) to keep it cheap enough for nightly runs.
- Evaluate with data-efficiency, not just accuracy: for each task report probe performance at 1%/10%/100% labels and an MDL/online-coding score; this directly certifies the project's design goal that decision-relevant information be linearly and cheaply accessible, and it predicts how the embedding will serve new consumers and sparse geo-zones.
- Periodically run a capacity-varying readout (readout model switching): the gap between linear-probe and small-MLP performance per task is the diagnostic for whether to add another auxiliary linear head to the encoder (information exists but is not linearly organized) versus accept a nonlinear consumer.
- Instrument encoder training with a label-free metric panel — RankMe effective rank, alpha eigenspectrum decay, alignment/uniformity on positive pairs (same zone adjacent buckets / perturbed views), and optionally LiDAR using those positive groups — as collapse alarms and for hyperparameter sweeps; but first validate each metric's correlation against the Eval Suite on marketplace data (Tsitsulin's reversal warning applies doubly to non-vision tabular/spatio-temporal data), and never optimize them directly.
- Train with Matryoshka nesting (e.g., 32/64/128/256 dims): low-latency pricing and dispatch consume a short prefix, forecasting consumes the full vector, and the prefix hierarchy is published as part of the embedding contract; use intrinsic-dimension estimates to pick the smallest prefix.
- Treat every encoder retrain as a compatibility event with explicit gates: (a) add a BCT-style influence loss against $v(n)$ probe heads — or better, adopt a fixed stationary target geometry (d-Simplex-style) so all future versions aim at the same coordinates; (b) gate releases on cross-version probe performance ($v(n)$ heads reading $v(n+1)$ embeddings) and per-consumer negative-flip rate (PC-training's metric), e.g., 'no zone-hour where surge decisions regress'; (c) for noisy/sparse zones prefer uncertainty-weighted alignment (2025 hyperbolic BCT) to avoid inheriting old-version errors.
- Preserve historical embeddings across versions cheaply: store FCT-style side-information alongside each logged zone-time embedding so backtesting and forecasting archives can be mapped forward with a learned transformation instead of re-encoding raw history; for frequent minor fine-tunes between major versions, use query-drift compensation so consumers stay on one interface.
- Deploy embedding drift monitoring from day one (Failing Loudly recipe, as productized by Evidently/Arize): per-dimension two-sample tests with multiple-testing correction on PCA-reduced embeddings, a domain classifier per region, centroid drift per geo-zone, plus monitoring of probe-head output distributions; log the embedding version with every downstream prediction for regression attribution.
- Because surge price/radius are control levers, add a causal triage layer to monitoring and evaluation: condition drift analysis on the logged action (so a surge-policy change is not misread as market drift), and

include action-conditioned probes in the Eval Suite (predict outcome given state embedding AND logged action) so the benchmark rewards causally useful state information rather than policy-artifact shortcuts.

- Operationalize versioning like an API: semantic-versioned embeddings in the feature store, dual-index/shadow rollout of each new version with consumer-specific regression dashboards before cutover, and a deprecation window during which both versions are served — compatibility training makes this window cheap, and negative-flip gates make it safe.

Annotated bibliography (19 sources)

2020 Wang, T., Isola, P. (2020). Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere. ICML 2020.

arxiv.org/abs/2005.10242

Proves that the InfoNCE contrastive loss asymptotically optimizes two measurable geometric properties of normalized embeddings: alignment (positive pairs map close together) and uniformity (features spread uniformly on the hypersphere, preserving maximal information). Introduces two closed-form, label-free metrics (L_{align} , L_{uniform}) computable from batches of embeddings, shows they strongly track downstream task performance, and shows directly optimizing them matches or beats contrastive learning.

Relevance to this project: Gives the project its first two cheap training-time health metrics: compute alignment over positive pairs (same zone, adjacent time buckets) and uniformity over the embedding batch to detect collapse or under-spread latents long before any downstream model is trained.

2020 Voita, E., Titov, I. (2020). Information-Theoretic Probing with Minimum Description Length. EMNLP 2020.

aclanthology.org/2020.emnlp-main.14/

Recasts probing as data transmission: instead of reporting probe accuracy, measures the description length of labels given representations, estimated via variational coding or online/prequential coding on top of a standard probing pipeline. MDL jointly captures probe quality and the effort (model size, sample count) needed to achieve it, and unlike raw accuracy it cleanly separates trained representations from random baselines and real tasks from control tasks.

Relevance to this project: Since the project deliberately constrains auxiliary heads to linear probes, MDL/prequential coding is the right acceptance metric: it certifies not just that surge-relevant or ETA-relevant information is in the embedding, but that it is cheaply extractable — the actual property downstream teams pay for.

2019 Zhai, X., Puigcerver, J., Kolesnikov, A., et al. (2019). A Large-scale Study of Representation Learning with the Visual Task Adaptation Benchmark (VTAB). arXiv:1910.04867.

arxiv.org/abs/1910.04867

Defines representation quality operationally: a good representation is one that yields good performance on unseen diverse tasks with limited task-specific data. VTAB fixes 19 frozen downstream tasks spanning natural, specialized, and structured domains, allows any pre-training, and scores encoders by mean transfer performance under a constrained adaptation budget.

Relevance to this project: The template for a 'market-state benchmark suite': a fixed, versioned battery of frozen downstream tasks (pricing, ETA, forecast, repositioning) with limited-label probes, so every candidate encoder gets one comparable scalar scorecard and the benchmark itself becomes the embedding's contract with consumers.

2025 Enevoldsen, K., Chung, I., Kerboua, I., et al. (2025). MMTEB: Massive Multilingual Text Embedding Benchmark. ICLR 2025 (extends Muennighoff et al., MTEB, EACL 2023).

arxiv.org/abs/2502.13595

Scales the MTEB embedding-benchmark pattern to 500+ quality-controlled tasks over 250+ languages, and — critically for practitioners — introduces correlation-based task downsampling that preserves model rankings while slashing evaluation cost, plus hard-negative sampling to keep small retrieval splits discriminative. Finds a 560M-parameter model tops much larger LLMs, showing benchmark-driven selection beats scale assumptions.

Relevance to this project: Two directly transferable ideas: (1) prune the internal benchmark via inter-task correlation so the eval suite stays cheap enough to run on every encoder candidate and nightly build; (2) maintain a public-leaderboard-style internal scoreboard so downstream teams trust version-over-version claims.

2023 Li, Y., Bornschein, J., Chen, T. (2023). Evaluating Representations with Readout Model Switching. ICLR 2023.

arxiv.org/abs/2302.09579

Treats representation evaluation as model selection under the MDL principle: instead of fixing a linear readout, it maintains a hybrid space of readout models of varying capacity and uses a Bayesian switching strategy over prequential predictions, yielding an MDL score that accounts for both model complexity and data efficiency. Computable online; ranks ResNet/ViT and supervised/SSL encoders consistently across downstream tasks.

Relevance to this project: The principled answer to 'is linear-probe-only evaluation blinding us?': periodically run the switching readout, and the gap between linear and higher-capacity readouts tells you whether decision-relevant market information exists but is not linearly accessible — the exact signal for adding or redesigning auxiliary heads.

2023 Garrido, Q., Balestrieri, R., Najman, L., LeCun, Y. (2023). RankMe: Assessing the Downstream Performance of Pretrained Self-Supervised Representations by Their Rank. ICML 2023.

arxiv.org/abs/2210.02885

Proposes the smooth effective rank of the embedding matrix (entropy of normalized singular values) as a zero-label, zero-hyperparameter criterion for joint-embedding SSL quality. Across hundreds of training runs, selecting hyperparameters by RankMe nearly matches label-based selection on both in-domain and OOD downstream tasks, because dimensional collapse is the dominant failure mode it detects.

Relevance to this project: A one-line metric to log every training epoch of the market-state encoder: collapsing effective rank flags a latent space that cannot support many heterogeneous downstream consumers, and it enables label-free hyperparameter sweeps before any downstream team is involved.

2022 Agrawal, K.K., Mondal, A.K., Ghosh, A., Richards, B. (2022). alpha-ReQ: Assessing Representation Quality in Self-Supervised Learning by Measuring Eigenspectrum Decay. NeurIPS 2022.

papers.nips.cc/paper_files/paper/2022/hash/70596d70542c51c8d9b4e423f4b...

Observes that the eigenspectrum of the feature covariance follows a power law and uses the decay coefficient α as a task-agnostic quality measure for linear-readout tasks: α near 1 balances expressivity (slow decay = more usable dimensions) against robustness (fast decay = noise suppression). Validated on CIFAR/STL10/MIT67/ImageNet with SSL methods like Barlow Twins for label-free model selection.

Relevance to this project: Complements RankMe with a finer-grained spectral diagnostic: for a market-state embedding it distinguishes 'high rank because rich structure' from 'high rank because noise', guarding against an encoder that inflates rank metrics without adding linearly decodable decision signal.

2024 Thilak, V., Huang, C., Saremi, O., Dinh, L., Goh, H., Nakkiran, P., Susskind, J., Littwin, E. (2024). LiDAR: Sensing Linear Probing Performance in Joint Embedding SSL Architectures. ICLR 2024.

arxiv.org/abs/2312.04000

Apple's refinement of rank-based metrics: computes the effective rank of the linear discriminant analysis matrix associated with the SSL surrogate task (between-'class' scatter over within-scatter, where a class is an augmentation group), thereby counting only discriminatively informative dimensions rather than all high-variance ones. Substantially outperforms covariance-rank methods (RankMe) at predicting linear-probe performance for hyperparameter selection.

Relevance to this project: The strongest current label-free predictor of linear-probe performance — well matched to this project since positive groups are naturally defined (perturbed views of the same zone-time market state) and downstream heads are constrained to be linear.

2019 Ansuini, A., Laio, A., Macke, J.H., Zoccolan, D. (2019). Intrinsic Dimension of Data Representations in Deep Neural Networks. NeurIPS 2019.

arxiv.org/abs/1905.12784

Estimates the intrinsic dimension (ID) of each layer's representation manifold with the TwoNN nearest-neighbor estimator, finding ID is orders of magnitude below ambient dimension, follows a 'hunchback' profile across depth, and — crucially — the ID of the last hidden layer predicts test generalization, with the representation manifold being strongly curved (linear estimators overestimate ID).

Relevance to this project: Establishes intrinsic dimension as a capacity diagnostic: tracking ID of the market-state embedding across versions reveals whether added encoder capacity is capturing genuinely new market structure or memorizing noise, and informs how small the Matryoshka prefix dimensions can safely go.

2023 Tsitsulin, A., Munkhoeva, M., Perozzi, B. (2023). Unsupervised Embedding Quality Evaluation. TAG-ML Workshop, ICML 2023.

arxiv.org/abs/2305.16562

Google systematically benchmarks label-free embedding quality metrics (spectral measures such as NESum and stable-rank, plus a new metric based on stable linear separability from SSL geometry) across models and datasets. Key finding: correlation with downstream quality is dataset-dependent and can fully reverse across settings — no single unsupervised metric is universal, so metrics must be validated per-domain.

Relevance to this project: The direct warning label for this project: before wiring RankMe/alpha/LiDAR into CI gates, empirically validate each against the company's own downstream task suite on marketplace data, and gate on a panel of metrics rather than any single one.

2025 Shestov, A., Klenitskiy, A., Denisova, D., et al. (2025). Topological Metric for Unsupervised Embedding Quality Evaluation. arXiv:2512.15285.

arxiv.org/abs/2512.15285

Proposes 'Persistence', a persistent-homology-based metric quantifying multi-scale geometric and topological richness of an embedding space, dropping the linear-separability and covariance-spectrum assumptions of prior label-free metrics. Reports top-tier correlation with downstream performance across diverse domains (including recommendation/tabular-adjacent settings) and effective model/hyperparameter selection.

Relevance to this project: The newest (Dec 2025) entrant in label-free evaluation and notable for working beyond vision SSL; market-state data (tabular, spatio-temporal, non-image) is exactly where covariance-spectrum metrics are least validated, making a topology-based alternative worth including in the metric panel.

2022 Kusupati, A., Bhatt, G., Rege, A., et al. (2022). Matryoshka Representation Learning. NeurIPS 2022.

papers.nips.cc/paper_files/paper/2022/hash/c32319f4868da7613d78af99931...

Trains one embedding whose nested prefixes (e.g., first 8, 16, ..., 2048 dims) are each independently useful, by applying the training loss to every prefix granularity simultaneously — a minimal pipeline change with no inference cost. Prefixes match independently trained low-dimensional embeddings, giving up to 14x smaller vectors at equal accuracy and 14x retrieval speedups; since adopted by OpenAI and Google (Gemini) embedding APIs as user-truncatable dimensions.

Relevance to this project: Resolves the project's one-embedding-many-consumers capacity tension: pricing and dispatch can consume a cheap 32-64d prefix at low latency while forecasting consumes the full vector, all from a single versioned artifact — and the prefix hierarchy itself becomes part of the compatibility contract.

2020 Shen, Y., Xiong, Y., Xia, W., Soatto, S. (2020). Towards Backward-Compatible Representation Learning. CVPR 2020 (Oral).

arxiv.org/abs/2003.11942

Defines embedding backward compatibility (new query embeddings must work against an old-model gallery at least as well as the old model did) and achieves it with Backward-Compatible Training (BCT): the new encoder is trained with an added 'influence loss' that passes new features through the old model's frozen classifier head, anchoring the new space to the old one. Enables backfill-free model upgrades without sacrificing new-model accuracy; open-sourced (openBCT), originated at AWS for face search.

Relevance to this project: The foundational technique for the project's stated versioning problem: retraining the market-state encoder with a BCT-style influence loss against the previous version's probe heads means downstream pricing/dispatch/ETA models keep working during rollout instead of all requiring simultaneous retraining.

2021 Yan, S., Xiong, Y., Kundu, K., Yang, S., Deng, S., Wang, M., Xia, W., Soatto, S. (2021). Positive-Congruent Training: Towards Regression-Free Model Updates. CVPR 2021.

arxiv.org/abs/2011.09161

Amazon formalizes 'negative flips' — samples the old model got right that the new model gets wrong — as the regression metric for model updates, and proposes Focal Distillation: distill from the old model with extra weight on samples the old model classified correctly, optionally using an ensemble reference. Reduces per-sample regression substantially at negligible accuracy cost.

Relevance to this project: Average-metric improvement can still break individual downstream consumers; negative-flip rate per consumer task (e.g., zones where surge decisions worsen after an embedding update) should be a release gate for every embedding version, and focal distillation is the training-side lever.

2022 Ramanujan, V., Vasu, P.K.A., Farhadi, A., Tuzel, O., Pouransari, H. (2022). Forward Compatible Training for Large-Scale Embedding Retrieval Systems. CVPR 2022.

arxiv.org/abs/2112.02805

Apple inverts BCT: rather than constraining the future model, the current model is trained alongside auxiliary 'side-information' per sample that facilitates future updates; when a new model arrives, a lightweight learned transformation maps old embedding + side-info into the new space. Avoids BCT's plasticity penalty on the new model and outperforms BCT on ImageNet/Places-365/VGGFace2 model-upgrade benchmarks (follow-up FastFill, ICLR 2023, makes the backfill efficient and ordered).

Relevance to this project: For the market-state platform this is the archival strategy: store side-information with each historical zone-time embedding so years of logged embeddings consumed by forecasting/backtesting can be mapped forward into each new version's space instead of re-encoding raw history.

2024 Biondi, N., Pernici, F., Ricci, S., Del Bimbo, A. (2024). Stationary Representations: Optimally Approximating Compatibility and Implications for Improved Model Replacements. CVPR 2024.

arxiv.org/abs/2405.02581

Proves that representations learned with a fixed d-Simplex classifier (a neural-collapse-inspired, non-learnable target geometry) optimally approximate the formal two-inequality definition of compatibility: class prototypes stay stationary across retrainings, so successive models land in mutually usable spaces by construction rather than via pairwise alignment losses. Demonstrates seamless asynchronous model replacement without gallery reprocessing.

Relevance to this project: Suggests the cleanest architecture for a company-wide embedding: fix the latent target geometry (or the linear-probe head structure) once, so every future encoder retrain aims at the same coordinates — turning compatibility from a per-release alignment chore into a structural property of the platform.

2025 Bui, N., Yang, M., Chen, R., Neves, L., Ju, M., Ying, R., Shah, N., Zhao, T. (2025). Learning Along the Arrow of Time: Hyperbolic Geometry for Backward-Compatible Representation Learning. arXiv:2506.05826.

arxiv.org/abs/2506.05826

Snap Research + Yale reframe compatibility in hyperbolic space: new embeddings are constrained to lie within the entailment cone of the old ones rather than to reconstruct them exactly, with a robust contrastive alignment loss that weights alignment strength by the old embedding's estimated quality. Treats model generations as an 'arrow of time', letting uncertain old embeddings constrain the new model less.

Relevance to this project: State of the art (2025) on softening the compatibility-plasticity trade-off: for noisy marketplace data where old embeddings of sparse zones are unreliable, uncertainty-weighted alignment prevents the new encoder from inheriting the old version's mistakes while staying consumable by old heads.

2025 Goswami, D., Wang, L., Twardowski, B., van de Weijer, J. (2025). Query Drift Compensation: Enabling Compatibility in Continual Learning of Retrieval Embedding Models. CoLLAs 2025.

arxiv.org/abs/2506.00037

For continually fine-tuned retrieval embedding models, estimates the embedding drift between the old and updated model and, at query time, projects new-model query embeddings back into the old embedding space (plus embedding distillation during training for stability), so the existing document index remains usable with no re-indexing. Approaches joint-training performance on a continual retrieval benchmark.

Relevance to this project: The lightweight alternative when full BCT retraining is too heavy: a cheap post-hoc drift-compensation transform lets the marketplace ship frequent encoder refreshes (e.g., weekly fine-tunes on new market data) while downstream consumers and historical embedding stores stay on a stable interface between major versions.

2019 Rabanser, S., Günnemann, S., Lipton, Z.C. (2019). Failing Loudly: An Empirical Study of Methods for Detecting Dataset Shift. NeurIPS 2019.

arxiv.org/abs/1810.11953

Systematically benchmarks shift-detection pipelines: dimensionality-reduce inputs (PCA, autoencoders, or the model's own learned representations / softmax outputs — 'black-box shift detection'), then apply multiple univariate two-sample tests with Bonferroni correction or MMD; separately train a domain classifier to characterize and assess the malignancy of the shift. BBSD on learned representations wins across shift types; foundation for industrial embedding-drift tooling (Evidently, Arize, WhyLabs all implement its recipes plus centroid-distance and drifted-component-share methods).

Relevance to this project: The blueprint for production monitoring of the market-state embedding: run per-dimension two-sample tests and a domain classifier on embedding streams per geo-zone to catch drift; but because surge is a control lever, alerts must be triaged into policy-induced vs. environment-induced shift before triggering retrains.

12 Causal Inference & Experiment Design Under Marketplace Interference

This chapter closes a gap identified by the completeness critic; it was researched by a dedicated gap-fill agent.

State of the art

Causal inference under marketplace interference has evolved from documenting the problem to prescribing designs and estimators that are now standard at ride-hailing and delivery platforms. The field's starting point is that surge-style levers violate SUTVA twice over: spatially (a price/radius change in one zone moves drivers and riders across zone boundaries — an equilibrium effect formalized by the wild-goose-chase and spatio-temporal-pricing literatures) and temporally (the platform state is a Markov chain, so treatments carry over). First-generation responses were design-side: switchback experiments that randomize the whole market over time blocks (Bojinov-Simchi-Levi-Zhao; refined by Hu-Wager's burn-in analysis showing naive switchbacks lose a $T^{1/6}$ rate factor to carryover bias) and two-sided randomization for cross-side interference (Johari et al., who prove the sign and regime-dependence of naive A/B bias). Second-generation work is estimator-side and model-based: Farias et al.'s Differences-in-Q's estimator uses a learned Q-function on the platform's Markov state to cut interference bias to second order with exponentially less variance than unbiased OPE; Luo et al.'s varying-coefficient decision process decomposes ATE into direct and indirect (spillover) effects on real ride-hailing experiments. Current SOTA (2024-2026) unifies space and time: clustered switchback designs with near-minimax Horvitz-Thompson guarantees (Jia-Kallus-Yu, extended in 2025 to non-stationary, treatment-independent random interference graphs motivated by driver movement), MSE-optimal spatial cluster construction via causal graph cuts validated on city-scale ridesharing simulators (Zhu et al., ICML 2025; related work shows spatially randomized designs can beat global rollouts for policy evaluation, Yang et al., AOAS 2026), permutation-invariant neural OPE that learns the mean-field interference function from data (Causal Deepsets), and design-based frameworks for arbitrary spillover/carryover with software support (Mukaigawara-Imai et al.'s geocausal). Open problems: jointly optimal spatio-temporal cluster/block sizing under unknown mixing times, interference-aware inference when the lever is continuous (dose-response rather than binary), endogenous interference graphs that respond to treatment, and integrating these estimators into always-on release gates rather than one-off studies. For this project the literature carries a blunt upstream message: the causal content of a market-state embedding is bounded by the randomization actually present in the logs, so the experimentation substrate is part of the representation-learning system, not an afterthought.

Key principles

- Interference bias is first-order and sign-varying, not noise: naive unit-level A/B in a two-sided market can overstate or understate effects depending on whether the market is demand- or supply-constrained (Johari et al.), so any embedding head trained or validated on naively analyzed logs encodes a biased dose-response. Causal usefulness downstream requires interference-aware estimands baked in from the start.
- There is a menu of estimands — direct (own-zone), indirect/spillover (neighbor-zone), total/global (market-wide policy contrast), and continuous dose-response — and different downstream consumers need different ones: pricing needs own-zone dose-response, repositioning and dispatch need spillover terms, and executive go/no-go decisions need the total effect. An embedding that only supports one estimand under-serves the platform.
- Identifiability is purchased by design, not by modeling: the randomization unit must match the interference structure — time blocks longer than the system's mixing time (with burn-in discarding) for temporal carryover, graph-cut spatial clusters that sever driver-flow edges for spatial spillover, and clustered switchbacks when both operate. Without such variation in the logs, action-conditioned representations are unidentifiable no matter the architecture.
- Learned value/Q-functions are the bridge between representation learning and interference-robust evaluation: the Differences-in-Q's insight is that a Q-head on the (embedded) market state converts interference bias into a second-order term while slashing variance relative to unbiased OPE — so a Q-head is simultaneously an auxiliary training signal and the platform's best experiment-analysis tool.
- Small, always-on randomized perturbations of continuous levers ('experimenting in equilibrium') identify local dose-response gradients at vanishing business cost, and are the cheapest permanent data source for causal heads — dithering is a logging requirement, not an experiment to be scheduled.
- Mean-field / permutation-invariant aggregation of neighbor states and actions is the right inductive bias for spatial interference: outcomes depend on one's own state plus an exchangeable summary of neighbors (Causal Deepsets), so the encoder should ingest neighborhood aggregates and spillover heads should pool neighbor embeddings permutation-invariantly rather than assuming a fixed adjacency ordering.
- Control design and inference design are coupled: spatially/temporally smooth pricing (Ma-Fang-Parkes) both prevents the equilibrium unraveling (surge chasing, wild-goose-chase regimes) that generates the worst spillovers and keeps the system in a locally smooth response regime where dose-response estimation and mean-field approximations are valid. Constraining the lever makes its effects learnable.
- Evaluation of the embedding itself is subject to the same interference as the policies it powers: an embedding-version rollout changes pricing/dispatch market-wide, so promotion gates must use switchback or cluster-randomized designs analyzed with design-based (truncated Horvitz-Thompson), Q-based (DQ), or decomposition-based (VCDP direct/indirect) estimators — never user-level difference-in-means.

Recommendations for the market-state embedding project

- LOGGING SPEC — record identification-critical fields per zone x time-bucket: realized surge multiplier and radius, the policy's intended (pre-dithering) value, the dithering draw and its seed, the assignment mechanism and cluster-map version, cluster-level propensities, and a permutation-invariant snapshot of neighbor zones' states and levers. Without logged propensities and neighbor levers, none of the interference-aware estimators (HT, DQ, VCDP, Causal Deepsets) can be computed later.
- LOGGING SPEC — run a permanent clustered-switchback substrate (Jia-Kallus-Yu design): spatial clusters derived by causal graph cuts (Zhu et al. 2025) on the empirically measured driver-flow / demand-correlation graph, re-derived quarterly as mobility patterns shift; time blocks sized above the estimated supply-mixing

time (start at 60-120 min, tuned via Bojinov et al.'s data-driven carryover procedure) with Hu-Wager burn-in periods flagged in the logs so training and estimation can discard post-flip transients.

- **LOGGING SPEC** — keep always-on small-amplitude randomized dithering of the continuous levers (Wager-Xu 'experimenting in equilibrium'): e.g., $\pm 2-5\%$ multiplicative jitter on surge price and ± 1 step on surge radius, independently randomized per zone-bucket within policy-safe bounds. This is the primary identification source for the dose-response heads and costs vanishingly little revenue; treat its removal as a breaking change to the embedding pipeline.
- **HEAD DESIGN** — four interference-aware auxiliary heads on the encoder, all trained only on randomized/dithered data windows: (1) an own-zone dose-response head predicting outcome deltas as a function of own lever (Munro-Wager-Xu direct effect); (2) a spillover head predicting neighbor-zone outcome response to own-zone lever using permutation-invariant pooling of neighbor embeddings (Causal Deepsets architecture; VCDP indirect effect); (3) a TD-trained zone-level Q/value head over the platform Markov state — this head doubles as the engine of the DQ estimator (Farias et al.) and should be a versioned, officially supported output of the embedding service; (4) a cluster-level propensity head reproducing the logging policy's lever distribution, for IPS/DR-style corrections at the correct (cluster, not unit) granularity.
- **HEAD DESIGN** — encode regime and smoothness structure: include distance-to-wild-goose-chase-regime indicators (idle-driver density, pickup ETA percentiles per Castillo et al.) as encoder inputs and probe targets, and add a spatial/temporal smoothness regularizer on any surge recommendations derived from the embedding, consistent with Ma-Fang-Parkes — this both curbs the driver-flow spillovers that inflate interference and keeps the response surface in the locally smooth regime the causal heads assume.
- **EVALUATION PROTOCOL** — embedding-version promotion gates must be interference-aware: online tests of a new embedding version (or any surge policy it powers) run as clustered switchbacks, never rider/driver-level splits (Johari et al. bias results); analyze every test with a three-estimator scorecard — truncated Horvitz-Thompson (design-based, Jia-Kallus-Yu), Differences-in-Q's via the Q-head (model-based, low variance), and VCDP direct/indirect decomposition — and gate on the total (global) effect while reporting DE and IE separately so consumers like repositioning see spillover impact explicitly. Divergence among the three estimators is itself a red flag to investigate before promotion.
- **EVALUATION PROTOCOL** — offline pre-gates use interference-aware OPE: score candidate embedding versions and surge policies from logs with Causal Deepsets / VCDP-based OPE (and geocausal as an assumption-light cross-check), reporting direct, indirect, and total estimands plus sensitivity to the assumed interference neighborhood; only candidates passing the offline scorecard graduate to the (expensive, market-level) clustered-switchback slot, integrating with the release-gate ladder from the evaluation/versioning workflow.
- **EVALUATION PROTOCOL** — for changes touching both sides of the market (rider-facing and driver-facing consumers of the embedding), use two-sided randomization (Johari et al.) as the fallback when a switchback slot is unavailable, and log market balance (demand- vs supply-constrained regime) at test time because the sign of residual bias depends on it.
- **COST AND GOVERNANCE** — budget the substrate explicitly: reserve a small standing fraction of zone-bucket cells for the clustered-switchback rotation and accept the dithering variance as a permanent data-collection cost; version the cluster maps, block lengths, and dithering amplitudes alongside embedding versions so that every embedding release records exactly which randomization regime generated its training data — this is the backward-compatibility analogue for causal validity.

Annotated bibliography (13 sources)

2023 Bojinov, I., Simchi-Levi, D., & Zhao, J. (2023). Design and Analysis of Switchback Experiments. *Management Science* 69(7); arXiv:2009.00148.

arxiv.org/abs/2009.00148

Formalizes switchback experiments — sequentially exposing the entire market to randomized treatments over time — as a minimax discrete optimization over assignment sequences under bounded carryover of known duration. Derives the optimal block structure, provides exact randomization-based p-values and a finite-population CLT for the Horvitz-Thompson effect estimator, plus a data-driven procedure for choosing the carryover length and robustness results under misspecification.

Relevance to this project: The canonical design for testing market-wide changes (surge policies, embedding versions) when within-market unit randomization is hopeless; its carryover-duration machinery tells the platform how to size time blocks for surge-lever experiments and version rollouts.

2022 Hu, Y., & Wager, S. (2022, rev. 2025). Switchback Experiments under Geometric Mixing. arXiv:2209.00197.

arxiv.org/abs/2209.00197

Analyzes switchbacks when the platform is a Markov chain with geometric mixing rather than finite carryover. Shows standard switchback estimators suffer carryover bias that degrades the error rate from $T^{(-1/2)}$ to $T^{(-1/3)}$, and proves that inserting burn-in periods — discarding data just after each treatment flip — restores a near-optimal $\log(T)^{1/2} T^{(-1/2)}$ rate.

Relevance to this project: Ride-hailing supply is exactly a mixing Markov system, so this paper dictates the block-length and burn-in spec for the platform's permanent switchback substrate and warns that un-burned switchback logs will bias any causal head trained on them.

2022 Johari, R., Li, H., Liskovich, I., & Weintraub, G. Y. (2022). Experimental Design in Two-Sided Platforms: An Analysis of Bias. *Management Science* 68(10); arXiv:2002.05670.

arxiv.org/abs/2002.05670

Builds a dynamic two-sided matching-market model and analytically characterizes the bias of customer-side randomization (CR), listing-side randomization (LR), and a novel two-sided randomization (TSR) design in which both sides are randomized and treatment applies only to treated-treated pairs. Shows CR is unbiased in demand-constrained regimes and LR in supply-constrained regimes, while TSR interpolates and can debias across regimes.

Relevance to this project: Establishes that naive rider- or driver-level A/B tests of anything touching matching give regime-dependent, sign-varying bias — the reason embedding-version gates cannot reuse the company's standard user-split pipeline; TSR is the fallback when switchbacks are too costly.

2021 Wager, S., & Xu, K. (2021). Experimenting in Equilibrium. Management Science 67(11); arXiv:1903.02124.

arxiv.org/abs/1903.02124

Proposes 'local experimentation': apply vanishingly small, unit-level randomized perturbations (dithering) to a continuous system parameter such as payments or prices while the market remains in mean-field equilibrium. Lightweight modeling of the equilibrium response lets the platform consistently estimate the gradient of steady-state outcomes with respect to the parameter and even run stochastic-gradient-style optimization, without the large shocks and interference bias of standard A/B tests.

Relevance to this project: Directly provides the 'lever dithering' logging requirement: continuous small randomized jitter on surge multiplier and radius identifies local dose-response gradients — precisely the signal the causal auxiliary heads must regress on — at near-zero revenue cost.

2021 Munro, E., Wager, S., & Xu, K. (2021, rev. 2025). Treatment Effects in Market Equilibrium. arXiv:2109.11647 (forthcoming, American Economic Review).

arxiv.org/abs/2109.11647

Studies treatment effects when spillovers are mediated by equilibrium prices of traded goods in a single large market. Combines a standard Bernoulli-randomized trial (identifying direct effects) with an augmented experimental design that perturbs prices to estimate demand/supply elasticities, enabling decomposition of the total policy effect into direct and equilibrium-mediated indirect components without needing many isolated markets.

Relevance to this project: Gives the estimand decomposition (direct vs. price-mediated indirect) that surge dose-response heads should target, and shows the platform can identify equilibrium effects inside one city by pairing unit randomization with lever perturbations rather than city-level experiments.

2022 Farias, V. F., Li, A. A., Peng, T., & Zheng, A. (2022). Markovian Interference in Experiments. NeurIPS 2022 (oral); arXiv:2206.02371.

arxiv.org/abs/2206.02371

Models experiments with interference through a shared system state (e.g., driver supply) as policy evaluation in a Markov decision process and proposes the Differences-in-Q's (DQ) estimator: the treatment-effect estimate is the difference of learned Q-values between treatment and control actions averaged over the experiment's state distribution. Proves bias is second-order in the intervention's impact while variance is exponentially smaller than unbiased off-policy evaluation, and validates on a city-scale ride-hailing simulator where naive and IPS estimators fail.

Relevance to this project: The single most direct connection between the embedding and experiment analysis: a TD-trained Q-head on the market-state embedding is exactly the object DQ needs, making the embedding service itself the engine for interference-corrected A/B readouts of surge policies and embedding versions.

2024 Luo, S., Yang, Y., Shi, C., Yao, F., Ye, J., & Zhu, H. (2024). Policy Evaluation for Temporal and/or Spatial Dependent Experiments. Journal of the Royal Statistical Society Series B; arXiv:2202.10887.

arxiv.org/abs/2202.10887

Introduces a varying-coefficient decision process (VCDP) model for experiments run by technology platforms in which treatment effects evolve over time and across space. Decomposes the average treatment effect into a direct effect (DE) and an indirect, system-state-mediated effect (IE), and provides estimation and inference procedures for both, validated in simulation and on real platform experiments (developed with Didi ride-hailing collaborators).

Relevance to this project: Supplies the DE/IE decomposition and inference toolkit for analyzing the platform's switchback logs, and defines the two quantities (own-lever direct response, state-mediated spillover) the encoder's causal heads should expose as separate linear probes.

2023 Jia, S., Kallus, N., & Yu, C. L. (2023, rev. 2025). Clustered Switchback Designs for Experimentation Under Spatio-temporal Interference. arXiv:2312.15574.

arxiv.org/abs/2312.15574

Combines spatial clustering with temporal blocking: units are grouped into graph clusters, time into blocks, and each cluster-block cell receives one random treatment. Under graph-neighborhood spatial interference plus fast-mixing Markovian temporal interference, a truncated Horvitz-Thompson estimator achieves $O(1/NT)$ MSE for well-clusterable graphs, matching lower bounds up to logs and improving on pure switchback and pure network-interference results.

Relevance to this project: This is the design template for the platform's permanent experimentation substrate — zone-clusters crossed with time blocks — handling surge's spatial and temporal spillovers simultaneously; its truncated HT estimator is the design-based leg of the evaluation scorecard.

2025 Zhu, J., Li, J., Zhou, H., Lin, Y., Lin, Z., & Shi, C. (2025). Balancing Interference and Correlation in Spatial Experimental Designs: A Causal Graph Cut Approach. ICML 2025; arXiv:2505.20130.

arxiv.org/abs/2505.20130

Learns the spatial clustering itself: derives a surrogate for the MSE of the causal-effect estimator that trades off interference (cut edges between clusters) against spatial correlation of outcomes, and optimizes it with classical graph-cut algorithms. Accommodates moderate-to-large spillovers and general spatial covariance, with a public Python implementation, and is validated on synthetic environments plus a city-scale ridesharing dispatch simulator.

Relevance to this project: The current SOTA answer to 'how do we draw the experiment clusters?': the platform should periodically re-derive surge-experiment clusters via causal graph cuts on its driver-flow/demand-correlation graph instead of using static hexagons or districts.

2024 Dai, R., Wang, J., Zhou, F., Luo, S., Qin, Z., Shi, C., & Zhu, H. (2024). Causal Deepsets for Off-policy Evaluation under Spatial or Spatio-temporal Interferences. arXiv:2407.17910.

arxiv.org/abs/2407.17910

Replaces the hand-specified mean-field averaging used in spatial OPE with a learned, permutation-invariant (DeepSets-style) aggregation of neighboring units' states and treatments, estimating the interference function directly from data. The resulting OPE estimators substantially outperform fixed mean-field baselines on spatio-temporal benchmarks derived from ride-sharing-type settings.

Relevance to this project: Provides both the offline evaluation tool for candidate surge policies from logs and the architectural blueprint for the spillover head: permutation-invariant pooling of neighbor-zone embeddings and levers to predict cross-zone outcome responses.

2022 Ma, H., Fang, F., & Parkes, D. C. (2022). Spatio-Temporal Pricing for Ridesharing Platforms. Operations Research 70(2); arXiv:1801.04015.

arxiv.org/abs/1801.04015

Game-theoretic mechanism design over a spatio-temporal graph of trips: the Spatio-Temporal Pricing (STP) mechanism computes competitive-equilibrium prices that are 'appropriately smooth in space and time,' making it a subgame-perfect equilibrium for drivers to accept dispatches instead of chasing surge, while achieving welfare-optimal, envy-free, budget-balanced, core-selecting outcomes (and proving impossibility of the same in dominant strategies).

Relevance to this project: Explains mechanistically why sharp spatial surge gradients create the cross-zone driver-flow spillovers that break SUTVA, and motivates a smoothness regularizer on the surge lever that simultaneously improves market efficiency and keeps the system in a regime where the embedding's dose-response heads are well-posed.

2017 Castillo, J. C., Knoepfle, D. T., & Weyl, E. G. (2017; journal version Management Science 2025). Surge Pricing Solves the Wild Goose Chase / Matching and Pricing in Ride Hailing: Wild Goose Chases and How to Solve Them. ACM EC 2017; Management Science (doi:10.1287/mnsc.2022.00096).

dl.acm.org/doi/10.1145/3033274.3085098

Equilibrium model of ride-hailing matching under a first-dispatch protocol showing that when prices are too low relative to demand, idle drivers become so scarce that matches occur at long pickup distances — the 'wild goose chase' — collapsing effective throughput; all market equilibria in that regime are WGCs. Surge pricing (and secondarily matching-rule changes) restores the market to the efficient branch of the supply curve, using Uber data for calibration.

Relevance to this project: Documents a bifurcation in the market's response surface: near the WGC regime, outcomes respond discontinuously to the surge lever, so the embedding should explicitly encode distance-to-WGC-regime state and experiments must be powered and analyzed regime-aware.

2025 Mukaigawara, M., Imai, K., Lyall, J., & Papadogeorgou, G. (2025). Spatiotemporal causal inference with arbitrary spillover and carryover effects. arXiv:2504.03464.

arxiv.org/abs/2504.03464

A design-based framework treating treatments and outcomes as spatiotemporal point patterns and estimating effects of treatment-strategy interventions without restricting spillover or carryover structure, sidestepping arbitrary panel aggregation into zones/periods. Supports heterogeneous-effect and mediation analysis and ships as the open-source geocausal R package; demonstrated on airstrike/insurgency data from the Iraq War.

Relevance to this project: Offers an assumption-light retrospective analysis path for the platform's logged surge interventions when zone/bucket aggregation is itself suspect, and a ready-made software baseline (geocausal) for validating the more parametric VCDP/DQ estimates.

13 Temporal Distribution Shift: Generalization & Continual Adaptation of the Encoder

This chapter closes a gap identified by the completeness critic; it was researched by a dedicated gap-fill agent.

State of the art

Temporal distribution shift research has moved through three overlapping waves. First came benchmarking: Wild-Time (NeurIPS 2022) established that on real timestamped data, models lose roughly 20% performance going from in-distribution to future-time evaluation, and that 13 methods spanning domain generalization, continual learning, SSL, and ensembling barely beat plain ERM — establishing temporally forward splits (Eval-Fix, Eval-Stream) as the mandatory evaluation contract. The second wave is temporal domain generalization (TDG) proper: treat time as a domain index and extrapolate the model to the near future — DRAIN forecasts next-period network parameters with a Bayesian recurrent graph generator; prompting-based TDG and TEA (EMNLP 2025) make this scale to large backbones by extrapolating only prompts or weight-averaging coefficients; FreKoo (2025) extrapolates only the low-frequency spectral component of parameter trajectories via Koopman operators; CODA sidesteps the model entirely by simulating future training data; Drift-Resilient TabPFN (NeurIPS 2024) bakes gradually-shifting structural causal models into a prior-fitted transformer so drift adaptation happens in-context in one forward pass. The third wave is a sobriety check plus operational adaptation: 'Temporal Generalization: A Reality Check' (2025) finds that across large-scale monthly/yearly tasks no interpolation or extrapolation method consistently beats simply using the latest warm-started model, echoing the LLM continual pre-training result (Ibrahim et al. 2024) that LR re-warming + re-decay + replay matches full retraining at a fraction of the compute — with Ash & Adams' shrink-and-perturb as the known fix for the warm-start generalization pathology. On the adaptation side, Tent-style test-time entropy minimization is powerful but the TTAB benchmark (ICML 2023) documents its instability (online batch dependency, hyperparameter fragility), ARM offers a safer meta-learned adapt-from-unlabeled-context alternative, Proceed (KDD 2025) maps estimated drift to parameter offsets via a learned generator, and CaSSLe shows continual SSL needs cross-version representation distillation to avoid forgetting. Open problems: no TDG method reliably wins over frequent warm-started retraining at scale; disentangling environment drift from self-induced (performative) drift is unsolved; and the interaction between fast adaptation and cross-version embedding compatibility is essentially unstudied.

Key principles

- The near future is the only test set that matters: model selection, hyperparameter tuning, and probe evaluation must all use temporally forward splits (Wild-Time Eval-Fix / Eval-Stream); random splits anywhere in the encoder pipeline silently overstate serving performance by ~20%.
- Simple frequent warm-started retraining is the champion to beat: across large-scale evaluations (Reality Check 2025; Ibrahim et al. 2024) no parameter interpolation/extrapolation method consistently outperforms 'serve the latest warm-started model' — any TDG machinery must first beat that baseline on your own forward splits.
- Warm-starting has a known generalization pathology and a cheap cure: apply shrink-and-perturb and/or LR re-warming + re-decay + replay at every scheduled retrain, or the encoder ossifies while training loss looks fine.
- Separate slow and fast parameters: keep a frozen or slowly-updated trunk (stable latent geometry, backward compatibility) and confine temporal adaptation to small fast components — prompts, normalization/affine layers, per-zone identity embeddings, low-rank adaptation offsets (Prompting-TDG, Tent, Proceed).
- Extrapolate structure, suppress noise: forecast only the low-frequency component of drift (seasonality, city growth) and regularize against chasing high-frequency transients (FreKoo, TEA, DRAIN) — trajectory-aware weight averaging of recent checkpoints is the scalable version of this.
- Continual self-supervised training forgets tail regimes unless you make it not forget: cross-version representation distillation (CaSSLe-style predictor from new to old latent space) plus replay of rare regimes doubles as the backward-compatibility mechanism.
- Label delay makes reactive online learning adapt to stale concepts: in systems where outcomes arrive after decisions, adaptation must be proactive (drift-estimated parameter adjustment, meta-learned adaptation from unlabeled context) rather than fine-tuning on the latest labels (Proceed, ARM).
- Unsupervised test-time adaptation is fragile in feedback-loop systems: entropy-minimization-style TTA suffers online batch dependency and impossible model selection (TTAB), and in a platform whose own surge policy shapes the data stream it can amplify performative drift — keep it out of the control loop.
- Evaluate aging, not just accuracy: the decay of frozen-encoder probe performance as a function of train-serve gap (a 'future-decay curve') is the operational quantity that sets retraining cadence and is itself a release metric.

Recommendations for the market-state embedding project

- Adopt Wild-Time's protocols as the encoder release contract: Eval-Fix (train on data through time T , gate the release on frozen-encoder linear-probe metrics over $(T, T+\delta]$) for every version, and Eval-Stream (rolling next- K -buckets evaluation) in CI to tune cadence. Ban random temporal splits everywhere in the embedding pipeline, including probe-head model selection.
- Make the future-decay curve a first-class release metric: for each release, freeze the encoder trained through T and plot each decision-relevant linear probe's metric at serve gaps of 1, 2, 4, 8, 12 weeks, per city; report the decay half-life. Set the retraining cadence per city where cumulative decay exceeds the downstream error budget — expect the Wild-Time-style ~20% cliff if this is never measured.
- Default retraining policy: scheduled warm-start retrains (monthly is a sane start) from the previous checkpoint with shrink-and-perturb plus LR re-warming/re-decay, and a replay buffer over historical data with deliberate oversampling of tail regimes (extreme weather, holidays, outages, past surge-cap events) so continual training does not forget them. Reserve full from-scratch retrains for architecture changes or when the warm-start lineage's Eval-Fix metrics degrade versus a from-scratch challenger trained annually.

- Do not adopt DRAIN-style full-parameter forecasting for the production encoder: it scales poorly and churns latent geometry, breaking backward compatibility. If you want future-proofing between retrains, evaluate TEA-style trajectory-aware weight averaging of the last k checkpoints against the 'latest checkpoint' baseline on Eval-Stream — it is cheap, works at full model scale, and weight averaging keeps successive versions geometrically close.
- Two-speed architecture (answers question 4): keep the trunk frozen between releases and designate STID-style per-hex/per-time-slot identity embeddings plus normalization/affine layers as the fast-adapting component, refreshed daily-to-weekly with a small learning rate and an L2 tether to their previous values (bounding drift so downstream consumers see smooth movement). Monitor identity-embedding displacement per hex as a cheap spatially-resolved drift telemetry signal.
- Add CaSSLe-style cross-version distillation to every warm-start retrain: a small predictor maps new embeddings to the previous version's space, trained jointly with the SSL/auxiliary objectives. This simultaneously (a) mitigates representation forgetting under continual training and (b) yields the alignment map the versioning workstream needs for backward compatibility — implement it once, own it jointly.
- Prohibit Tent-style unsupervised test-time adaptation inside the serving loop: entropy minimization on a data stream your own surge policy shapes risks confidence collapse and performative feedback (TTAB failure modes). If serve-time adaptation is needed, use mechanisms validated offline: ARM-style meta-learned adaptation from an unlabeled context window of the current market, or Proceed-style drift-to-parameter-offset generators trained on historical drift pairs — both respect label delay by not requiring fresh labels.
- Operationally disentangle environment drift from policy-induced drift (question 3): log policy version, surge parameters, and experiment flags as first-class fields in the training corpus; maintain frozen-policy shadow evaluations (holdout zones or switchback windows) so probe decay can be attributed. Trigger scheduled warm-start retrains on environment drift (decay curves degrade under frozen policy too); on policy changes, do NOT trigger an immediate encoder retrain — the encoder should be action-conditioned per the causal workstream, and post-change data should be folded in only after adequate state-action coverage exists, otherwise the encoder memorizes the new policy rather than the market.
- Sequence the evidence before adding TDG complexity: run a bake-off on your own Eval-Stream between (a) latest warm-started checkpoint, (b) TEA-style averaged checkpoints, (c) prompting/identity-embedding fast adaptation, and (d) a FreKoo-style low-frequency-only extrapolation regularizer. The Reality Check result predicts (a) is hard to beat — require a sustained win across at least two cities and two quarters before promoting anything fancier to production.
- For small per-zone tabular probe models (not the main encoder), pilot Drift-Resilient TabPFN-style in-context adaptation: it is the only tabular approach shown to consistently beat drift-ignoring supervised learning and needs no retraining infrastructure, making it a low-cost hedge for long-tail cities with sparse data.

Annotated bibliography (15 sources)

2022 Yao, H., Choi, C., Cao, B., Lee, Y., Koh, P. W., & Finn, C. (2022). **Wild-Time: A Benchmark of in-the-Wild Distribution Shift over Time**. *NeurIPS 2022 Datasets & Benchmarks*. **arXiv:2211.14238**.

arxiv.org/abs/2211.14238

Benchmark of 5 real timestamped datasets (patient prognosis, news, drug discovery, etc.) with two evaluation protocols: Eval-Fix (train on data up to time T, test on a fixed future window) and Eval-Stream (rolling evaluation on the next K timestamps). Systematically benchmarks 13 approaches from domain generalization, continual learning, self-supervised learning, and ensembling, finding an average ~20% performance drop from in-distribution to future-time data and that no method category reliably closes the gap over ERM.

Relevance to this project: Directly supplies the temporal-split evaluation contract the project should adopt for every encoder release: Eval-Fix as the release gate and Eval-Stream for tuning retrain cadence; the ~20% ID-to-future decay is exactly the failure mode a market-state encoder must be measured against.

2023 Bai, G., Ling, C., & Zhao, L. (2023). **Temporal Domain Generalization with Drift-Aware Dynamic Neural Networks (DRAIN)**. *ICLR 2023*. **arXiv:2205.10664**.

arxiv.org/abs/2205.10664

Formulates TDG in a Bayesian framework that jointly models data and model dynamics: the network's parameters across historical time points are treated as a sequence of dynamic graphs, and a recurrent graph generator learns the drift so it can synthesize the parameters of a future-time model without seeing future data. Provides uncertainty and generalization-error analysis for the extrapolated model.

Relevance to this project: The canonical 'forecast the encoder's future parameters' approach; a candidate for making the market encoder future-proof between retrains, but its per-timestep parameter synthesis scales poorly to large encoders and would churn latent geometry, conflicting with the project's backward-compatibility constraint.

2023 Hosseini, S., Zhai, M., Hajimirsadeghi, H., & Tung, F. (2023). **Prompting-based Temporal Domain Generalization**. **arXiv:2310.02473** (extended as 'Scalable Temporal Domain Generalization via Prompting', RBC Borealis).

arxiv.org/abs/2310.02473

Keeps a pre-trained backbone frozen and learns three kinds of prompts — global, per-time-domain, and drift-aware prompts that extrapolate temporal dynamics — so only a tiny parameter set adapts to future time periods. Parameter- and time-efficient, requires no future data, and works for classification, regression, and forecasting.

Relevance to this project: Template for the project's two-speed design: freeze the encoder trunk (preserving cross-version compatibility) and confine temporal adaptation to small fast-moving parameters, analogous to letting per-hex/per-slot identity embeddings absorb drift.

2025 Chang, C.-Y., Chuang, Y.-N., Jiang, Z., Lai, K.-H., Jiang, A., & Zou, N. (2025). CODA: Temporal Domain Generalization via Concept Drift Simulator. KDD 2025 (arXiv:2310.01508). arxiv.org/abs/2310.01508

Data-centric, model-agnostic TDG: a concept-drift simulator learns how feature correlations evolve over time and generates simulated future-timestamp training data; any architecture is then trained on the simulated future data instead of stale historical data. Evaluated on temporally evolving tabular classification and regression datasets.

Relevance to this project: Attractive for the project because it leaves the encoder architecture and training pipeline untouched — a drift simulator over zone-level demand/supply features could future-shift the training corpus before each scheduled retrain, independent of the encoder's design.

2025 Zhao, L., & Shen, Y. (2025). Proactive Model Adaptation Against Concept Drift for Online Time Series Forecasting (Proceed). KDD 2025. arXiv:2412.08435. arxiv.org/abs/2412.08435

Observes that in online forecasting, ground truth arrives only after the forecast horizon, so naive online learning always adapts to outdated concepts. Proceed estimates the drift between the recently used training samples and the current test sample, then a learned adaptation generator translates the estimated drift into low-rank parameter adjustments applied before predicting. Outperforms SOTA online learning across five real datasets and multiple forecast backbones.

Relevance to this project: Market data has exactly this label-delay structure (realized demand/ETA outcomes arrive after decisions), so any online adaptation of the encoder or its probes should be proactive drift-conditioned adjustment, not reactive fine-tuning on stale labels.

2025 Madaan, D., Chopra, S., & Cho, K. (2025). Temporal Generalization: A Reality Check. arXiv:2509.23487. arxiv.org/abs/2509.23487

Large-scale evaluation of temporal generalization under a strict no-future-access constraint: convex parameter interpolation and explicit parameter extrapolation across monthly and yearly datasets (language modeling, news tagging, paper categorization, satellite land use, yearbook photos) and multiple architectures. Finding: no evaluated method consistently beats the trivial baseline of just using the latest available model parameters.

Relevance to this project: The most important negative result for the project's design decision (1): sophisticated TDG machinery must demonstrably beat 'retrain frequently, serve the latest warm-started encoder' on your own temporally forward splits before earning its complexity.

2025 Liu, A., Miller, K., Saligrama, V., Saenko, K., Gong, B., Lim, S.-N., & Plummer, B. A. (2025). **Scaling Up Temporal Domain Generalization via Temporal Experts Averaging (TEA)**. EMNLP 2025. **arXiv:2509.26045**.

arxiv.org/abs/2509.26045

Fine-tunes a domain-agnostic base model on each temporal domain with constrained weight changes to create parameter-similar but functionally diverse 'temporal experts', then forecasts the weight trajectory in a PCA subspace and adaptively weight-averages the experts by projected proximity to the future domain. Up to 69% gains and 60x more efficient than parameter-forecasting TDG, across 7 benchmarks and 5 models.

Relevance to this project: The scalable middle ground for the project: keep the last k monthly encoder checkpoints as experts and serve a trajectory-aware weight average — cheap, applicable to a full-size encoder, and weight averaging inherently keeps successive versions geometrically close, aiding backward compatibility.

2025 Yu, E., Lu, J., Yang, X., Zhang, G., & Fang, Z. (2025). **Learning Robust Spectral Dynamics for Temporal Domain Generalization (FreKoo)**. **arXiv:2505.12585**.

arxiv.org/abs/2505.12585

Fourier-decomposes the trajectory of model parameters over time: low-frequency components (long-term structure, periodicity) are extrapolated with a Koopman operator, while high-frequency components (transient noise) are suppressed via temporal regularization with Bayesian justification and stability guarantees. Outperforms TDG baselines especially in streaming settings with complex drift.

Relevance to this project: Encodes the right inductive bias for a market encoder: extrapolate the slow structural drift (city growth, seasonality) and explicitly refuse to chase high-frequency noise (weather shocks, one-off events) — a principle worth adopting even if only as regularization on how fast fast-parameters may move.

2024 Helli, K., Schnurr, D., Hollmann, N., Müller, S., & Hutter, F. (2024). **Drift-Resilient TabPFN: In-Context Learning Temporal Distribution Shifts on Tabular Data**. NeurIPS 2024. **arXiv:2411.10634**.

arxiv.org/abs/2411.10634

A prior-data-fitted network trained on synthetic datasets generated from structural causal models that gradually shift over time; at deployment it ingests the entire (timestamped) training set in-context and predicts the future test set in one forward pass, approximating Bayesian inference under drift. Beats XGBoost, CatBoost, TabPFN, and Wild-Time-applicable methods (accuracy 0.688 to 0.744) with no hyperparameter tuning.

Relevance to this project: Notably, the first tabular method to consistently beat drift-ignoring supervised learning — evidence that modeling drift as gradual SCM change works; its in-context 'adapt without retraining' paradigm suits small per-zone probe models even if the main encoder stays amortized.

2021 Zhang, M., Marklund, H., Dhawan, N., Gupta, A., Levine, S., & Finn, C. (2021). Adaptive Risk Minimization: Learning to Adapt to Domain Shift. NeurIPS 2021. arXiv:2007.02931.
proceedings.neurips.cc/paper/2021/hash/c705112d1ec18b97acac7e2d6397342...

Meta-learns adaptation instead of invariance: training data are structured into groups/domains, and the model is trained so that a small adaptation procedure using an unlabeled batch of test-time context (e.g., a context network producing an embedding of the current batch) provably improves post-adaptation risk. At test time the model adapts to the new domain from unlabeled data alone.

Relevance to this project: The safe alternative to naive test-time adaptation for the project: the encoder can be meta-trained so that a context window of the current day's unlabeled market data adapts it, with the adaptation mechanism validated offline rather than improvised in the serving loop.

2021 Wang, D., Shelhamer, E., Liu, S., Olshausen, B., & Darrell, T. (2021). Tent: Fully Test-Time Adaptation by Entropy Minimization. ICLR 2021 (Spotlight). arXiv:2006.10726.
arxiv.org/abs/2006.10726

Adapts a deployed model using only unlabeled test batches by minimizing prediction entropy, updating only batch-norm statistics and channel-wise affine parameters, one gradient step per batch. Set SOTA on ImageNet-C and works for source-free domain adaptation.

Relevance to this project: The archetype of lightweight serve-time adaptation (update only normalization/affine parameters — analogous to updating only per-hex identity embeddings), but its confidence-seeking objective is dangerous in a system whose own surge decisions shape the data it would adapt on.

2023 Zhao, H., Liu, Y., Alahi, A., & Lin, T. (2023). On Pitfalls of Test-Time Adaptation. ICML 2023. arXiv:2306.03536.
arxiv.org/abs/2306.03536

Introduces TTAB, a benchmark of 10 SOTA test-time adaptation algorithms under diverse shifts with rigorous protocols. Identifies three systematic failure modes: hyperparameter/model selection is nearly impossible under online batch dependency; TTA effectiveness depends heavily on the quality of the pre-trained model; and even under ideal conditions no method handles all shift types.

Relevance to this project: The documented risk register for question (3)-adjacent decisions: it justifies gating Tent-style online adaptation out of the pricing/dispatch feedback loop and restricting serve-time adaptation to mechanisms validated with temporally forward offline protocols.

2022 Fini, E., da Costa, V. G. T., Alameda-Pineda, X., Ricci, E., Alahari, K., & Mairal, J. (2022). Self-Supervised Models are Continual Learners (CaSSLe). CVPR 2022. arXiv:2112.04215.
arxiv.org/abs/2112.04215

Shows that any self-supervised loss can be converted into a continual-learning distillation mechanism by adding a predictor network that maps current representations onto their past state, so the new encoder is trained jointly on the SSL objective for new data and consistency with its predecessor. Works across six SSL methods with minimal tuning and substantially reduces representation forgetting.

Relevance to this project: Answers 'does continual SSL forget?' (yes) and gives the fix — and the predictor-based old-to-new alignment is mechanically the same tool the versioning survey needs for backward-compatible embedding upgrades, so one component serves both goals.

2024 Ibrahim, A., Thérien, B., Gupta, K., Richter, M. L., Anthony, Q., Lesort, T., Belilovsky, E., & Rish, I. (2024). Simple and Scalable Strategies to Continually Pre-train Large Language Models. **arXiv:2403.08763 (TMLR).**

arxiv.org/abs/2403.08763

Empirically establishes that continual pre-training with three simple ingredients — learning-rate re-warming, LR re-decaying, and replay of previous data — matches the performance of fully retraining from scratch on the union of datasets, under weak (English-to-English) and strong (English-to-German) shifts at 405M and 10B parameters over hundreds of billions of tokens.

Relevance to this project: The strongest evidence for the project's retraining-policy question: a warm-started encoder with LR re-warming and a historical replay buffer can match full retrains at a fraction of the compute, making frequent warm-start retraining the default over exotic TDG objectives.

2020 Ash, J. T., & Adams, R. P. (2020). On Warm-Starting Neural Network Training. **NeurIPS 2020. arXiv:1910.08475.**

arxiv.org/abs/1910.08475

Documents the warm-start pathology: initializing from a previously trained network on incrementally grown data reaches similar training loss but systematically worse generalization than random re-initialization. Proposes shrink-and-perturb — shrink old weights toward zero and add small noise before resuming training — which closes the generalization gap while retaining most of the compute savings.

Relevance to this project: The must-know caveat for a continuously warm-started market encoder: every scheduled retrain should apply shrink-and-perturb (or LR re-warming per Ibrahim et al.) or the encoder will slowly ossify against new market regimes despite fitting the training data.

14 Temporal Distribution Shift: Generalization & Continual Adaptation of the Encoder

This chapter closes a gap identified by the completeness critic; it was researched by a dedicated gap-fill agent.

State of the art

Continuous-time representation learning treats the raw data — an asynchronous, marked event stream on an evolving graph — as the object to encode, rather than pre-aggregating it into per-zone-per-bucket tensors. The field grew from two roots that merged around 2019-2020: (1) neural temporal point processes (Neural Hawkes 2017, Transformer Hawkes 2020), which model the conditional intensity $\lambda(t)$ of the event process and thus represent 'demand' as an arrival rate rather than a count — the principled framing under censoring, since a censored likelihood can recover latent demand suppressed by price or supply caps; and (2) continuous-time dynamic graph embeddings (DyRep 2019, JODIE 2019, TGAT 2020, unified by TGN 2020), which maintain per-node embeddings updated event-by-event, with JODIE's projection operator making the embedding queryable at ANY timestamp between updates. Since 2023 the field has been reshaped by rigorous benchmarking (Temporal Graph Benchmark, NeurIPS 2023) which exposed that on dynamic node-property prediction — the task closest to estimating a zone's market state — simple heuristics and memory-free sequence models (DyGFormer 2023, CRAFT 2025) often match or beat complex memory-based models at a fraction of the serving cost; industrial frameworks (TGL at VLDB 2022, LasTGL 2023) show billion-edge training is feasible but confirms that per-node memory imposes sequential-update and state-management burdens. Open problems most relevant here: principled evaluation beyond link prediction, long-range dependency modeling, action-

conditioned/counterfactual intensity under policy-endogenous logging, and how stateful memories interact with model versioning. The pragmatic 2026 consensus is a hybrid: heavy event-level encoding offline or as auxiliary training signal, cheap stateless queries online — not wholesale replacement of aggregated pipelines.

Key principles

- Make time a first-class input, not an index: functional time encodings (TGAT's Bochner features, THP's temporal encodings) let one embedding be a queryable function of continuous time, eliminating the bucket width as a hand-engineered hyperparameter at the representation level even if serving stays bucketed.
- There are exactly two mechanisms for any-timestamp queryability, with opposite ops profiles: (a) stored state + learned projection/decay (JODIE, TGN memory) — $O(1)$ at query time but introduces version-specific mutable state; (b) stateless recomputation over a recent event window (TGAT, DyGFormer, CRAFT) — pure function of data, replayable and version-portable, at higher per-query compute. Choose per consumer latency budget.
- Likelihood of the event process (TPP intensity) is the principled training signal for demand: it models the generative arrival process rather than a bucketed statistic, supports censored likelihoods that recover demand suppressed by the surge lever, and conditioning the intensity on the action gives an action-conditioned representation — directly addressing policy-endogeneity of logged data.
- Auxiliary event-level objectives transfer even when serving stays aggregated: DyRep's core insight — parameterize event intensities BY the embedding — means a bucketed encoder trained with an intensity head is forced to encode the underlying arrival dynamics without any event-level serving infrastructure.
- Benchmark before believing: TGB shows complex temporal-graph models often lose to persistence/moving-average baselines on node property prediction (the task class closest to zone market state), and rankings flip across datasets — event-level lift must be demonstrated on the company's own downstream tasks, not assumed.
- Memory-free is winning on the efficiency-accuracy frontier (DyGFormer 2023, CRAFT 2025): per-node memories create sequential update dependencies, train-serve skew risk, and version-coupled state; recent-history attention restores parallelism and determinism.
- Split compute offline-heavy / online-cheap: expensive encoding of long event histories offline (or in training-time auxiliary heads), cheap projection or short-window recomputation online — the same pattern as BigST in the spatiotemporal survey, mirrored by JODIE's stored-trajectory + projection design.

Recommendations for the market-state embedding project

- BUILD NOW — TPP intensity auxiliary head on the bucketed encoder (answers Q2: yes, and it is the highest value-per-cost item in this survey). Add a head that models the conditional intensity of request/driver-session arrivals within each zone as a function of the market-state latent z AND the concurrently active surge action a : $\lambda(t | z, a)$. Use a censored likelihood (Gammelli et al.) so bookings suppressed or abandoned under high surge are treated as censored observations of latent demand, not as low demand. This forces the latent to encode the un-censored arrival process and its price sensitivity — the action-conditioned, decision-relevant structure that reconstruction loss cannot induce. Architecture: THP-style attention head or a Shchur-style mixture-density head (closed-form likelihood, no numerical integration) to keep training cheap.
- PILOT — stateless event-window encoder as an input pathway (answers Q1: bucketed aggregates are a strong baseline; event-level wins are task- and regime-dependent). Replace hand-binned within-bucket counts with a DyGFormer/CRAFT-style transformer over the last N raw events per zone (with Bochner time encodings), feeding the bucketed spine. Evaluate strictly against the persistence and bucketed-count baselines per TGB's findings, on downstream linear-probe metrics, and expect gains concentrated in bursty/fine-timescale regimes (event onset, surge transitions) rather than uniformly. Budget per TGL: full event-level training at

billions of events needs dedicated infrastructure — use LasTGL/TGL-style tooling rather than a bespoke stack for the pilot.

- **BUILD** — JODIE-style time-of-query projection on the served embedding (answers Q3: yes, with a caveat). Store per-zone state at each refresh tick plus a learned drift/decay head $g(z, \delta_t)$ so any consumer can request the embedding at its own timestamp; this serves heterogeneous freshness needs (dispatch wants seconds-fresh, forecasting is fine with the tick) from one artifact with no extra refresh infrastructure. Caveat: projection accuracy decays with elapsed time — publish the measured staleness-vs-accuracy curve as part of the embedding contract so consumers can decide whether projected values meet their SLO.
- **AVOID** for the market-state backbone — per-entity TGN memory (answers Q4). Memory vectors are encoder-version-specific hidden state: they are meaningless under a new encoder version and must be rebuilt by replaying a bounded event history at every version bump, complicating the backward-compatibility contract and creating train-serve skew risk. Prefer memory-free designs where the embedding is a deterministic function of (recent event window, model weights): version migration is then just re-running the new encoder over the same window, and shadow-serving two versions is trivial. If a memory-based component is ever adopted, mandate (a) a bounded replay window sufficient to reconstruct memory from raw events, and (b) replay-based warm-up as part of the version-rollout runbook.
- **NET BUILD/SKIP VERDICT**: build the event-level layer as **TRAINING SIGNAL** and **INPUT PATHWAY**, skip it as **SERVING BACKBONE**. Concretely: (1) action-conditioned censored TPP intensity head — build now; (2) stateless per-zone event-window encoder feeding the bucketed spine — pilot behind an ablation gate; (3) time-of-query projection on served embeddings — build; (4) full per-entity continuous-time dynamic-graph system (memories over all drivers/riders) — skip for market state; revisit only for entity-level products (driver churn, fraud, matching affinity) where per-entity trajectories are the actual object of interest.
- **Evaluation add-ons**: adopt TGB-style protocol discipline (temporal splits, strong naive baselines, node-property tasks) for the event-level pilot, and use the EasyTPP benchmark toolkit (ICLR 2024, <https://arxiv.org/abs/2307.08097>) for standardized TPP-head goodness-of-fit checks; report intensity calibration under held-out surge levels to verify the action-conditioning actually generalizes off-policy.

Annotated bibliography (15 sources)

2019 Kumar, S., Zhang, X., & Leskovec, J. (2019). Predicting Dynamic Embedding Trajectory in Temporal Interaction Networks (JODIE). KDD 2019.

kdd.org/kdd2019/accepted-papers/view/predicting-dynamic-embedding-traj...

Two coupled RNNs update the embeddings of the two endpoints of every user-item interaction event, so each entity's embedding is a trajectory in latent space rather than a snapshot. The key novelty is a projection operator — an attention-scaled temporal drift — that estimates an entity's embedding at any future time t after its last event, trained end-to-end so projected embeddings predict the next interaction. A t-Batch algorithm builds time-consistent minibatches for 9x faster training.

Relevance to this project: The projection operator is the canonical answer to the staleness-between-refresh-ticks serving problem: one stored state per zone plus a cheap learned drift yields an embedding at any query timestamp, letting consumers with different freshness needs share one artifact without extra refresh infrastructure.

2020 Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., & Bronstein, M. (2020). Temporal Graph Networks for Deep Learning on Dynamic Graphs. arXiv:2006.10637.

arxiv.org/abs/2006.10637

TGN is the unifying framework for continuous-time dynamic graphs represented as event sequences: each node holds a memory vector updated by message functions when events touch it, and an embedding module (e.g., temporal graph attention over recent neighbors) computes the served embedding from memory plus recent events. JODIE, DyRep, and TGAT are shown to be specific instances. A training strategy decouples memory updates from gradient computation to keep parallelism while respecting event order.

Relevance to this project: This is the reference architecture if the project builds a per-zone event-level encoder: zone memory updated by request/driver-session events, with the embedding module and projection choices as design knobs. It also crystallizes the versioning problem — memory is learned-parameter-dependent state, so a new encoder version requires memory rebuild by event replay.

2020 Xu, D., Ruan, C., Korpeoglu, E., Kumar, S., & Achan, K. (2020). Inductive Representation Learning on Temporal Graphs (TGAT). ICLR 2020.

arxiv.org/abs/2002.07962

TGAT replaces positional encoding in self-attention with a functional time encoding derived from Bochner's theorem (learned random Fourier features of time deltas), then attends over a node's temporal neighborhood to produce an embedding as an explicit function of query time. It is fully inductive and memoryless: the embedding is recomputed from the recent event history at any requested timestamp.

Relevance to this project: Demonstrates the stateless alternative to JODIE-style projection: time-of-query is just an input feature, so any timestamp is servable by recomputation over a bounded event window — attractive under the versioning contract because there is no encoder-version-specific state. Its Bochner time encoding is now the standard way to make time a first-class input, directly reusable in the bucketed encoder.

2019 Trivedi, R., Farajtabar, M., Biswal, P., & Zha, H. (2019). DyRep: Learning Representations over Dynamic Graphs. ICLR 2019.

dblp.org/rec/conf/iclr/TrivediFBZ19.html

DyRep couples representation learning with a two-time-scale deep temporal point process: fast 'communication' events and slow 'association' (topology) events each have conditional intensities parameterized by the node embeddings, and embeddings are updated by a temporally-attentive network when events occur. Training maximizes the likelihood of the observed event stream under these intensities.

Relevance to this project: The direct precedent for the project's most valuable idea: making embeddings the parameters of an event-intensity model, so the latent space is organized by 'what drives arrivals' rather than by reconstruction. It is the bridge between the dynamic-graph and TPP halves of this survey.

2017 Mei, H., & Eisner, J. (2017). The Neural Hawkes Process: A Neurally Self-Modulating Multivariate Point Process. NeurIPS 2017.

arxiv.org/abs/1612.09328

A continuous-time LSTM whose hidden state decays between events defines the conditional intensity of each event type at every instant; past events excite or inhibit future intensities in learned, nonlinear ways, generalizing the classical Hawkes process. Trained by maximizing the exact continuous-time log-likelihood of the marked event sequence.

Relevance to this project: The seminal neural TPP — establishes that a latent state evolving continuously between events, with intensity as the training signal, is the principled representation of an asynchronous arrival process like ride requests, independent of any bucket width.

2020 Zuo, S., Jiang, H., Li, Z., Zhao, T., & Zha, H. (2020). Transformer Hawkes Process. ICML 2020 (PMLR v119).

proceedings.mlr.press/v119/zuo20a.html

Replaces the RNN in neural TPPs with self-attention plus sinusoidal-style temporal encodings: the transformer encodes the history of marked events, and the intensity at any time is a softplus function of the encoded state, capturing long-range dependencies that RNN-based TPPs miss. Outperforms prior TPPs in likelihood and event prediction, and can incorporate relational structure to model multiple interacting point processes.

Relevance to this project: The standard high-capacity intensity head: a THP-style head on the market-state latent (conditioned additionally on the surge action) is the concrete architecture for the proposed auxiliary intensity objective, and its multi-process extension matches interacting per-zone demand streams.

2021 Shchur, O., Turkmen, A. C., Januschowski, T., & Gunnemann, S. (2021). Neural Temporal Point Processes: A Review. IJCAI 2021.

ijcai.org/proceedings/2021/623

Systematizes neural TPP design: how to encode event history (RNN vs attention), how to parameterize the conditional distribution (intensity, density, or CDF/mixture heads — the latter giving closed-form likelihoods without numerical integration), and how to condition on marks and covariates. Also surveys applications and evaluation pitfalls.

Relevance to this project: The design manual for the proposed TPP auxiliary head; in particular its mixture-density parameterizations avoid intensity-integral computation, which matters if the head must run in a production training loop at company scale.

2022 Zhou, Z., Yang, X., Rossi, R., Zhao, H., & Yu, R. (2022). Neural Point Process for Learning Spatiotemporal Event Dynamics (DeepSTPP). L4DC 2022 (PMLR v168).

proceedings.mlr.press/v168/zhou22a/zhou22a.pdf

Models a joint space-time intensity as a mixture of kernels governed by a latent process inferred with amortized variational inference; the construction gives closed-form integration of the intensity, so training and forecasting avoid numerical integration entirely. Forecasts irregularly-sampled events over continuous space and time.

Relevance to this project: Shows how to drop the hand-drawn zone partition as well as the time bucket: demand becomes a continuous spatial intensity surface, and hexagon-level features become queries against it. A candidate teacher/auxiliary target even if serving stays zone-indexed.

2020 Gammelli, D., Peled, I., Rodrigues, F., Pacino, D., Kurtaran, H. A., & Pereira, F. C. (2020). Estimating Latent Demand of Shared Mobility through Censored Gaussian Processes. Transportation Research Part C.

arxiv.org/abs/2001.07402

Treats observed mobility demand as right-censored by available supply and fits a Gaussian Process with a censored likelihood (Tobit-style), recovering the latent demand curve rather than the supply-capped observation. Demonstrated on shared-mobility data where naive models systematically underestimate true demand.

Relevance to this project: The censoring-aware likelihood is exactly what a bucketed count inherits silently: when surge suppresses or aborts requests, observed bookings understate demand. Combining this censored-likelihood idea with a neural intensity head conditioned on the surge action yields the action-conditioned, causally-meaningful demand representation the project needs for its pricing lever.

2023 Huang, S., Poursafaei, F., Danovitch, J., Fey, M., Hu, W., Rossi, E., Leskovec, J., Bronstein, M., Rabusseau, G., & Rabbany, R. (2023). Temporal Graph Benchmark for Machine Learning on Temporal Graphs. NeurIPS 2023 Datasets & Benchmarks.

arxiv.org/abs/2307.01026

Large-scale, multi-year benchmark datasets (social, trade, transaction, transport) with standardized loaders, realistic negative sampling, and leaderboards for both dynamic link prediction and dynamic node property prediction. Headline finding: model rankings vary drastically across datasets, and on node property prediction simple baselines (e.g., moving averages / persistence) often beat sophisticated temporal GNNs.

Relevance to this project: Directly calibrates question 1 of the brief: for zone-state (node-property-like) targets, event-level models are NOT reliably better than simple aggregates, so any event-level stage must be validated against a persistence/bucketed baseline on the company's own downstream tasks before adoption.

2023 Yu, L., Sun, L., Du, B., & Lv, W. (2023). Towards Better Dynamic Graph Learning: New Architecture and Unified Library (DyGFormer / DyGLib). NeurIPS 2023.

arxiv.org/abs/2303.13047

DyGFormer is a transformer over a node's first-hop interaction history only: a neighbor co-occurrence encoding captures source-destination correlation, and a patching trick compresses long histories so attention scales to thousands of past events. No node memory, no multi-hop aggregation; released with DyGLib, a unified training/evaluation library that standardized comparisons across 9 prior methods.

Relevance to this project: Evidence that a memory-free sequence encoder over recent events matches or beats memory-based TGNs — meaning the project can get event-level signal as a pure function of a sliding event window: deterministic, replayable, and free of version-specific memory state.

2025 Yi, L., Lei, R., Mo, F., Zheng, Y., Wei, Z., & Ye, Y. (2025). Future Link Prediction Without Memory or Aggregation (CRAFT). NeurIPS 2025; arXiv:2505.19408.

arxiv.org/abs/2505.19408

CRAFT discards both memory and message-passing aggregation: learnable node identifier embeddings plus cross-attention between a candidate destination and the source's recent interaction sequence perform target-aware matching. Achieves superior accuracy and efficiency across diverse temporal-graph datasets, including strong generalization to unseen edges.

Relevance to this project: The 2025 SOTA data point confirming the field's drift away from stateful per-event memories toward cheap stateless architectures — reinforcing the recommendation that any event-level component in the market-state encoder be stateless for serving simplicity and version portability.

2022 Zhou, H., Zheng, D., Nisa, I., Ioannidis, V., Song, X., & Karypis, G. (2022). TGL: A General Framework for Temporal GNN Training on Billion-Scale Graphs. VLDB 2022.

arxiv.org/abs/2203.14883

A production training framework decomposing temporal GNNs into temporal sampler, mailbox, node memory, memory updater, and message-passing engine; a Temporal-CSR structure, parallel sampling, and random chunk scheduling deliver ~13x speedup and one-epoch throughput of over a billion temporal edges on multi-GPU machines.

Relevance to this project: Establishes the compute envelope for question 1: event-level training at ride-hailing scale (billions of events) is feasible but requires dedicated infrastructure of exactly this shape — a real cost that must be weighed against the measured downstream lift.

2023 Li, J., et al. (2023). LasTGL: An Industrial Framework for Large-Scale Temporal Graph Learning. arXiv:2311.16605.

arxiv.org/abs/2311.16605

An industrial framework (Ant Group lineage) offering unified, extensible implementations of temporal GNNs — both snapshot-based and continuous-time — with datasets, tutorials, and production-oriented components for prototyping through deployment.

Relevance to this project: Shows what industrial adoption of temporal-graph learning actually looks like and provides an off-the-shelf starting point for the pilot, avoiding a bespoke event-encoder stack.

2025 Jiao, P., Chen, H., Guo, X., Zhao, Z., He, D., & Jin, D. (2025). A Survey on Temporal Interaction Graph Representation Learning: Progress, Challenges, and Opportunities. IJCAI 2025 Survey Track.

ijcai.org/proceedings/2025/1166

The most recent field survey: a taxonomy of temporal interaction graph representation learning by information type (temporal, structural, semantic), covering memory-based, attention-based, and prompt/foundation-model directions, plus open challenges in scalability, evaluation, and long-range dependencies.

Relevance to this project: The up-to-date map of the area (through 2025) confirming the survey's framing — memory vs memory-free, evaluation beyond link prediction, and foundation-model directions — and a citation anchor for the project's literature appendix.

Additional sources contributed by the completeness critic

2021 Wager, S., Xu, K. (2021). Experimenting in Equilibrium. Management Science 67(11). arXiv:1903.02124.

arxiv.org/abs/1903.02124

Designs experiments for large stochastic systems with cross-unit interference captured by mean-field modeling: instead of A/B splitting units, the platform applies small random local perturbations ('dithering') to a global lever (e.g., supply-side incentives on a ride-sharing platform) while the system stays in equilibrium. Combining this unobtrusive randomization with lightweight structural modeling yields consistent estimates of the derivative of steady-state outcomes with respect to the lever — the marginal policy effect — which naive A/B comparisons get wrong under interference.

Relevance to this project: Gives the mathematically grounded recipe for the permanent lever randomization the project needs so that surge dose-response is identifiable from logs, and shows why zone-randomized A/Bs of pricing are biased in equilibrium — directly shaping both logging spec and causal-head training data.

2022 Johari, R., Li, H., Liskovich, I., Weintraub, G.Y. (2022). Experimental Design in Two-Sided Platforms: An Analysis of Bias. Management Science 68(10). arXiv:2002.05670.

arxiv.org/pdf/2002.05670

Builds a stochastic two-sided market model (customers arriving to a shared, limited supply of listings/drivers) and analytically characterizes the bias of customer-side, listing-side, and two-sided randomized experiments when treatment changes market balance. Shows bias can be as large as the treatment effect itself and that two-sided randomization reduces but does not eliminate it, with bias formulas depending on market tightness.

Relevance to this project: The canonical statement of why naive A/B tests on a ride-hailing/delivery platform mis-measure intervention effects — any evaluation of embedding-driven pricing/dispatch policies, and any engagement-supervised training label, inherits this bias unless corrected.

2023 Bojinov, I., Simchi-Levi, D., Zhao, J. (2023). Design and Analysis of Switchback Experiments. Management Science 69(7). arXiv:2009.00148.

arxiv.org/pdf/2009.00148

Formalizes switchback experiments — randomly alternating a whole market between treatment and control over time — under carryover effects of bounded order, casting optimal switch-interval design as a minimax discrete optimization and solving it via continuous relaxation, with exact randomization-based inference. Provides the standard design/analysis toolkit used by ride-hailing and delivery platforms for market-level interventions where unit-level randomization is impossible.

Relevance to this project: Switchbacks are the workhorse for generating exogenous variation in the surge lever at market level; the resulting logs are exactly the action-randomized data the action-conditioned encoder heads and OPE scorecards require.

2022 Farias, V., Li, A.A., Peng, T., Zheng, A. (2022). Markovian Interference in Experiments. NeurIPS 2022 (oral). arXiv:2206.02371.

arxiv.org/abs/2206.02371

Models experiments in dynamical systems where interventions on some units affect others through a shared state (e.g., limited driver inventory) as a Markov decision process, and proposes the Differences-In-Q's (DQ) estimator: estimate Q-functions from logged data and difference them between treatment and control states. DQ has bias second-order in the intervention size but exponentially smaller variance than full off-policy evaluation, empirically dominating naive and OPE estimators in a ride-hailing simulator.

Relevance to this project: Connects the project's learned value/Q auxiliary heads directly to unbiased-ish experiment readouts under interference: the same embedding-based Q heads used for dispatch can power DQ estimation of new-policy and new-embedding-version effects.

2022 Luo, S., Yang, Y., Shi, C., Yao, F., Ye, J., Zhu, H. (2022, rev. 2023). Policy Evaluation for Temporal and/or Spatial Dependent Experiments. arXiv:2202.10887.

arxiv.org/abs/2202.10887

Introduces a Varying Coefficient Decision Process model for experiments run in ride-sourcing markets where outcomes depend on treatments at other times and neighboring regions; treatment effects are decomposed into direct and indirect (spillover) components with estimation and inference procedures, validated on simulations and real ride-hailing applications.

Relevance to this project: This is exactly the project's setting — spatially and temporally dependent lever experiments in ride-hailing — and its direct/indirect decomposition defines the estimands the spillover-aware auxiliary heads should target.

2025 Zhu, J., Li, J., Zhou, H., Lin, Y., Lin, Z., Shi, C. (2025). Balancing Interference and Correlation in Spatial Experimental Designs: A Causal Graph Cut Approach. ICML 2025. arXiv:2505.20130.

arxiv.org/abs/2505.20130

Optimizes the assignment of spatial units to treatment clusters by minimizing a surrogate of the MSE of the global treatment effect estimator via graph-cut algorithms, jointly trading off interference across cluster boundaries against outcome correlation within clusters, and accommodating moderate-to-large interference and general covariance structures. Validated on synthetic environments and a ridesharing-market simulator.

Relevance to this project: State-of-the-art 2025 answer to 'how should we partition hexes into experiment clusters for surge experiments' — the design layer that determines the quality of all causal training signal on the zone graph.

2022 Ma, H., Fang, F., Parkes, D.C. (2022). Spatio-Temporal Pricing for Ridesharing Platforms. *Operations Research* 70(2). arXiv:1801.04015.

arxiv.org/pdf/1801.04015

Develops the Spatio-Temporal Pricing (STP) mechanism in a discrete-time, multi-location model of a ridesharing market: prices computed from a minimum-cost-flow planning problem are smooth in space and time, making it a subgame-perfect equilibrium for strategic drivers to accept dispatches rather than chase surge into neighboring zones; the mechanism is welfare-optimal, envy-free, and budget balanced from any history onward.

Relevance to this project: Supplies the structural/economic model of WHY surge creates spatial spillovers (driver strategic repositioning) and what smoothness constraints prevent them — grounding for simulator-based counterfactual training data and for spatial-smoothness regularization of lever-response heads.

2022 Yao, H., Choi, C., Cao, B., Lee, Y., Koh, P.W., Finn, C. (2022). Wild-Time: A Benchmark of in-the-Wild Distribution Shift over Time. *NeurIPS 2022 Datasets & Benchmarks*. arXiv:2211.14238.

arxiv.org/abs/2211.14238

A benchmark of five real datasets with natural temporal distribution shift, defining two evaluation protocols — Eval-Fix (train on past, test on a fixed future window) and Eval-Stream (continual evaluation over a data stream) — and adapting 13 methods from domain generalization, continual learning, SSL, and ensembling by treating time windows as domains. Finds an average ~20% performance drop from in-distribution to future out-of-distribution data with no method reliably winning.

Relevance to this project: Provides the evaluation protocol the market-state embedding needs but no survey proposed: probe metrics must be reported on temporally forward splits, since serving-time data is always the future of the training data.

2023 Bai, G., Ling, C., Zhao, L. (2023). Temporal Domain Generalization with Drift-Aware Dynamic Neural Networks (DRAIN). *ICLR 2023 (oral)*. arXiv:2205.10664.

arxiv.org/abs/2205.10664

Formulates temporal domain generalization as a Bayesian sequence problem over both data and model parameters: a recurrent graph-generation mechanism treats the network's weights as a dynamic graph, learns the drift of parameters across sequential training domains, and extrapolates the model that will fit the near future without seeing future data, with uncertainty and generalization-error analysis.

Relevance to this project: Represents the method family for making a periodically retrained encoder anticipate drift rather than lag it — an alternative or complement to frequent warm-start retraining, with direct implications for retraining cadence and version churn.

2019 Kumar, S., Zhang, X., Leskovec, J. (2019). Predicting Dynamic Embedding Trajectory in Temporal Interaction Networks (JODIE). KDD 2019, pp. 1269-1278.

kdd.org/kdd2019/accepted-papers/view/predicting-dynamic-embedding-traj...

Learns per-entity embedding trajectories from a raw stream of timestamped interactions using two coupled RNNs that mutually update user and item embeddings at every event, plus a learned projection operator that estimates an entity's embedding at any future query time (embedding drift between events). A t-Batch algorithm creates time-consistent training batches for 9x faster training; trained on future-interaction and state-change prediction objectives.

Relevance to this project: The projection-operator idea solves a serving problem none of the surveys addressed: consumers querying the market-state embedding between refresh ticks can get a time-projected embedding instead of a stale one, and the event-level updates remove dependence on hand-chosen aggregation buckets.

2020 Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., Bronstein, M. (2020). Temporal Graph Networks for Deep Learning on Dynamic Graphs. ICML 2020 GRL+ Workshop. arXiv:2006.10637.

arxiv.org/abs/2006.10637

A general framework for inductive representation learning on continuous-time dynamic graphs: each node keeps a memory vector updated by message functions when timestamped interaction events arrive, and a graph-attention embedding module computes up-to-date node embeddings from memories and temporal neighborhoods; trained self-supervised on future edge prediction. Subsumes JODIE, DyRep, and TGAT as special cases.

Relevance to this project: The canonical architecture for maintaining continuously updated per-zone state from raw trip/driver event streams on the hex adjacency graph — the dynamic counterpart to the static region embeddings the spatiotemporal survey covered, and the subfield it admitted was missing.

2020 Zuo, S., Jiang, H., Li, Z., Zhao, T., Zha, H. (2020). Transformer Hawkes Process. ICML 2020, PMLR 119.

proceedings.mlr.press/v119/zuo20a/zuo20a.pdf

Models marked temporal point processes with self-attention: event history is encoded by a transformer with temporal encodings, producing hidden states that parameterize a continuous conditional intensity function trained by maximum likelihood plus event time/type prediction losses. Captures long-range dependencies in asynchronous event sequences better than RNN-based neural Hawkes models, with large likelihood and accuracy gains.

Relevance to this project: Ride requests, cancellations, and driver sign-ons are marked point processes; an intensity head on the market-state embedding (conditioned on the surge action) is the principled way to represent demand including censoring effects that bucketed counts hide, and doubles as a decision-relevant auxiliary objective.

15 Master Source Index (197 unique sources, chronological)

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